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Clinical Management of the Adult Kidney Transplant Recipient

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CHAPTER OUTLINE

THE TRANSPLANT SURGERY PROCEDURE

The kidney allograft is placed in the extra peritoneal iliac fossa (Fig. 70.1). A curvilinear incision is made in the right or left lower quadrant, the retroperitoneal space is widened, and the iliac vessels exposed. The external iliac artery and vein are mobilized and surrounding lymphatic vessels are ligated and divided. End-to-side anastomoses are performed between the renal vein and the external iliac vein, followed by the renal artery and the external iliac artery. Alternate techniques for the renal artery are end-to-side to the common iliac artery or end-to-end to the mobilized internal iliac artery. The site of anastomosis is chosen after examining the length, size, and quality of the donor's and recipient's vessels. The Lich-Gregoir implantation of the ureter to the bladder, a technique designed to minimize urinary reflux into the ureter, has become the preferred technique for neo-ureterocystostomy. A double J ureteric stent can be inserted either routinely, or selectively for higher risk candidates. It is usually retrieved 2 to 4 weeks after transplantation in the outpatient setting. A Cochrane review suggested that the incidence of urine leak and ureteric stenosis were significantly reduced with prophylactic uretic stenting.[1,2] A low-pressure suction drain can also be used and retrieved within the first week after surgery. While the surgical techniques for deceased- and living-donor kidney transplant are similar, with deceased donor allografts attention is given to preserving longer stretches of the donor vessels. The renal artery or arteries are procured along with an aortic patch; this technique facilitates the vascular anastomosis and reduces the risk of renal artery stenosis.

LIVE DONOR NEPHRECTOMY

Traditionally, the procurement of kidneys from live donors has been performed through a right or left flank incision. In 1995, a laparoscopic approach for kidney donation was introduced as an alternative that would reduce postoperative pain, wound morbidity, and reduce recovery time associated with the traditional donor nephrectomy.[3] The laparoscopic approach is now the procedure of choice, and is performed in over 90% of live kidney donations in the United States.[4] Initial concerns regarding ureteral complications and graft dysfunction have mostly subsided with improvements in surgical technique and experience.[5,6] Pure laparoscopic, hand-assisted, robotic-assisted, and single-port access are variants of the laparoscopic donor nephrectomy. All of these techniques avoid flank incision by intraabdominal mobilization of the kidney after establishing the pneumoperitoneum. A periumbilical or infra-umbilical small incision, which spares transection of muscle tissue, is used to retrieve the kidney. The hand-assisted variant is the most popular approach. A small periumbilical or infra-umbilical incision of 8 cm allows

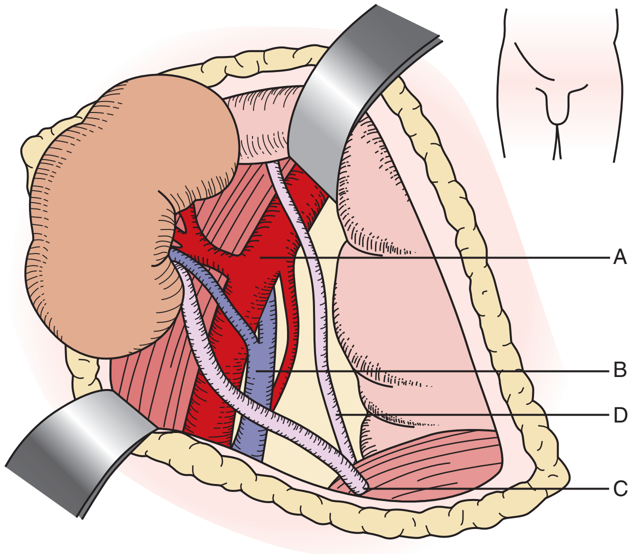

Fig. 70.1 The anatomy of a typical first kidney transplant. A, external iliac artery, B, external iliac vein, C, implanted donor ureter, D, native ureter.

the surgeon to insert one hand inside the abdomen with the help of a device that seals the pneumoperitoneum. The hand is used to help with retraction of the kidney and also allows faster retrieval of the kidney, reducing the warm ischemia by 50% compared to a pure laparoscopic approach. The left kidney is preferred even more so than with the open nephrectomy, because the laparoscopic instrument used to secure the renal vein effectively reduces its length by ½ to 1 cm. When an open donor nephrectomy is performed, an incision is made from the rectus muscle in the direction of the twelfth right or left rib tip. This operation is retroperitoneal as opposed to the laparoscopic approach, which is transperitoneal. Smaller muscle sparing incisions are now used to improve the postoperative recovery.

HANDLING AND PRESERVATION OF THE DONOR KIDNEY

The basic principle of organ preservation is to prevent damage from ischemia by replacing the circulating blood with a preservation solution, and retrieving and cooling the organs expediently. The composition of preservation solutions vary, but all have ingredients designed to a) minimize intracellular edema, b) preserve the integrity of cells and tissue, and c) buffer free radicals.

The two cold storage preservation solutions most widely used in the United States are the University of Wisconsin (Viaspan, UW) and the Histidine-Tryptophan Ketoglutarate (Custodiol, HTK) solutions. The UW solution has an electrolyte composition similar to that of the intracellular fluid, with a high concentration of potassium (120 mmol/L). Lactobionate and raffinose prevent cellular edema, and hydroxyethyl starch buffers free radicals. Its viscosity is three times that of water, making it relatively difficult to flush. The HTK solution is a low viscosity, low sodium (15 mmol/L) solution. The concentration of potassium is also low (15 mmol/L), and this reduced the risk of hyperkalemia after transplantation. Histidine serves as a buffer, and tryptophan and mannitol are free radical scavengers.

The UW solution had been the preservation solution of choice for multiorgan deceased donor recovery since its introduction in 1988. However, HTK is used by a growing number of Organ Procurement Organizations (OPO). One of the major advantages of HTK is its relatively low cost. Two small randomized studies comparing UW to HTK preservation solutions found no difference in delayed graft function or graft survival between groups.[7,8] However, a large retrospective (non-randomized) study comparing HTK to UW preservation solution found that HTK use was associated with an increased risk of late death-censored graft loss.[9] For living donor kidneys with short ischemia times, the use of simple and inexpensive solutions such as heparinized lactated Ringer's with procaine has been proposed; however, we prefer to use a preservation solution such as HKT.[10]

Hypothermic machine perfusion is an alternative to static cold preservation for deceased donor kidneys. Following standard retrieval and flushing of the kidney allograft, the renal artery is connected to a perfusion pump that circulates a preservation solution, maintained at 1°C to 10°C. Perfusion parameters can be used to determine the quality of the graft. Hypothermic machine perfusion is associated with a reduction in delayed graft function compared to static cold storage and some, but not all, studies have shown an improvement of graft survival at 1 year.[11] There is increasing interest in ex vivo normothermic perfusion as a preservation method; one potential advantage of this technique is that it allows ex vivo assessment of graft function.[12]

SURGICAL COMPLICATIONS OF THE TRANSPLANT PROCEDURE

Advances in donor and recipient surgical technique, anesthesia, organ preservation, and perioperative care have all likely contributed to improvements in 1-year patient and graft survival rates.[13] However, early recognition and management of surgical complications remains important. Specific vascular and urologic complications are described more fully later in the section on Management of Allograft Dysfunction. A brief description of these complications is discussed here.

VASCULAR COMPLICATIONS

HEMORRHAGE

Acute and life-threatening hemorrhagic complications in the immediate postoperative period usually involve the anastomotic site. Bleeding can be brisk and immediate surgical exploration is required. Perirenal hematomas can result from either venous or small artery bleeding and may be related to the incision or retroperitoneal dissection. Unless small and stable, perirenal hematomas require surgical exploration to ensure adequate hemostasis. Large hematomas should be evacuated in order to reduce the risk for subsequent infection.[14]

ARTERIAL THROMBOSIS

Renal artery thrombosis of the allograft is rare (0.5% to 1%). Acute thrombosis is usually related to an anastomotic problem or kink in the renal artery. Recipient arteriosclerosis, multiple arteries, vasospasm, and hypotension are risk factors. Sudden anuria is a clue to arterial thrombosis. Arterial thrombosis results in immediate warm ischemia. Delays inherently associated with confirming the diagnosis and preparing the patient for surgical exploration usually exceed the time required to reestablish arterial flow to the kidney, resulting in prolonged warm ischemia and hypoxia and often permanent loss of function.

VENOUS THROMBOSIS

Renal vein thrombosis usually presents with local swelling, pain, and hematuria. Ultrasound shows decreased or absent blood flow in the renal vein and either absent or reversed diastolic arterial flow. The causes of venous thrombosis include problems with the surgical anastomosis, extrinsic compression by a lymphocele or a hematoma, and a deep venous thrombosis that extends in the iliac vein at the level of the venous anastomosis. Surgical exploration to attempt thrombectomy followed by anticoagulation can be performed; however, it is rarely successful. Contributing factors such as thrombophilia, should be evaluated and corrected if possible.[15]

Pseudoaneurysm of the arterial anastomosis is a rare infectious complication that leads almost invariably to graft loss and is associated with significant mortality and morbidity. Depending on the severity, noninvasive treatment with covered stenting has been attempted; however, transplant nephrectomy, vascular reconstruction and/or excision with extraanatomical bypass is usually required.[16] The loss of a lower limb due to distal thrombosis of the femoral artery or arterial dissection at the time of transplant is a very rare but devastating complication.

LYMPHOCELE

A lymphocele is a lymphatic fluid collection originating from either the severed iliac lymphatics or lymphatic drainage of the renal allograft itself. Most lymphoceles are asymptomatic. However, some may compress the ureter causing hydronephrosis, or obstruct lower limb venous return resulting in unilateral edema. Large lymphoceles may present as an abdominal mass. Analysis of aspirated fluid will typically show a high lymphocyte count and a creatinine concentration similar to serum. This contrasts with a urinoma, which has a fluid creatinine concentration much higher than serum. Lymphoceles frequently reaccumulate following percutaneous drainage, although injection of sclerosing agent following aspiration may reduce the risk of recurrence. The preferred and more definitive treatment is internal drainage of the lymphocele into the peritoneal cavity. In many centers, a laparoscopic transabdominal approach has replaced the traditional open approach that utilizes the kidney transplant incision.[17]

CURRENTLY USED IMMUNOSUPPRESSIVE AGENTS IN KIDNEY TRANSPLANTATION

OVERVIEW

The immunosuppressive drugs commonly used in clinical transplantation are summarized in Table 70.1 and their mechanisms of action are illustrated in Fig. 70.2. T lymphocytes play a central role in the recognition of the allograft as foreign and in the initiation of the rejection process. The T-cell immune response is described as requiring three distinct signaling events (the three signal model). Briefly, antigen-presenting cells (APCs; dendritic cells, macrophages, activated endothelium), most likely of donor origin, migrate to the secondary lymphoid organs of the recipient where foreign antigen/major histocompatibility (MHC) complex is presented to the (recipient) T-cell receptor (**signal 1**). A costimulatory T cell-APC interaction (**signal 2**) is then required

Table 70.1 Drugs Used in Maintenance Immunosuppression

Drug	Mechanism of Action	Adverse Effects
Corticosteroids	Block synthesis of several cytokines including IL-2; multiple antiinflammatory effects	Glucose intolerance, hypertension, hyperlipidemia, osteoporosis, osteonecrosis, myopathy, cosmetic defects; growth suppression in children
Cyclosporine	Inhibits calcineurin: synthesis of IL-2 and other molecules critical for T-cell activation thereby inhibited	Nephrotoxicity (acute and chronic), hyperlipidemia, hypertension, glucose intolerance, cosmetic defects
Tacrolimus	Similar to cyclosporine, although binds to different cytoplasmic protein (FKBP)	Broadly similar to cyclosporine; diabetes mellitus more common; hypertension, hyperlipidemia, and cosmetic defects less common
Azathioprine	Inhibits purine biosynthesis; lymphocyte replication therefore inhibited	Bone marrow suppression; rarely pancreatitis, hepatitis
Mycophenolate mofetil (MMF)	Inhibits de novo pathway of purine biosynthesis (relatively lymphocyte selective); lymphocyte replication therefore inhibited	Bone marrow suppression, gastrointestinal upset; invasive. CMV disease more common than with azathioprine
Sirolimus	Sirolimus-FKBP complex inhibits TOR blocking lymphocyte proliferative response	Bone marrow suppression, proteinuria, mouth ulcers, hyperlipidemia, interstitial pneumonitis; edema; enhanced nephrotoxicity of cyclosporine/tacrolimus
Belatacept	Blocks T-cell costimulation	PTLD in EBV seronegative, PML (rare), reactivation of TB

CMV, Cytomegalovirus; *EBV,* Epstein-Barr virus; *FKBP,* FK-binding protein; *IL-2,* interleukin-2; *MMF,* mycophenolate mofetil; *PTLD,* posttransplant lymphoproliferative disorder; *TB,* tuberculosis; *TOR,* target of rapamycin.

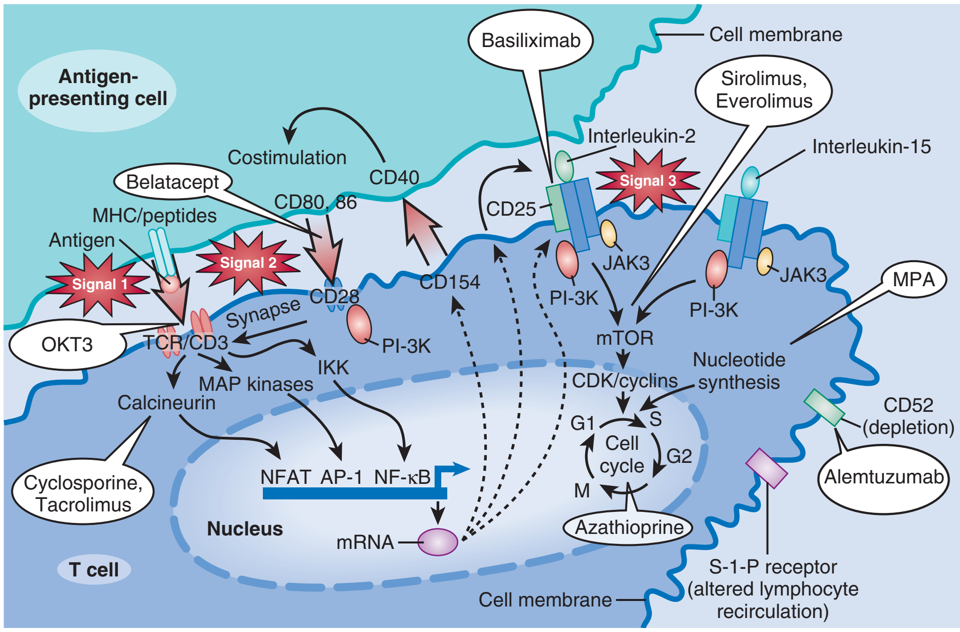

Fig. 70.2 Stages of T-cell activation: multiple targets for immunosuppressive agents. Signal 1: The Ca^{++}-dependent signal induced by T-cell receptor (TCR) stimulation results in calcineurin activation, a process inhibited by the calcineurin inhibitors (CNIs). Calcineurin dephosphorylates nuclear factor of activated T cells (NFAT), enabling it to enter the nucleus and bind to the interleukin-2 (IL-2) promoter. Corticosteroids bind to cytoplasmic receptors, enter the nucleus, and inhibit cytokine gene transcription in both the T cell and antigen-presenting cell (APC). Corticosteroids also inhibit activation of the transcription factor, nuclear factor-κB (not shown). Signal 2: Costimulatory signals, such as that between CD28 on the T cell and B7 on the APC, are necessary to optimize T-cell transcription of the IL-2 gene, prevent T-cell anergy, and inhibit T-cell apoptosis. Signal 3: IL-2 receptor stimulation induces the cell to enter the cell cycle and proliferate. IL-2 and related cytokines have both autocrine and paracrine effects. Signal 3 may be blocked by IL-2 receptor antibodies or by sirolimus. Further downstream, azathioprine and mycophenolate mofetil (MMF) inhibit progression into the cell cycle by inhibiting purine and therefore DNA synthesis. (From Halloran PF: Immunosuppressive drugs for kidney transplantation. *N Engl J Med* 351:2715-2729, 2004.)

for downstream signal transduction to occur. The ensuing activation of the T-cell calcineurin-NFAT, MAP kinase and NFκB signaling pathways results in the production of cytokines (including interleukin-2, interleukin-15) and surface molecules (including the interleukin-2 receptor). Interleukin-2 (and other cytokines) then stimulate T-cell proliferation (**signal 3**) via the PI3K/Akt/mTOR, JAK3/STAT5 and MAP kinase signaling pathways.[18] Most immunosuppressive agents either deplete lymphocytes (depleting agents) or act at the level of, or downstream from, one or more of the three T-cell immune activation signals.

The B cell immune response is increasingly being targeted by newer therapies in kidney transplantation. Key roles for B cells in the alloimmune response include 1) the production of anti-HLA antibody, 2) differentiation into HLA-antibody producing plasma cells, 3) the development of long-lasting immune memory through the generation of B memory cells, 4) antigen presentation, and 5) cytokine production.[19]

Immunosuppressive strategies can also be divided into: 1) induction and 2) maintenance therapy. Induction of immunosuppression is defined as the rapid achievement of profound immunosuppression, usually at the time of transplant, with the use of depleting agents. Maintenance immunosuppression is achieved by the combination of oral agents that take advantage of additive or synergistic immunosuppressive effects of different drug categories to minimize their non-immunosuppressive side effects. Dosage is usually greater during the first three months after transplantation and decreases afterwards. A combination of calcineurin inhibitor (CNI), antiproliferative agent, and corticosteroids is the most common regimen.

INDUCTION THERAPIES

LYMPHOCYTE-DEPLETING AGENTS

Rabbit antithymocyte globulin (Thymoglobulin) and horse antithymocyte globulin (Atgam) are approved in the United States for treatment of acute rejection. Muromonab-CD3 (Orthoclone OKT 3) is no longer commercially available. Alemtuzumab (Campath) was initially approved for the treatment of B-cell chronic lymphocytic leukemia, but has been used off-label for the induction of immunosuppression.

POLYCLONAL T-CELL DEPLETING ANTIBODIES

Polyclonal antibodies against human T cells, antithymocyte globulins (ATG), are prepared by immunization of animals with human lymphoid cells; available products include Thymoglobulin (globulin derived from rabbits inoculated with human thymocytes) ATG-Fresenius (globulin derived from rabbits inoculated with Jurkat T-cells, [not available in the United States]), and Atgam (gamma globulin derived from horses inoculated with human thymocytes). For Thymoglobulin, the resulting purified globulin includes antibodies that target more than 20 different T-cell epitopes. Antibodies at higher concentrations include TCR, CD2, CD3, CD5, CD6, CD8, CD11A, CD49, and β2 microglobulin.[20,21] The mechanism of T-cell depletion is thought to involve complement-dependent lysis, mostly in the blood compartment, and apoptosis and phagocytosis in the peripheral lymphoid tissue. Antibodies against adhesion molecules that are present in the ATG preparation may also play a role by modulating leukocyte function.[22]

A number of randomized controlled trials have compared ATG induction with other strategies. A study from the late 1970s compared a one-month induction using horse ATG with no induction treatment in patients treated with azathioprine and steroid-based immunosuppression regimen and found a significantly improved 2-year graft survival in the ATG treated group.[23] Rabbit ATG (ATG-Fresenius) induction resulted equivalent 1-year graft survival but a lower rate of acute rejection- and infection- and noninfection-related complications compared to OKT3 in patients treated with cyclosporine, azathioprine, and steroid maintenance immunosuppression.[24] A long-running randomized trial comparing horse ATG to rabbit ATG induction therapy demonstrated the superiority of rabbit ATG in terms of reduction of the composite of death, graft loss, or rejection.[25] In sensitized patients, defined as a current or peak PRA >30% or >50%, respectively, rabbit ATG induction was associated with significantly lower rate of biopsy proven acute rejection and steroid resistant rejection in the first year posttransplant but no difference in 1-year survival compared to daclizumab induction.[26] In another study of kidney transplant recipients with risk factors for delayed graft function or acute rejection, rabbit ATG resulted in a lower acute rejection rate than basiliximab induction but no difference in delayed graft function or 1-year graft survival; 5-year follow-up of this study showed sustained superiority of rabbit ATG in terms of acute rejection.[27,28] Notably, infectious complications were more common in the recipients of rabbit ATG (versus IL-2 receptor antibodies) in both aforementioned trials. In immunologically high-risk kidney transplant recipients randomly assigned to either rabbit ATG or alemtuzumab induction, no differences were seen in acute rejection rates between the groups followed out to 3-years posttransplant,[29] although infectious complications were more common in rabbit ATG-treated subjects. A recent retrospective analysis of US kidney transplant recipients that compared outcomes in matched patients receiving induction with 1) ATG versus alemtuzumab and 2) ATG versus basiliximab found that ATG use was associated with improved outcomes (death or allograft failure versus alemtuzumab and death *versus* basiliximab).[30]

ATG also has an important role in the treatment of severe or steroid-resistant acute cellular rejection. A randomized study showed rabbit ATG superior to horse ATG in the treatment of acute rejection in terms of rejection reversal and 90-day recurrence rates.[31] In another study, rabbit ATG and OKT3 were shown similarly efficacious in the treatment of steroid-resistant acute rejection; however, rabbit ATG had the advantage of a more favorable side-effect profile.[32] Given a more favorable side-effect profile, rabbit ATG has now replaced OKT3 as the second-line therapy for acute rejection.

Rabbit ATG is given at 1 to 1.5 mg/kg per day. The initial regimen of 7 to 14 days of treatment is rarely used nowadays, as a shorter course of treatment (5 days) has been demonstrated to be efficacious.[27] Infusion-related side effects, such as fever, chills, hypotension and, less frequently, cardiovascular events, are usually mild, particularly with adequate steroid and antihistamine premedication and a slow infusion rate. These reactions are more likely to occur during the first few infusions and become rare with subsequent infusions. Serum sickness, characterized by fever, rash and arthralgia, occurring 10 to 15 days after treatment have been reported as well, possibly more frequently in patients not receiving steroid prophylaxis.[33]

MONOCLONAL T-CELL DEPLETING ANTIBODY

Muromonab-CD3 (Orthoclone OKT3) is a mouse anti-human monoclonal antibody against the T-cell receptor-associated CD3 antigen that was first approved for clinical use in 1985. OKT3 results in apoptosis and rapid depletion of T cells from the circulation. The cytotoxic effect is preceded by a transient antibody-induced T-cell activation and cytokine surge, an effect that is responsible for many of the undesired effects associated with OKT3. T-cell number and function usually return to normal limits one week after the completion of the treatment. Cytokine release syndrome is present usually after the first infusion. It is most frequently reported to be a mild, self-limited flu-like illness; however, severe life-threatening reactions, such as serious cardiovascular and central nervous system manifestations, have been reported.[34] Non-cardiac pulmonary edema has also been seen, particularly if the patient is fluid overloaded pretransplant. Patients may develop anti-mouse neutralizing antibodies rapidly, and this may limit the efficacy of the treatment and prevent retreatment.[35] The efficacy of OKT3 as an induction agent and in the treatment of acute rejection is well established.[34,36] As a result of its side effects, the use of OKT3 decreased considerably following the emergence of alternative immunosuppressive agents, such as ATG and IL-2 blockers.[37] OKT3 was the first monoclonal antibody to be approved by the U.S. Food and Drug Administration (FDA); however it role is now consigned to the history books as its manufacture was discontinued by Janssen-Cilag in 2010.

ANTI-CD52 ANTIBODY

Alemtuzumab (Campath) is a humanized monoclonal antibody against CD52, and was originally developed to treat refractory B-cell chronic lymphocytic leukemia. Alemtuzumab treatment results in depletion of both T and B lymphocytes. Evidence for the efficacy of alemtuzumab as an induction agent was provided by a randomized controlled trial that compared a single 30-mg dose of alemtuzumab to induction with basiliximab in immunologically low-risk patients and with rabbit ATG in immunologically high-risk patients. Alemtuzumab treatment proved superior to basiliximab and equivalent to rabbit ATG in terms of biopsy-confirmed acute rejection followed out to 3 years posttransplant, although the overall rate of late biopsy-proven rejection (between 1 and 3 years posttransplant) was significantly higher in the alemtuzumab group.[29] Infectious complications were higher with alemtuzumab than basiliximab but lower with alemtuzumab than rabbit ATG. Alemtuzumab may cause an infusion first dose reaction, which can be avoided if the subcutaneous route is used.[38] Neutropenia and anemia are also seen. Concern about the development of autoimmune disease (especially thyroid-related) following alemtuzumab treatment was initially raised after publication of data from a study comparing the agent with interferon beta for the treatment of multiple sclerosis.[39] The development of autoimmune disease has also been reported in solid organ transplant recipients treated with alemtuzumab induction.[40,41] Potential advantages of alemtuzumab include the simplicity of a single dose treatment and lower cost compared to other induction agents. Since 2012, alemtuzumab has not been widely commercially available, although the manufacturer does offer ongoing access to the drug for use as induction immunosuppression through the 'Campath Distribution Program'.[42]

INTERLEUKIN-2 RECEPTOR ANTAGONIST (IL-2-RA)

Daclizumab is a humanized monoclonal antibody and basiliximab is a chimeric monoclonal antibody. Both are IgG1 antibodies directed against the alpha chain of the IL-2 receptor (CD25 antigen), which is expressed on activated T-lymphocytes. Blockade of the IL-2 receptor inhibits a key T-lymphocyte proliferation signaling pathway, thereby blunting the cellular immune response. These agents reduce the rate of rejection by approximately 30% to 40% compared to placebo when used in combination with conventional immunosuppression.[43–45] Rabbit ATG and alemtuzumab are both more effective in reducing the rate of rejection, but incur a higher infectious risk than the IL-2 receptor blockers.[26,27,29] Advantages of IL-2 receptor blockers include minimization of injection-related side effects and lack of risk of infection or cancer compared to placebo.[44,45] Daclizumab was withdrawn from the market by the manufacturer in 2009 and thus basiliximab is the only IL-2 blocker available in the United States. The regimen of the treatment consists of two infusions of 20 mg; the first at the time of transplantation, and the second 3 to 4 days posttransplantation. The pharmacokinetics of that dose regimen provides prophylaxis for 30 days posttransplant. Few data exist regarding the use of IL-2 receptor antagonists in the treatment of acute rejection.

ANTI-CD20 ANTIBODIES

Rituximab (Rituxan) is a chimeric anti-CD20 cytolytic monoclonal antibody that has been approved for the treatment of non-Hodgkin's lymphoma, chronic lymphocytic leukemia, and rheumatoid arthritis. Rituximab was initially used in the transplant population for the treatment of post-transplant lymphoproliferative disease.[46] It interferes with the humoral allo-response by targeting B-lymphocytes. Many transplant programs, ourselves included, now use rituximab as part of an induction regimen for selected immunologically high-risk patients based on evidence that rituximab treatment in such patients has a favorable safety profile and is associated with a reduction in rejection and antibody-mediated graft dysfunction.[47–49] Rituximab is commonly employed in protocols for the treatment of frequently in combination with steroids, plasmapheresis, and/or intravenous immunoglobulin (IVIg).[50,51] Rituximab is also an important element in many pretransplant desensitization protocols (see section on desensitization)[52,53] and has been used in the treatment of recurrent or de novo posttransplant glomerular diseases such as membranous glomerulonephritis and focal segmental glomerulosclerosis (FSGS).[54,55] In patients receiving rituximab, we routinely premedicate with steroid, acetaminophen and an anti-histamine in order to decrease infusion-related side effects. We very carefully weigh the risk-benefit of rituximab treatment in those patients with positive hepatitis B surface antigen/detectable hepatitis B DNA.[56] We initiate hepatitis B prophylaxis in all patients treated with rituximab who have evidence of prior hepatitis B infection (positive hepatitis core antibody/negative hepatitis surface antigen) irrespective of hepatitis surface antibody titer. Obintuzumab, a next generation anti-CD20 antibody, is currently under clinical trial for pre-transplant desensitization.[57]

INTRAVENOUS IMMUNOGLOBULIN (IVIg)

Various pooled human immunoglobulin products are commercially available. The increasingly widespread use of IVIg in the kidney transplant population parallels a greater awareness in recent years of antibod-mediated rejection (ABMR) and its risk factors. IVIg is used pretransplant as part of both HLA- or ABO incompatible-desensitization protocols, at induction in patients with pre-formed donor-specific antibody (DSA) and for treatment of ABMR.[47,58–60] Its mechanism of action is not fully understood but may include neutralization of circulating (anti-HLA) antibody, inhibition of complement, modulation of B-cell and antigen presenting cell function, and cytokine inhibition.[57,61,62] IVIg has also been used in the treatment of posttransplant viral infections such as BK virus and parvovirus.[60] Side effects include infusion reaction, headache (common and troublesome), aseptic meningitis, hemolysis (especially in patients who are blood group A) and, rarely, thrombosis.

MAINTENANCE IMMUNOSUPPRESSIVE AGENTS

CALCINEURIN INHIBITORS (CNIS)

Calcineurin is a calcium dependent serine/threonine phosphatase that is involved in a diverse range of cellular functions including T-cell signal transduction.[63] Binding of foreign antigen to the T-cell receptor, when accompanied by a co-stimulatory signal, triggers the cytosolic influx of calcium and the downstream activation of calcineurin. Activated calcineurin dephosphorylates the transcription factor, nuclear factor of activated T-cells (NFAT), which then translocates to the nucleus and activates a host of target genes, including the cytokine IL-2.[64] IL-2 then binds to its receptor and initiates T-cell expansion.[18]

The excellent posttransplant outcomes seen with the advent of the CNIs in the 1980s made the widespread expansion of solid organ transplantation possible, heralding the modern era of transplantation. CNIs, first in the form of cyclosporine and then later tacrolimus, have remained the cornerstone of transplant immunosuppression. However, the CNIs are not without side effects. CNI-associated nephrotoxicity, first described in the heart transplant population certainly contributes to a reduction in long-term graft survival in a proportion of kidney transplant recipients.[65–67] However, the magnitude kidney transplant failure attributable to CNI toxicity remains hotly debated.

CYCLOSPORINE

Cyclosporine is a lipophilic amino acid cyclic peptide that binds to cyclophilin, a cytoplasmic protein, and forms a complex that inhibits calcineurin. Cyclosporine came into clinical use in the early 1980s. Both European and American clinical trials demonstrated the superiority of cyclosporine either alone or with steroid over the standard immunosuppression regimen of azathioprine and steroid.[68,69] The original oil-based formulation of cyclosporine (Sandimmune) was associated with erratic gastrointestinal absorption and highly variable bioavailability. The subsequent development of a microemulsion formulation of cyclosporine (Neoral) significantly improved the drug's absorption and the pharmacokinetic profile.[70] Cyclosporine is now also available in a generic microemulsion formulation. Gengraf is an alternative water-based microemulsion that many patients find more palatable.[71] Care should be taken to avoid non-microemulsion generic cyclosporine formulations without the prescriber's knowledge, as they are associated with unpredictable absorption and may expose the patient to an increased risk of rejection or toxicity.

Therapeutic drug monitoring of cyclosporine is most commonly performed using 12-hour trough levels, although monitoring blood levels two hours after ingestion of cyclosporine (C2 level) actually has a better correlation with drug exposure.[72] The side effects of cyclosporine include hypertension, hyperlipidemia, gingival hyperplasia, hypertrichosis, tremors, and nephrotoxicity. Cyclosporine is also associated with the development of posttransplant diabetes mellitus (PTDM) and rarely hemolytic uremic syndrome.

TACROLIMUS

Tacrolimus (Prograf) is a macrolide antibiotic that binds to the cytosolic FK506 binding protein (FKBP12). The resulting drug-protein complex inhibits calcineurin activity. Many trials have demonstrated reduced rates of rejection compared to cyclosporine,[73] particularly the original formulation of cyclosporine. The toxicity profile is slightly different than cyclosporine; tacrolimus is not associated with gingival hyperplasia, hypertrichosis, or hyperlipidemia. However, it is associated with greater neurotoxicity, a higher incidence of PTDM, and gastrointestinal toxicity. Hypertension and nephrotoxicity appear to be milder than with cyclosporine. Mycophenolate acid exposure is reduced by approximately 40% when combined with cyclosporine versus tacrolimus (see below), an effect that likely contributes to the relatively greater immunosuppressive potency of tacrolimus compared to cyclosporine-based CNI-MMF regimens.[74] Tacrolimus is now also available in generic formulation. Two extended-release formulations of tacrolimus (Astagraf and Envarsus XR) have also been approved in the US. Once daily tacrolimus may be beneficial in terms of compliance and may mitigate neurological side effects.[75,76] Close monitoring of levels, especially in patients early posttransplant, is recommended following formulation changeover.[77]

CALCINEURIN-INHIBITOR TARGET LEVELS

In patients treated with CNIs, the dangers of drug overexposure must be weighed against those of underexposure. CNI overexposure puts the patient at heightened risk of opportunistic infections and cancer. In addition, good evidence exists that chronic exposure to CNI is itself nephrotoxic and may result in progressive graft failure, so-called calcineurin-inhibitor nephrotoxicity.[78] Concern regarding CNI nephrotoxicity has, in recent decades, led to a greater focus on strategies to minimize CNI exposure. Data from two randomized controlled trials from the late 2000s supported the efficacy and safety of lower dose cyclosporine and tacrolimus-based regimens in immunologically low-risk patients.[79,80] However, over the past decade antibody mediated allograft injury has been increasingly recognized as an important cause of late graft loss.[81] There is increasing awareness that chronic CNI underexposure may permit allorecognition and chronic ABMR to occur. Targeting very low CNI levels or CNI withdrawal does appear to increase the risk of de novo donor-specific antibody (DSA) formation.[82,83] The development of posttransplant de novo DSA is in turn associated with reduced graft survival.[84,85] To date, belatacept (see below) probably represents the best available option for calcineurin avoidance.[86]

ANTIPROLIFERATIVE AGENTS

AZATHIOPRINE

Azathioprine (Imuran) is a pro-drug whose metabolites exert a number of actions including 1) the incorporation of thioguanine purine analogues into DNA and RNA resulting in cell death, 2) inhibition of de novo purine synthesis by methylthioinosine monophosphate, and 3) T-cell apoptosis via inhibition of the Rho family GTPase, Rac1.[87] Azathioprine, in combination with prednisone, was the mainstay of transplant immunosuppression from the early 1960s until the introduction of cyclosporine in the early 1980s, at which point it maintained its role as an adjuvant medication to cyclosporine. Azathioprine has now been largely superseded by mycophenolate mofetil (MMF) in most new immunosuppressive protocols,[88] although many longer vintage kidney transplant recipients remain on the drug. Additionally, azathioprine is commonly substituted for MMF in female transplant recipients who are planning to conceive. The most important side effect of azathioprine is myelosuppression. Individuals with reduced thiopurine methyltransferase enzyme activity (TPMT) tend to accumulate active drug metabolites and are predisposed to hematologic toxicity. By some estimates, 10% of the population have reduced and 0.3% absent TPMT activity.[89] Notably, coadministration of allopurinol with azathioprine also shifts azathioprine metabolism toward the production of metabolically active substrates and may lead to potentially serious toxicity.

MYCOPHENOLATE ACID

Mycophenolate mofetil (MMF) is a pro-drug that releases mycophenolate acid, an inhibitor of inosine monophosphate dehydrogenase (IMPDH), which is required for the de novo pathway synthesis of guanosine from inosine. The effects of MMF are relatively lymphocyte-specific as, lacking a purine salvage pathway, T and B cells rely exclusively on de novo purine synthesis. By limiting the pool of available guanine triphosphate, MMF prevents T and B lymphocyte replication and suppresses both the cellular and humoral immune responses. Clinical trials have demonstrated that MMF reduces acute rejection rates by 50% compared to azathioprine or placebo.[90] A meta-analysis of 19 trials has shown that MMF was associated with a reduction of acute rejection and improved graft survival compared to azathioprine.[88] Cyclosporine blocks the enterohepatic recirculation of MMF reducing the exposure of the drug by approximately 40%. In contrast, tacrolimus and sirolimus do not interfere with the metabolism of MMF.[74] The most troublesome MMF side effects are gastrointestinal: diarrhea, bloating, epigastric pain and nausea, and such side effects may require dose reduction. MMF can cause neutropenia and, less frequently, anemia. Alopecia may also occur with MMF treatment and can be very bothersome. However, it is not associated with nephrotoxicity or hypertension. MMF is associated with an increased risk of major fetal malformations and patients who are planning pregnancy should be switched to azathioprine at least 6 weeks before attempting to conceive.[91] Despite the high inter- and intra-individual variability of its pharmacokinetics, routine therapeutic drug monitoring of MMF has not been established. MMF is approved at a dose of 1 g bid when used in conjunction with a calcineurin inhibitor and steroids. The excellent results obtained with a standard dosing regimen and the poor correlation of a single time point to area under the curve (AUC) (particularly when used in combination with CsA) supports a fixed dose regimen.[18,92] Mycophenolate mofetil is now also available in generic formulation in the US.

Mycophenolate sodium (Myfortic) is an enteric-coated slow-release formulation of mycophenolic acid that was developed to decrease the gastrointestinal side effects of MMF. Randomized trials have demonstrated a similar efficacy and side effect profile to MMF.[93] The dose conversion for mycophenolate mofetil to mycophenolate sodium is 250 mg to 180 mg.

TARGET-OF-RAPAMYCIN INHIBITORS

Sirolimus (Rapamune) is a macrocyclic antibiotic that has immunosuppressive properties. Even though it shares its cytoplasmic binding protein with tacrolimus (FKBP12), the resulting complex does not interfere with calcineurin. Rather, it binds with the mammalian target of rapamycin (mTOR). The resulting inhibition of mTOR prevents the propagation of IL-2-mediated cell proliferation signaling through the PI3K/AKT/mTOR pathway.[94] Sirolimus inhibits cellular proliferation of T and B lymphocytes and reduces antibody production.[18] Its immunosuppressive effects have been demonstrated in preclinical studies to be synergistic with CsA and tacrolimus.[95,96] Side effects of sirolimus include hyperlipidemia, thrombocytopenia, anemia, poor wound healing, diarrhea, pneumonitis, mouth ulcers, proteinuria, and peripheral edema. Indeed, high drug withdrawal rates due to side effects are a common theme in many of the sirolimus trials.

Sirolimus has been studied in a number of different immunosuppression protocols including as de novo maintenance immunosuppression in combination with either a CNI and steroid or an anti-metabolite and steroid, in early CNI withdrawal protocols or as replacement for a CNI in patients with chronic allograft dysfunction or cancer. Sirolimus/MMF versus cyclosporine/MMF combined with rabbit ATG induction and 6 months of steroids resulted in similar 12-month survival and rejection rates, although adverse drug events and discontinuation rates were significantly higher in sirolimus-treated subjects.[97] Another study that compared sirolimus/tacrolimus with tacrolimus withdrawn at 3 months, sirolimus/MMF and tacrolimus/MMF in patients treated with daclizumab induction and steroid maintenance reported significantly higher acute rejection rates with sirolimus/MMF treatment, prompting early withdrawal of the group from the trial; impaired wound healing and dyslipidemia were both more common in the sirolimus treated groups, as was drug withdrawal.[98] In another study, switching from tacrolimus to sirolimus at 3 months posttransplant (versus staying on tacrolimus) was associated with similar graft function but a higher rate of acute rejection.[99]

A randomized controlled trial with median follow-up of 8 years reported increased acute rejection rates and lower glomerular filtration rate (GFR) in cyclosporine/sirolimus and tacrolimus/sirolimus versus tacrolimus/MMF treated subjects. Tacrolimus/sirolimus treatment was also associated with an excess in death with a functioning graft compared to the other groups. A trend toward worse GFR in CNI/sirolimus treatment combinations (especially cyclosporine) is consistent with animal studies that show that sirolimus co-treatment potentiates CNI nephrotoxicity.[100,101]

Late conversion from CNI to sirolimus has shown some promise in improving/stabilizing kidney function in patients with chronic allograft dysfunction, with the caveat that the development of nephrotic or sub-nephrotic range proteinuria following conversion to sirolimus is relatively common and requires cessation of the treatment. The presence of proteinuria before conversion is predictive of a poor response and should be screened for prior to considering a switch to sirolimus.[102] Proposed mechanisms for sirolimus-associated proteinuria include reduced tubular protein reabsorption (often exacerbated by the glomerular hemodynamic effects of concurrent CNI withdrawal) and sirolimus induced podocyte injury.[103,104] Finally, conversion from CNI to sirolimus is associated with regression of Kaposi's sarcoma and reduction in the development of squamous cell skin cancers.[105,106] Sirolimus use peaked at just under 20% of kidney transplants in the early 2000s but its use has since declined to the low single digits.

Everolimus (Zortress) is derived from sirolimus and has a shorter half-life. It has been approved in the United States for prophylaxis against rejection in kidney and heart transplant recipients. A recent randomized controlled trial compared everolimus and low-dose tacrolimus (EVR/Tac) with MMF and standard dose tacrolimus (MMF/Tac) in de novo kidney transplant recipients. At 12-months posttransplant the EVR/Tac group was noninferior compared to MMF/Tac for the composite outcome of biopsy proven acute rejection or an eGFR < 50 mL/min. The rates of death, graft loss, and rejection were similar in the two groups at 12 months. CMV and BK infection occurred less frequently in EVR/Tac treated subjects. As with many mTor studies, drug discontinuation due to adverse events was more frequent in the everolimus vs MMF treatment groups (23% vs 12%).[107] Everolimus has also been approved for the treatment of advanced renal cell carcinoma (Affinitor).

CO-STIMULATORY SIGNAL BLOCKERS

Belatacept is a selective co-stimulation blocker. This human fusion protein binds the ligands CD80 and CD86 on antigen presenting cells and prevents their interaction with the co-stimulation receptor on the T cells (CD28). Blockade of the co-stimulation signal (signal 2) prevents T-cell activation. Belatacept is given as an IV infusion. Two studies randomized recipients of living or standard deceased donor (BENEFIT) and expanded criteria deceased donor kidneys (BENEFIT-EXT) to higher dose belatacept, lower dose belatacept or cyclosporine, with all subjects receiving IL-2 antagonist induction, MMF, and steroid concurrently. One-year results showed similar graft and patient survival across the groups but significantly higher GFR in belatacept treated subjects. Acute rejection was significantly higher in those treated with belatacept in the BENEFIT study but not BENEFIT-EXT.[86,108] Estimated GFR remained approximately 25 and 19 mL/min higher with belatacept versus cyclosporine treatment at 7-year follow-up of the BENEFIT and BENEFIT-EXT studies, respectively.[109,110] In addition, graft and patient survival were significantly higher at 7 years posttransplant in the belatacept versus cyclosporine treated group in the BENEFIT Study.[109] A recent analysis of a subset of patients from the BENEFIT and BENEFIT-EXT trials showed a lower rate of development of de novo DSA in belatacept versus cyclosporine at 3 years posttransplant, a finding that may bode well for the long-term outcome of belatacept patients.[111] The incidence of (mostly central nervous system) PTLD was higher in patients treated with belatacept in both trials and was associated with pretransplant EBV seronegativity. The current package insert recommends use of belatacept only in kidney transplant patients who are documented anti-EBV antibody seropositive because of the risk of PTLD. Posttransplant conversion from CNI- to belatacept-based therapy is also a viable strategy and may help to preserve GFR.[112] While the BENEFIT studies certainly show belatacept to be a viable alternative to CNI-based immunosuppression regimens it must be remembered that these trials compared belatacept to cyclosporine and not to the modern era 'standard of care' CNI, tacrolimus; tacrolimus is itself associated with superior graft survival when compared to cyclosporine (see above).

CORTICOSTEROIDS

Since the early days of clinical transplantation, corticosteroids have played an important role in the immunosuppressive management of transplant recipients. It remains the first line of treatment for acute rejection. Corticosteroids have both antiinflammatory and immunosuppressive properties. The antiinflammatory effects of steroids are mediated by reduction of pro-inflammatory molecules including the platelet activated factor (PAF), prostaglandins, and leukotrienes and reduction of the release of tumor necrosis factor-α (TNF-α). Immunosuppressive properties of steroids include prevention of T-cell proliferation, inhibition of cytokine production including IL-2 and interference with antigen presentation. Some of these effects are mediated through the inhibition of the transcription factor nuclear factor kappa B (NF-κB). Chronic use of steroids potentiates lymphocyte apoptosis and results in lymphopenia and interferes with the leukocyte trafficking.[14] These various mechanisms acting in concert make steroids a potent and versatile immunosuppressive drug. However, the side effects related to its chronic use are also extensive (e.g., diabetes, hypertension, hyperlipidemia, osteoporosis, avascular necrosis, truncal obesity, hypertrichosis, acne, and cataracts). The emergence of new, more potent immunosuppressive drugs has allowed the successful implementation of steroid sparing protocols (avoidance, minimization, or withdrawal).[113,114] This is reflected by a significant trend toward decreased steroid utilization, with just over 60% compared to 90% of US kidney transplant recipients maintained on the drug in the years 2011 and 1995, respectively.

The reported benefits of steroid sparing strategies are manifold and include improvements in blood pressure, glycemic and lipid control, reduced posttransplant weight gain, better bone mineral density, growth, and physical appearance.[115–118] A Cochrane review of 30 randomized trials of steroid avoidance (less than 2 weeks of exposure) and steroid withdrawal (greater than 2 weeks exposure) demonstrated that such steroid sparing strategies are not associated with an increase in mortality or all-cause graft loss but were associated with an increase in death censored graft loss and acute rejection. However, the increased risk of acute rejection was seen in those treated with cyclosporine but not tacrolimus-based immunosuppression regimens.[119] Steroid sparing strategies are a reasonable option for low immunological risk patients who are treated with modern era (tacrolimus/MMF) immunosuppression regimens. Most steroid withdrawal protocols wean steroids within 3 to 6 months of transplant. Late withdrawal from steroid (greater than 1 year posttransplant) has been associated with increased acute rejection rates and deterioration in graft function in studies from the cyclosporine/azathioprine era.[120,121] While late withdrawal may be safer in the tacrolimus/MMF era, ongoing caution is recommended with this approach.[122]

IMMUNOSUPPRESSION AT OUR INSTITUTION

The vast majority of kidney transplant recipients in the US are maintained on tacrolimus and MMF and around two thirds of patients currently receive maintenance steroid.[123] The greatest source of intra-program variability in immunosuppressive regimen is in the choice of induction therapy and in patient selection for, and method of, steroid withdrawal.

At our institution, we aim to tailor the immunosuppressive regimen to strike the best balance between over- and under-immunosuppression based on patient characteristics detailed below. Our options for induction immunosuppression, used either alone or in various combinations, include intravenous glucocorticoid (methylprednisolone), IL-2 receptor antagonist (basiliximab), anti-thymocyte globulin (Thymoglobulin), IVIg and rituximab. Maintenance immunosuppression usually consists of tacrolimus and MMF with or without steroid. Around one third of our patients meet the criteria for early steroid withdrawal. We have a protocol for rapid steroid withdrawal (steroid free by hospital discharge) but most commonly employ a slower steroid taper that results in the patient being off steroid by around 6 months posttransplant. We take a number of factors into account when deciding about a patient's induction and maintenance immunosuppression.

Flow crossmatch. We consider a positive flow crossmatch (in the presence of a DSA) to be a contraindication to kidney transplantation in the absence of pretransplant desensitization (see desensitization section).

Calculated panel reactive antibody (cPRA). We consider those patients with a cPRA ≥20% to be sensitized and use

Thymoglobulin induction therapy. For ‘hypersensitized’ patients (cPRA >98%) or previously desensitized patients we additionally administer a single dose of rituximab (usually 500 mg) prior to discharge in the hope of preventing de novo DSA formation and acute rejection.[48,49,124] All sensitized patients (cPRA ≥20%) are maintained on lifelong triple immunosuppression with CNI, MMF, and steroid. For unsensitized patients we use either steroid alone or IL-2 receptor antagonist at induction. Many unsensitized patients are eligible for early steroid withdrawal.

PREFORMED DONOR-SPECIFIC ANTIBODY WITH A NEGATIVE FLOW CROSSMATCH

The presence of preformed DSA (with or without high cPRA) versus a high cPRA (without DSA) has much greater prognostic significance for graft survival.[125] We try to avoid crossing immunological barriers; however, in the absence of a better alternative option, we do not consider the presence of low level preformed DSA to be a contraindication to transplant. Depending on the strength of the DSA at the time of transplant we either monitor the antibody closely posttransplant or administer high dose IVIg in the immediate posttransplant period and repeat another dose around 4 weeks posttransplant. We also take the presence of preformed ‘historic’ DSA, present on prior but not current HLA antibody screening sera, into account when planning the posttransplant immunosuppression and immune monitoring regimen.

HLA mismatch. We take donor-recipient HLA mismatch into account in assessing immunological risk in all patients. The recipients of two haplotype matched sibling donated kidney transplants merit a special mention as these patients can expect excellent outcomes with relatively little immunosuppression. Our practice for these patients has been to use steroid-only induction and low dose dual immunosuppression usually with a CNI and antimetabolite thereafter.

Prior Kidney Transplant. We treat these patients as sensitized irrespective of their cPRA and therefore give Thymoglobulin for induction and maintain on chronic triple immunosuppression.

ESRD from Primary or Secondary GN. We do not usually withdraw prednisone in patients with ESRD from lupus nephritis because of concern regarding a higher rejection risk and lupus nephritis recurrence.[126,127] We discuss the retrospective study that showed a significantly reduced risk of IgA recurrence in steroid-treated versus steroid-free transplant recipients with ESRD from IgA nephropathy. For each patient with IgA nephropathy, we weigh this potential benefit of remaining on steroid against the advantages of steroid withdrawal.[128] There is little firm evidence on what to do with steroid in patients with ESRD from other GNs, although we will continue steroid therapy if it was being taken chronically pretransplant.

African-American Ethnicity. We usually maintain African-American kidney transplant recipients on chronic maintenance steroid even if unsensitized. This stems from data showing that African-Americans, perhaps because of an intrinsically more robust alloimmune response, appear to be at higher rejection risk in the absence of steroid.[129] However, a newer study suggests that early steroid withdrawal may be safe in unsensitized African-American recipients when transplant is performed with lymphocyte depleting induction.[130]

Other important factors that must be taken into consideration are the baseline quality of the allograft, and the anticipated vulnerability profile to specific side effects of immunosuppression.

DESENSITIZATION

Finding a compatible match for highly sensitized patients can be very challenging.[4,131] The last 15 years has seen significant growth in desensitization protocols that aim to lower anti-HLA antibody titers in sensitized kidney transplant candidates sufficiently to create an immunological window for successful transplantation. Desensitization protocols vary from center to center but fall into two broad categories 1) high-dose IVIg/anti-CD20-based,[53] and 2) low-dose IVIg/plasmapharesis-based[131] regimens. High-dose IVIg has been employed as a monotherapy and does effectively lower DSA titers and permit successful transplantation;[58] however, a number of studies have suggested that IVIg alone (versus IVIg plus plasmapharesis and rituximab or IVIg plus rituximab) is associated with a posttransplant rebound in DSA titres and increased incidence of ABMR.[132] More recent reports have also described successful desensitization using newer agents such as the anti-IL-6 receptor antibody, tocilizumab, and the IgG-degrading enzyme (IdeS).[133,134]

Desensitization may be performed in patients who have an identified living donor or may be timed to coincide with the candidate being positioned at the top of the deceased donor waiting list. Desensitization increases the prospect of highly sensitized patients receiving a transplant and is associated with good medium term graft survival rates and a survival advantage compared to remaining on dialysis.[52,131,135] However, ABMR rates are high[52,136] and few data are available about long-term graft survival in desensitized patients.

We believe that crossing immunological barriers in transplant should be a last resort. At our institution we explore alternative avenues to transplant before embarking on desensitization. Like most other transplant centers in the United States, we have had great success using donor exchange programs to help recipients with immunologically incompatible living donors get transplanted.[137] The new U.S. Kidney Allocation System, which offers significant allocation score bonuses (the equivalent of waiting time) and high priority access to kidney transplants across the country for the most sensitized patients on the list, was implement in December 2014. This has resulted in a significant increase in transplantation rates of highly sensitized patients (cPRA >98%).[138,139] Indeed, absent the new Kidney Allocation System, many of these highly sensitized patients would have lingered on the waiting list and perhaps been offered desensitization. There remains a group of hypersensitized patients (often with cPRAs of 99.8% and above) who in the absence of some intervention remain unlikely to get transplanted.[140] We believe that it is this select group of patients who should be offered desensitization therapy.

ABO INCOMPATIBLE TRANSPLANTATION

The other important immunological barrier in kidney transplantation is the ABO blood group barrier. Blood group antigens are expressed on endothelial cells and, with certain exceptions, ABO-incompatible transplantation in the absence of desensitization will result in ABMR.[141] Much of the pioneering work on desensitization for ABO-incompatible transplant has been done in Japan where reliance on living donation created a need to eliminate ABO-incompatibility as a barrier to transplant.[142] ABO-incompatible protocols now mostly entail a relatively short but intensive pretransplant regimen consisting of rituximab, IVIg and plasmaphereis – continued until anti-A/B titres fall below a prespecified safe threshold (usually 1 : 8 or 1 : 16), some protocols also include additional posttransplant plasmapheresis or IVIg.[143–145] The long-term outcome of desensitized recipients of blood group incompatible organs is good although perhaps slightly inferior to that of ABO-compatible kidney transplant recipients, much of the excess risk associated with ABO-incompatible (versus compatible) kidney transplantation being concentrated in the early posttransplant period.[146,147] In patients with low baseline anti-A/B titers ABO-incompatible transplant has been shown to be safe with minimal pretreatment (1-2 weeks MMF).[148] The transplantation of blood group A2 donors (about 20% of the blood group A population who express A antigen at low level) into B donors with low anti-A titres is safe without desensitization.[148–150] The new Kidney Allocation System in the United States has permitted that A2 or A2B deceased donor kidneys be allocated to selected blood group B recipients in an attempt to reduce the long waiting times faced by blood group B patients.

EVALUATION OF THE RECIPIENT IMMEDIATELY BEFORE THE TRANSPLANT

ASSESSING COMPATIBILITY AND IMMUNOLOGICAL RISK

DONOR IMMUNE STATUS AND CROSSMATCH

The presence of preformed recipient antibody against either the donor ABO blood group or donor human leukocyte antigen (HLA) will frequently result in hyperacute rejection. The past 50 years has seen huge advances in the laboratory detection of recipient anti-HLA antibody. The complement dependent lymphocyte cytotoxicity (CDC) assay, where donor lymphocytes were incubated with recipient serum, was first developed in the 1960s.[151] Preformed recipient antibody to donor lymphocytes (subsequently identified as anti-HLA antibody) resulted in lymphocyte death in vitro and predicted rapid (hyperacute) graft failure in vivo. The test was subsequently refined and divided into two separate assays 1) with donor T lymphocytes, which exclusively express HLA class I antigen, and 2) with donor B lymphocytes, which express both HLA class I and II antigens. The addition of antihuman globulin to the assay increased its sensitivity.

A positive CDC T-cell crossmatch is an absolute contraindication to transplantation. A positive CDC B cell crossmatch (in the setting of a negative T-cell flow crossmatch) may indicate the presence of preformed antibody to class II HLA antigen or low level antibodies against class I antigens; however, this scenario (negative CDC T/positive CDC B crossmatch) is always best interpreted in combination with other HLA antibody detection techniques as the binding of (clinically irrelevant) non-HLA antibodies may result in a false positive B cell crossmatches. The CDC assays are now being phased out in many US laboratories having been largely supplanted by flow crossmatch and solid phase technologies.

The flow crossmatch (FCXM) is a more sensitive assay for detecting donor anti-HLA antibody. Donor T lymphocytes (T-cell flow) or B lymphocytes (B cell flow) are mixed with recipient serum and a fluorescent anti-IgG antibody. Any recipient antibody that binds to donor HLA will be tagged by the fluorescent IgG and subsequently detected using flow cytometry. Flow crossmatch detects low-titer and noncomplement binding anti-HLA antibody and may be positive when a CDC crossmatch is negative. A negative CDC crossmatch but positive T-cell FCXM is associated with poor short-term transplant outcomes and is a (relative) contraindication to transplantation.[152]

Solid phase technologies such as the Luminex single antigen bead (SAB) anti-HLA antibody assay permit detection and quantification of a wide array of anti-HLA antibodies. Multiple color-coded beads coated with a broad range of HLA antigens are incubated with recipient serum. Binding of antibody to a bead coated with a specific HLA antigen results in a distinct fluorometric signal that is then detected by flow cytometry. Molecular (polymerase chain reaction) HLA typing techniques have now superseded serologic identification of HLA-type for both donors and recipients. Accurate molecular donor HLA-typing and SAB-based recipient anti-HLA antibody screening can be combined to perform a virtual crossmatch. Many centers (ours included) will now proceed with a deceased donor transplant in low immunological risk patients based on a negative virtual crossmatch. For high immunological risk donors, virtual crossmatching has assisted in the identification of potentially compatible donors and has improved transplant rates for sensitized donors.[153] Computerized virtual crossmatch algorithms have also been central to the success of kidney paired donor exchange and transplant chains.[154,155] In the US, wait-listed kidney transplant candidates usually have SAB-based HLA-antibody screening performed every three months. Transplant centers regularly update each candidate's antibody status on UNet, a centralized computer database that virtually matches candidates with potential donors. HLA antigens against which a candidate has antibody are listed as unacceptable HLA antigens or 'avoids' on the system.

The calculated panel reactive antibody (cPRA) replaced the measured PRA in the US in 2009 as the main measure of immune sensitization. The cPRA is calculated using an algorithm that correlates donor anti-HLA antibody (as measured by SAB assay) with the population frequency of HLA antigens to generate a percent score. A candidate with no anti-HLA antibody will have a cPRA of 0%. While a sensitized candidate with a cPRA of 85% will be immunologically incompatible with 85% of the donor population. The choice of predonation crossmatch depends mostly on a donor's immunological risk (see Table 70.2). Intended recipients should be specifically asked if they have recently received a blood transfusion because this could cause a surge in alloantibodies.[156]

Donor-specific anti-HLA antibody (DSA) may be present pretransplant or develop de novo posttransplant. De novo DSA development frequently follows TCMR and is also associated with patient noncompliance.[157] Preformed and de novo DSA are both associated with an increased risk

Table 70.2 Recipient Risk Factors for Acute Rejection

Previous blood transfusions, particularly if recent
Previous pregnancies, particularly if multiple
Previous allograft, particularly if rejected early
African-American ethnicity
cPRA > 20%
Donor-specific antibody (current or historic)

cPRA, Calculated panel reactive antibody.

of ABMR and poor graft outcomes.[125,158] DSAs may also be measured using a variation on the SAB technique, the Luminex C1q assay, which detects anti-HLA antibodies that bind and activate complement. The development of post-transplant C1q-binding DSA is strongly associated with poor graft outcome.[159] DSA IgG subtyping also yields similarly helpful prognostic information with DSAs of the IgG3 subclass being associated with the worst clinical prognosis.[160]

The use of HLA epitope matching is a new way at looking at donor-recipient tissue matching, though not yet in widespread clinical use. Briefly, each HLA antigen can be broken into a series of epitopes – (3-dimensional) regions that are capable of binding antibody. These HLA epitopes may be shared across different HLA molecules. The regions of the epitope that are targets for antibody binding are short polymorphic (often nonlinear) amino acid sequences on the HLA molecule that are clustered together in the 3-dimensional structure of the HLA molecule and are known as 'eplets'.[161] Donor-recipient compatibility is expressed as the number of eplet-mismatches. Minimizing donor-recipient epitope mismatch reduces the number of viable 'foreign' targets to which the recipient may make antibody and represents a more relevant way quantify tissue mismatch. Certain eplet mismatches may be especially unfavorable and should be avoided if possible.

MEDICAL STATUS

The potential recipient should be evaluated to ensure that there are no new contraindications to general anaesthesia, and transplant surgery. This is more relevant to recipients of deceased donor allografts, for whom the date of surgery cannot be planned. Special emphasis should be placed on cardiovascular and infectious risk. With longer waiting periods for transplantation, it is common for a potential recipient to have increased cardiovascular risks since the time of last evaluation. The need for hemodialysis prior to surgery should be assessed. In general, preoperative hemodialysis is advisable if the plasma K is greater than 5.5 mmol/L or severe volume overload is present. The threshold for preoperative dialysis should be lower when delayed or slow graft function is anticipated or in patients with higher cardiovascular risk. A short session of 1½ to 2 hours without anticoagulation usually suffices. Patients on peritoneal dialysis need only drain out instilled fluid before surgery; if the patient is hyperkalemic, several rapid exchanges can be performed.

EVALUATION OF THE RECIPIENT IMMEDIATELY AFTER THE TRANSPLANT

The nephrologist should carefully review the donor history and the operating room records, noting all intraoperative fluids given, intra- and postoperative blood pressures and urine output, and any technical difficulties encountered. Warm and cold ischemia times should be established. Between donor nephrectomy and reperfusion there are two distinct 'warm ischemia' periods: 1) donor warm ischemia time/extraction time is the period during donor surgery between aortic cross clamping or asystole and the establishment of cold preservation and 2) graft warm ischemia time is the period between the removal of the kidney from cold storage to reperfusion of the graft in the recipient.[162,163] Cold ischemia time (CIT) is the time that the kidney spends in cold preservation and is sandwiched between the two warm ischemia periods. Prolonged warm and cold ischemia times increase the risk of ischemic graft injury and delayed graft function (DGF). Immediate excellent urine output, which should always be the case with living donor transplants, greatly simplifies management. The management of the oligoanuric recipient can be complicated and is discussed later.

MANAGEMENT OF ALLOGRAFT DYSFUNCTION

In the current era of surgical technique and immunosuppression, major postoperative complications are rare. The focus of clinical management has shifted from improving short-term outcomes to long-term outcomes. Early detection and treatment of early graft dysfunction is an important factor in preserving long-term allograft function. Because early signs of graft dysfunction rarely manifest as detectable symptoms, routine surveillance laboratory testing is a key element of posttransplant management. Surveillance laboratory testing should be performed regardless of the difficulty of the immediate post-operative course. In general, surveillance is performed frequently in the early posttransplant period, and progressively less frequently with time. While there may be variations to this schedule among different centers, a typical schedule for routine surveillance laboratory testing is shown in Table 70.3.

Management is discussed here under three time periods: immediate, early, and late post-transplant.

IMMEDIATE POSTTRANSPLANT PERIOD (FIRST WEEK)

Patients can be divided into three groups based on allograft function in the first posttransplant week: those with excellent graft function, DGF, and slow graft function (SGF). Excellent allograft function is manifest by an ample urine output and rapidly falling plasma creatinine concentration. Management of patients with excellent allograft function (almost all living donor recipients and a highly variable percentage of deceased donor allograft recipients) is relatively straightforward. Routine imaging studies are not required.

DGF is usually defined by the need for one or more dialysis treatments within the first posttransplant week. SGF defines

Table 70.3 Routine Surveillance Laboratory Testing After Transplantation

	< 1 Month	1-2 Months	2-6 Months	6-24 Months	>24 Months
CBC	Twice Weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
BMP/glucose/phosphorus	Twice Weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Drug level[a]	Twice Weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Liver enzymes	Weekly	Weekly	Every 2 weeks	Monthly	Every 6-12 months
Urinalysis	Twice Weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Lipid profile			Annually	Annually	Annually
UPCR[b]			3 Monthly	6 Monthly	Annually
BK PCR	Monthly	Monthly	3 Monthly	3/6 Monthly	
PTH[c]			3 Monthly[c]	6 Monthly[c]	Annually[c]
DSA[d]	[d]	[d]	[d]	[d]	[d]

[a]Depending on the immunosuppressive regimen, to include tacrolimus, cyclosporine, MPA.
[b]Frequent UPCR monitoring for patients with ESRD from FSGS.
[c]Posttransplant parathyroid hormone monitoring frequency determined by individual PTH and calcium levels.
[d]DSA monitoring frequency determined by patient risk profile.

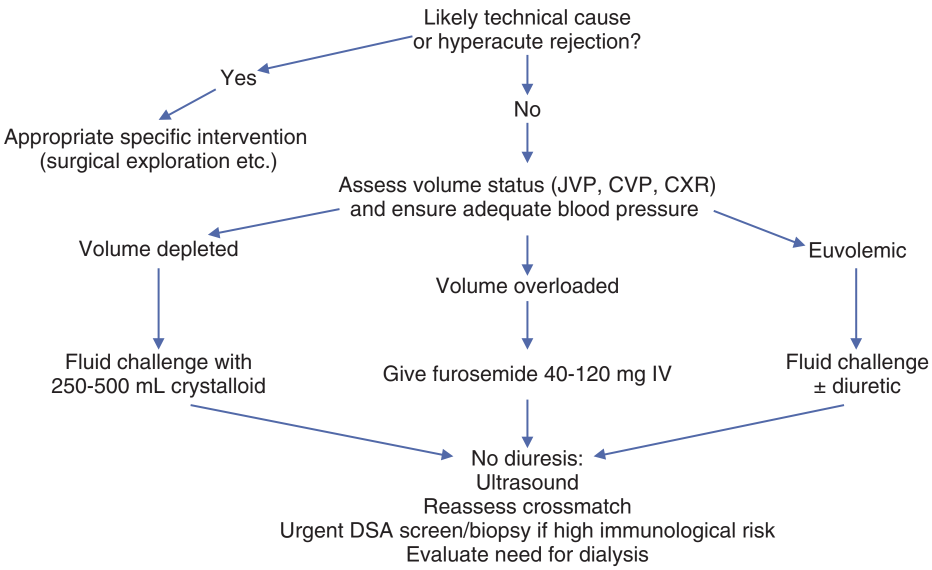

Fig. 70.3 Management of kidney allograft non-function/oliguria immediately after transplant.

a group of recipients with moderate early graft dysfunction. One commonly employed definition of SGF is a plasma creatinine level greater than 3 mg/dL at 1 week post-transplant.[164] The causes, management, and outcomes of SGF are similar to those of DGF (see below).[164] Interventions that simply convert DGF to SGF appear to have little effect on allograft outcomes. A scheme for managing allograft dysfunction in the immediate post-transplant period is depicted in Fig. 70.3.

DELAYED GRAFT FUNCTION

The definition of DGF as the requirement for dialysis within one week of transplantation is somewhat arbitrary. The designation excludes some patients with residual native kidney function, such as those undergoing preemptive transplantation, who have DGF per se, but do not require dialysis. There is also likely significant intra-center variation in the use of dialysis posttransplant. The incidence of DGF, by the conventional definition, varies widely depending on the donor source. Scientific Registry of Transplant Recipients data (up to 2012) shows a 3%, 23%, and 31% incidence of DGF in living donor, standard deceased donor, and expanded–criteria donor (ECD) allografts, respectively.[4] Although the causes of DGF include prerenal, intrarenal, and postrenal insults, ischemic acute tubular necrosis (ATN) is by far the most common cause of DGF.

Risk factors for DGF include recipient factors such as male sex, black race, longer dialysis vintage, higher panel-reactive antigen (PRA) status, and greater degree of HLA mismatching. Recipient hypotension requiring pretransplant midodrine use is associated with an increased risk of DGF, graft failure, and death.[165] Donor factors include use of deceased donor kidneys (especially with ECD or DCD), older donor age, and longer cold ischemia time. Most of these factors mediate their effects through ischemia-reperfusion injury as well as

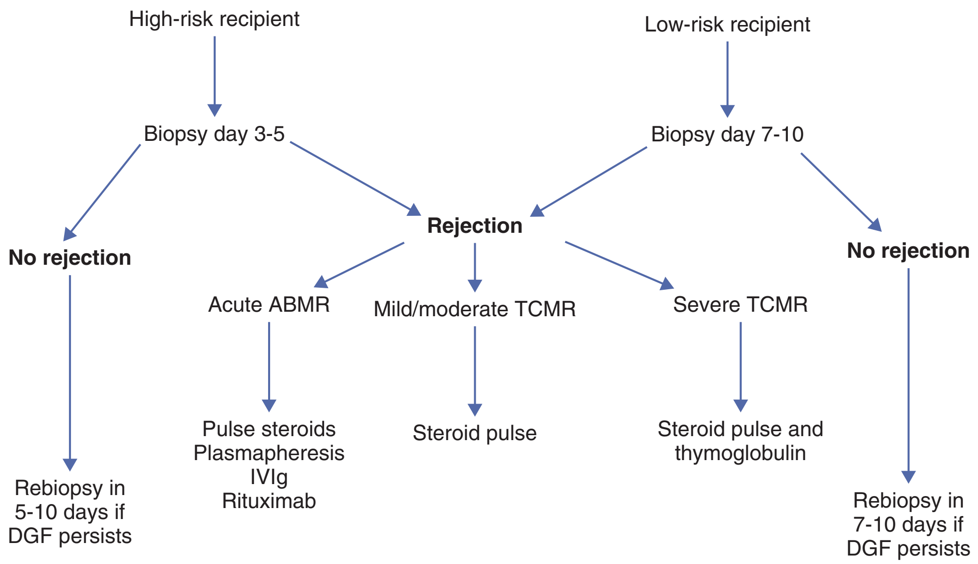

Fig. 70.4 Algorithm for management of persistent delayed graft function. The presence of anti-donor human leukocyte antigen (HLA) antibodies should prompt immediate biopsy in this setting.

immunologic mechanisms. Older studies suggested that CNIs prolonged or worsened DGF; however, a French randomized controlled trial comparing delayed versus immediate introduction of cyclosporine posttransplant showed no difference in DGF rates or graft function at 3 months but did show a numerical trend toward increased acute rejection in the delayed cyclosporine group.[166]

The diagnosis of the underlying cause of DGF is based on clinical, radiologic, and histologic findings. Careful review of the donor history and of the retrieval and transplantation procedure provides clues about the etiology of DGF. Notably, the interpretation of posttransplant urine output requires knowledge of the residual (native) urine output. Prerenal and postrenal causes (such as volume depletion and urinary catheter malposition or obstruction) should be excluded. If such simple steps fail to improve urine output, further investigation is warranted. Standard ultrasonography is commonly used to assess potential surgical complications in the immediate postoperative period. Ultrasonography can be performed quickly, is inexpensive and noninvasive, and is usually effective in identifying postrenal causes of kidney failure. Duplex sonography is also useful in assessing the graft's arterial and venous blood flow. The resistive index (RI) is often reported in transplant kidney ultrasounds and is elevated in the setting of intrarenal graft dysfunction. However, a raised RI does not discriminate between ATN and rejection and is therefore of limited diagnostic utility.[167]

The evaluation and management of DGF is patient context dependent. Persistent oliguria in a living donor kidney is much more likely to be a major surgical complication than ATN. However, in many cases, both prerenal and postrenal causes of DGF are excluded in deceased donor kidney transplant recipients who are at high risk of ischemic ATN. In the absence of a high suspicion for an alternative diagnosis (such as rejection), expectant management for 7 to 10 days posttransplant is a reasonable approach. A typical algorithm for managing persistent DGF is shown in Fig. 70.4.

Table 70.4 Risk Factors for DGF

Prerenal
Severe hypovolemia/hypotension Renal vessel thrombosis
Intrarenal
Ischemic ATN Hyperacute rejection Accelerated or acute rejection superimposed on ATN Acute CNI nephrotoxicity superimposed on ATN
Postrenal
Urinary tract obstruction/leakage

ATN, Acute tubular necrosis; *CNI,* calcineurin inhibitor; *DGF,* delayed graft function.

CAUSES OF DELAYED GRAFT FUNCTION

ISCHEMIC ACUTE TUBULAR NECROSIS

Ischemic ATN is the most common cause of DGF in deceased donor kidney recipients. At multiple steps during the surgical transplantation procedure, the allograft is at risk of ischemia-reperfusion injury (Table 70.4).[168]

There are no clinical or radiologic features unique to early posttransplant ATN. As is the case with acute kidney injury (AKI) in native kidneys, transplant ATN should be a diagnosis of exclusion. Several of the risk factors identified in Table 70.5 may be present. Intact allograft perfusion and the absence of obstruction should be confirmed with renal imaging. Histology, if available, shows tubular cell damage and necrosis. Patchy interstitial mononuclear cell infiltrates, but not tubulitis, may be present. The natural history of uncomplicated

Table 70.5 Causes of Ischemic Damage to the Deceased Donor Kidney Allograft

Donor State Prior to Organ Procurement
Shock syndromes
Endogenous and exogenous catecholamines
Brain injury[a]
Nephrotoxic drugs
Organ Procurement Surgery
Hypotension
Trauma to renal vessels
Organ Transport and Storage
Prolonged storage (cold ischemia time)
Pulsatile perfusion injury
Transplantation of Recipient
Prolonged second warm ischemia time
Trauma to renal vessels
Hypovolemia/hypotension
Postoperative Period
Cyclosporine/tacrolimus
Acute heart failure/venous congestion
Hemodialysis[a]

[a]Some evidence.

ATN is spontaneous resolution. Usually, improvements in urine output begin from 5 to 10 days after transplant, but ATN may persist for weeks.

Management of the patient during this period is supportive. When early hemodialysis is required, minimal anticoagulation should be used to reduce the risk of postsurgical bleeding. Intradialytic hypotension should also be avoided in order to prevent further renal ischemic injury. Peritoneal dialysis may be successfully continued posttransplant, although should be avoided if the peritoneum was opened at the time of surgery. Early postoperative treatments should be performed with low volume exchanges.

A major concern in the management of the patient with posttransplant ATN is that the concurrent diagnosis of new surgical or medical complications involving the allograft is difficult. Rejection, for example, may be easily missed. In fact, acute rejection occurs more frequently in allografts with delayed as versus immediate function. The postulated mechanism is that ischemia-reperfusion injury increases the immunogenicity of the allograft, thereby predisposing to acute rejection. Indeed, experimental animal models have demonstrated that ischemic ATN is associated with increased expression/production of class I and II MHC molecules, co-stimulatory molecules, proinflammatory cytokines, and adhesion molecules within the kidney parenchyma.[168] Such an altered local milieu could amplify the alloimmune response. Therefore, a high degree of suspicion for additional complications related to the allograft must be maintained. The possibility of accelerated acute rejection must be considered, particularly in the high-risk recipient. We recommend a low threshold for performing a graft biopsy in patients with DGF. Radiologic evaluation of the graft should also be repeated if there is suspicion of new urinary or vascular complications.

HYPERACUTE REJECTION

Hyperacute rejection has become a rare cause of immediate graft non-function because of improved tissue-typing technology. However, the increased prevalence of desensitization protocols has made this diagnosis regain clinical relevancy. Hyperacute rejection is caused by preformed recipient antibodies reacting with antigens on the endothelium of the allograft, resulting in activation of the complement and coagulation cascades. These antibodies are usually directed against antigens of the ABO blood group system or against HLA class I antigens. Anti-HLA class I antibodies are formed in response to previous transplantation, blood transfusion, or pregnancy. Less commonly, hyperacute rejection is caused by antibodies directed against donor HLA class II antigens or endothelial or monocyte antigens (the last two are not detected in the standard crossmatch). In classic hyperacute rejection, cyanosis and mottling of the kidney and anuria occur minutes after the vascular anastomosis is established. Disseminated intravascular coagulopathy may occur. Histology shows widespread small vessel endothelial damage and thrombosis, usually with neutrophils incorporated into the thrombus. There is no effective treatment and transplant nephrectomy is indicated. Screening for recipient-donor ABO or class I HLA incompatibility (the presence of the latter is often referred to as a “positive T-cell crossmatch”) has ensured that hyperacute rejection is now uncommon. Rare cases occur because of clerical errors or due to the presence of the other preformed antibodies that are not detected by routine screening methods.

ACCELERATED REJECTION SUPERIMPOSED ON ACUTE TUBULAR NECROSIS

Accelerated acute rejection refers to rejection occurring roughly 2 to 5 days after transplant. Accelerated rejection occurs in recipients with pretransplant sensitization to donor alloantigen and is frequently associated with the presence of historic or low-titer pretransplant anti-donor antibody. Rapid post-transplant antibody production by memory B cells may represent an important mechanism for this phenomenon.[169]

Accelerated acute rejection may be superimposed on ischemic ATN, in which case there may be no signs of rejection, or it may occur in an initially functioning allograft. Diagnosis is made by kidney biopsy in conjunction with crossmatch findings and DSA titers. Histology usually shows evidence of predominantly antibody rather than cell-mediated immune damage. The diagnosis and management of these two forms of rejection are discussed in detail below.

ACUTE CYCLOSPORINE OR TACROLIMUS NEPHROTOXICITY SUPERIMPOSED ON ACUTE TUBULAR NECROSIS

Cyclosporine and tacrolimus, especially in high doses or by intravenous route of administration, may result in an acute decrease in GFR through renal vasoconstriction, particularly of the afferent glomerular arteriole. Potentially, such vasomotor effects could exacerbate ischemic ATN. Acute CNI toxicity is now rare with the targeting of lower CNI levels. Caution should, however, be taken in the context of drug interactions that raise CNI levels (discussed later).

VASCULAR AND UROLOGIC COMPLICATIONS OF SURGERY

Renal vessel thrombosis, urinary leaks, and obstruction are rare but important causes of DGF. These complications may also cause allograft dysfunction in the early postoperative period and are discussed later in this chapter.

OUTCOME AND SIGNIFICANCE OF DELAYED GRAFT FUNCTION

In most cases, recovery of kidney function is sufficient to become independent of dialysis. There is no renal recovery in less than 5% of cases, resulting in primary non-function (PNF). The majority of studies suggest that DGF has a negative impact on long-term kidney allograft survival.[168] Patients with DGF require longer hospitalization and more investigations, and are at higher risk of occult rejection. Postoperative fluid and electrolyte management is more difficult.

Therefore, measures that limit the incidence and duration of DGF are important. Graft injury may occur 1) prior to donation, 2) at retrieval, 3) during transport, 4) during transplant surgery, or 5) postoperatively. Prior to donation heart-beating donors are potentially exposed to various renal insults that may impact future graft function. Good ICU management of the potential deceased donor is of great importance in mitigating many of these factors. Treating the donor with dopamine prior to organ retrieval has been shown to reduce the rate of DGF.[170] However, other specific donor interventions, including desmopressin and steroids, have not been shown to improve subsequent graft function.[171,172]

Meticulous surgical technique, rapid transport of retrieved allografts and use of optimum preservation solutions are also of extreme importance. Data regarding machine perfusion has been somewhat conflicting. A study that randomly assigned one kidney from 336 consecutive deceased donors to machine perfusion and the other to cold storage, showed a reduction in DGF and an improvement in 1-year graft survival in the machine perfusion group.[173] However, machine perfusion was associated with no benefit compared to cold storage in kidneys donated after cardiac death in a UK study.[174] The major disadvantages of machine perfusion are its relative expense and complexity compared to standard cold perfusion techniques.

Cold ischemia time is an important risk factor for DGF.[175] Measures that may decrease CIT include faster identification of potential recipients, the establishment of a list of patients in each transplant region who would quickly accept ECD kidneys and a consensus on allocation and management of DCD organs.[176,177] Nationwide organ sharing of zero HLA-mismatched organs also results in prolongation of cold ischemia time in some cases. However, the small increase in CIT, associated with national organ sharing, is offset by the substantial immunologic benefits of zero-HLA mismatched transplantation.[178] Similarly, the new Kidney Allocation System in the United States has resulted in greater regional and national sharing of organs for highly sensitized patients at the expense of a longer CIT.[138]

Intraoperative mean arterial pressure should be maintained greater than 70 mm Hg in the recipient. In cases in which DGF is expected, antilymphocyte antibody preparations are often used and may have benefits beyond acute rejection prophylaxis. In experimental models, anti-thymocyte globulins directly ameliorate ischemia reperfusion injury through modulation adhesion molecule expression and the inflammatory response.[179] Indeed, intra-operative (versus postoperative) thymoglobulin administration was associated with reduced DGF in one randomized study.[180] Another study demonstrated reduced DGF with thymoglobulin over daclizumab.[26]

Despite theoretical benefits, peri-transplant erythropoitin treatment showed no effect on DGF in two randomized controlled trials.[181,182] Calcium channel blockers have been shown in experimental models to prevent ischemic injury and attenuate CNI-mediated renal vasoconstriction. These properties suggested that their administration to recipients or to the donor before organ retrieval might reduce the incidence and duration of ischemic ATN. A Cochrane review that included 13 trials suggested a reduced risk of ATN and delayed graft function associated with perioperative calcium channel blocker use but no difference in graft loss or mortality.[183] A number of interventional trials have attempted to target patients at risk of DGF with various treatments to prevent IRI in the immediate post-transplant period. Experimental treatments have included C1 esterase inhibition and p53 siRNA.[184,185] However, data are currently insufficient to recommend their inclusion into the transplant therapeutic repertoire.

EARLY POSTTRANSPLANT PERIOD (FIRST SIX MONTHS)

Table 70.6 shows the causes of allograft dysfunction during the early post-transplant period. Despite its known limitations, the primary measure of early and late transplant function remains the plasma creatinine concentration. Prerenal and postrenal causes of graft dysfunction should be systematically excluded. An algorithm for the management of allograft dysfunction in the early posttransplant period is shown in Fig. 70.5.

Table 70.6 Causes of Allograft Dysfunction in the Early Postoperative Period

Prerenal
Hypovolemia/hypotension
Renal vessel thrombosis
Hemodynamic drug effects: ACE inhibitors, NSAIDs, acute CNI nephrotoxicity
Transplant renal artery stenosis
Intrarenal
Acute rejection – TCMR or ABMR
CNI-induced thrombotic microangiopathy
Recurrence of primary disease
Acute pyelonephritis
Acute interstitial nephritis
Postrenal
Urinary tract obstruction/leakage

ACE, Angiotensin-converting enzyme; *CNI,* calcineurin inhibitor; *NSAIDS,* nonsteroidal antiinflammatory drugs.

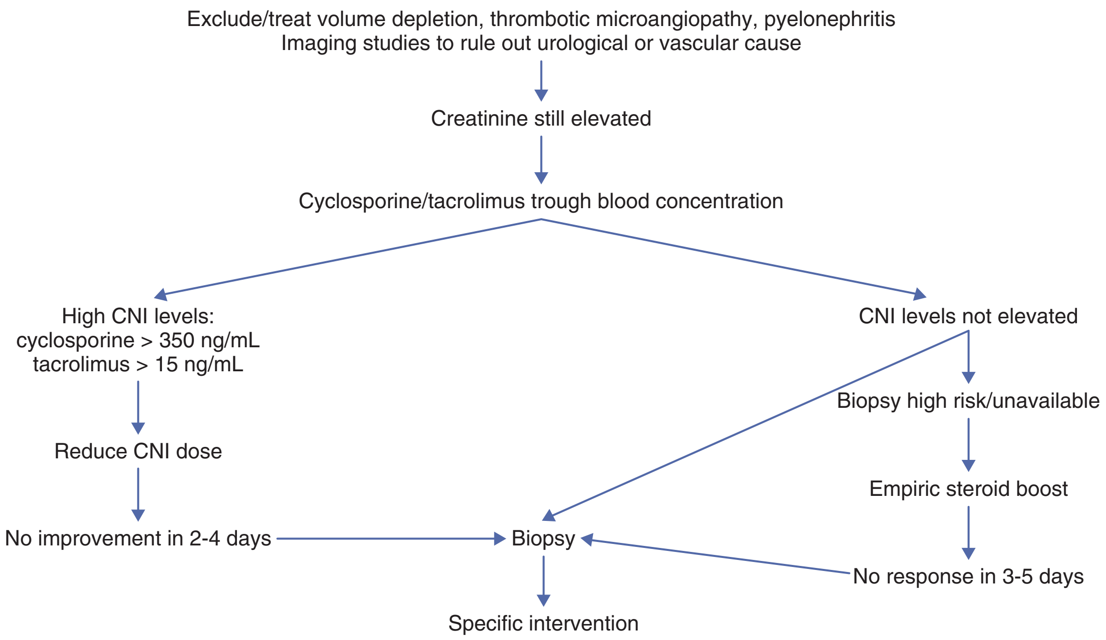

Fig. 70.5 Algorithm for management of allograft dysfunction in early posttransplant period. *CNI,* Calcineurin inhibitor.

PRERENAL RENAL DYSFUNCTION IN THE EARLY POSTTRANSPLANT PERIOD

Hypovolemia and Drugs

Hypovolemia may develop after excessive diuresis from the transplanted kidney or from diarrhea. Diarrhea is a common adverse effect of the MMF, especially when used with tacrolimus. Angiotensin-converting enzyme (ACE) inhibitors (ACE-Is), angiotensin-receptor blockers (ARBs), and nonsteroidal antiinflammatory drugs (NSAIDs) should be avoided in the early posttransplant period because of the risk of functional prerenal failure, a risk that may be potentiated by the renal vasoconstrictive effects of CNIs.

Renal Vessel Thrombosis

Transplant renal artery or renal vein thrombosis usually occurs in the first 72 hours but may be delayed for up to 10 weeks. Acute vascular thrombosis is the most common cause of allograft loss in the first week. Although surgical technique is a factor in some cases, there is now greater appreciation of the role of hypercoagulable states.[15]

Renal artery thrombosis presents with abrupt onset of anuria (unless there is a native urine output) and rapidly rising plasma creatinine, but often with negligible graft pain. Duplex studies show absent arterial and venous blood flow. Renography or magnetic resonance (MR) angiography shows absent perfusion of the transplanted kidney. Removal of the infarcted kidney is indicated.

Renal vein thrombosis also manifests with anuria and rapidly increasing plasma creatinine. Pain, tenderness, swelling in the graft, and hematuria are more pronounced than in renal artery thrombosis. Severe complications such as embolization or graft rupture and hemorrhage may occur. Duplex studies show absent renal venous blood flow and characteristic highly abnormal renal arterial waveforms. MR venography demonstrates thrombus in the vein. Transplant nephrectomy is usually indicated. If the venous thrombosis extends beyond the renal vein, anticoagulation is necessary to reduce the risk of embolization. There are reports of salvaging kidney function after early diagnosis of renal vessel thrombosis and its treatment with thrombolysis or thrombectomy. In almost all cases, however, infarction occurs too quickly to make this treatment worthwhile. Furthermore, thrombolysis is relatively contraindicated in the early posttransplant period because of the high risk of graft-related bleeding.

INTRARENAL DYSFUNCTION IN THE EARLY POSTTRANSPLANT PERIOD

Acute Rejection

Acute rejection is characterized by a decline in kidney function mediated by a recipient immune reaction against the allograft. Although acute rejection can occur at any time, it is most common in the first 6 months posttransplant. Fortunately, the incidence of acute rejection has decreased dramatically in the past 25 years; it is now around 10% in the first 12 months in the United States.[123]

In the era of modern immunosuppression, symptoms and signs of acute rejection are rarely pronounced, but low-grade fever, oliguria, and graft pain or tenderness may occur. Most cases of acute rejection are identified through surveillance monitoring of graft function. However, creatinine is a rather late and insensitive marker of renal injury. There is, therefore, a growing interest in development of early biomarkers of immune system activation. However, for now, the definitive diagnosis of acute rejection requires transplant biopsy.

Acute rejection involves both cellular and/or humoral immune mechanisms (TCMR and ABMR). Clinical transplantation has traditionally been focused on cell-mediated responses. However, ABMR has been receiving more attention because of improvements in diagnostic techniques and increased transplantation of immunological high-risk candidates. Differences between TCMR and ABMR are summarized in Table 70.7.

Table 70.7 Differences Between Pure Forms of Acute TCMR and Active ABMR

	Acute TCMR	Active ABMR
Timing of clinical onset	>5 days	>0 days
Tubulitis/interstitial inflammation	Present	Absent
Microvasular inflammation: peritubular capillaritis and/or glomerulitis	Absent	Present
C4d staining of peritubular capillaries	Absent	Present
Donor-specific antibody in serum	Usually absent	Present
Therapy	Pulse steroids/ATG	Pulsed steroids, plasmapheresis, IVIg, rituximab steroids, rituximab

T-Cell Mediated Rejection. The Banff classification[186,187] (Table 70.8) is a widely used schema for classifying rejection. Histologic findings characteristic of TCMR include 1) mononuclear cell infiltration of the interstitium, mainly with T cells but also with some macrophages and plasma cells; 2) tubulitis (infiltration of tubule epithelium by lymphocytes); and 3) arteritis which manifests as infiltration of mononuclear cells beneath the endothelium. Vascular involvement reflects more severe rejection.

Borderline TCMR, where histologic findings fall below the threshold for Banff 1A TCMR, is a fairly frequent finding. We usually treat borderline rejection.[188] Conversely, histologic evidence of rejection can also be seen in the presence of stable allograft function (subclinical TCMR). Some studies have reported improvement of graft function with treatment of subclinical TCMR,[189] but no benefit was found in a larger multicenter trial.[190] The presence of eosinophils in the infiltrate suggests severe rejection, but allergic interstitial nephritis should also be considered. Note that polyoma virus infection may also cause tubulointerstitial nephritis.

Uncomplicated TCMR is generally treated with a short course of high-dose steroids. There is a 60% to 70% response rate to this regimen, but the dose and duration of treatment has not been standardized. Typically, methylprednisolone 250 to 500 mg is given intravenously for 3 to 5 days. After completion of pulse therapy, the dose of oral steroids can be tapered back or resumed immediately to the maintenance dose. If the patient has been on a steroid-free regimen, adding a maintenance dose should be considered. An episode of acute rejection implies that prior immunosuppression has been inadequate. The patient's compliance with prescribed medications should be reviewed. If there are no contraindications, baseline immunosuppression should be increased or changed, at least in the short term. Lymphocyte-depleting antibodies are highly effective in treating first rejection episodes but because of toxicity and cost, these agents are usually reserved for steroid-resistant cases or when there is severe rejection on the initial biopsy.[191]

Steroid-resistant TCMR, defined somewhat arbitrarily as failure of improvement in urine output or plasma creatinine within 5 days of starting pulse treatment, is usually treated with depleting antibodies. If steroid treatment was based on an empiric rather than a histologic diagnosis, allograft biopsy should be performed to confirm this diagnosis before starting treatment with depleting antibody agents. A higher grade of TCMR with endothelial involvement (Banff II or III) is more likely to be steroid-resistant, and many centers use depleting antibody therapy as an initial treatment. The advantage of this approach is prompt and effective treatment of the TCMR in higher-risk patients. The disadvantage is cost, inconvenience, and exposure of the patients to potentially serious complications of therapy such as infection and cancer. However, in steroid-resistant rejection, the benefits of lymphocyte depleting agents outweigh their risks. ATG is the most commonly employed depleting agent.[32]

Refractory T-Cell Mediated Rejection. Refractory TCMR is generally defined as TCMR resistant to treatment with anti-lymphocyte antibody. By definition, the patient has already received aggressive immunosuppression; the risks and benefits of further amplifying immunosuppression should be very carefully considered. Renal histology is helpful in this regard. Therapeutic options include 1) continuing maintenance immunosuppression in the hope that kidney function will slowly improve, 2) repeating a course of antilymphocyte antibody therapy, or 3) switching from cyclosporine to tacrolimus if not already done.[192] If there is a component of active ABMR, this can be treated as discussed below.

Active Antibody-Mediated Rejection. ABMR is increasingly recognized as a cause of allograft dysfunction and is reported in 3.5% to 9% of for-cause biopsies.[193,194] Increased recognition of ABMR is partly a result of improved diagnostics (in particular, assays for C4d and DSA[195]) and in the expansion of immunologically high-risk and incompatible transplantation.[59] Diagnosis of active ABMR requires the presence of all of the following: 1) characteristic histological features which include microvascular inflammation (peritubular capillaritis and/or glomerulitis), intimal or transmural arteritis, thrombotic microangiopathy (TMA), and acute tubular necrosis; 2) evidence of recent antibody-endothelial interaction, usually identified using C4d staining of peritubular capillaries; and 3) serologic evidence of antibody against donor HLA or other antigens.[196] Active ABMR typically occurs early after transplantation but can also occur late, especially in the setting of reduced immunosuppression or noncompliance. Active ABMR may occur alone or with TCMR. In addition, subclinical active ABMR is commonly present on surveillance biopsies of immunologically high-risk kidney transplant recipients and is associated with poor graft outcome.[197]

The prognosis of active ABMR is poorer than TCMR. The optimal treatment of ABMR is as yet unestablished.[188,198] The most commonly employed regimens for active ABMR

Table 70.8 Banff Diagnostic Categories for Kidney Transplant Histopathology (Including 2017 Update)

Category 1: Normal biopsy or *nonspecific changes*

Category 2: Antibody-mediated changes

Active ABMR; all 3 criteria must be met for diagnosis

1. Histologic evidence of acute tissue injury, including 1 or more of the following:
   - Microvascular inflammation (g > 0 and/or ptc > 0), in the absence of recurrent or de novo glomerulonephritis, although in the presence of acute TCMR, borderline infiltrate, or infection, ptc ≥1 alone is not sufficient and g must be ≥1
   - Intimal or transmural arteritis (v > 0)c
   - Acute thrombotic microangiopathy, in the absence of any other cause
   - Acute tubular injury, in the absence of any other apparent cause
2. Evidence of current/recent antibody interaction with vascular endothelium, *including 1 or more of the following*:
   - Linear C4d staining in peritubular capillaries (C4d2 or C4d3 by IF on frozen sections, or C4d >0 by IHC on paraffin sections)
   - At least moderate microvascular inflammation ([g + ptc] ≥2) in the absence of recurrent or de novo glomerulonephritis, although in the presence of acute TCMR, borderline infiltrate, or infection, ptc ≥2 alone is not sufficient and g must be ≥1
   - Increased expression of gene transcripts/classifiers in the biopsy tissue strongly associated with ABMR, *if thoroughly validated*
3. Serologic evidence of donor-specific antibodies (DSA to HLA or other antigens). C4d staining or expression of validated transcripts/classifiers as noted above in criterion 2 may substitute for DSA; however, thorough DSA testing, including testing for non-HLA antibodies if HLA antibody testing is negative, is strongly advised whenever criteria 1 and 2 are met.

Chronic active ABMR; all 3 criteria must be met for diagnosis

1. Morphologic evidence of chronic tissue injury, including 1 or more of the following:
   - Transplant glomerulopathy (cg >0) if no evidence of chronic TMA or chronic recurrent/de novo glomerulonephritis; includes changes evident by electron microscopy (EM) alone (cg1a)
   - Severe peritubular capillary basement membrane multilayering (requires EM)
   - Arterial intimal fibrosis of new onset, excluding other causes; leukocytes within the sclerotic intima favor chronic ABMR if there is no prior history of TCMR, but are not required
2. Identical to criterion 2 for active ABMR, above
3. Identical to criterion 3 for active ABMR, above, including strong recommendation for DSA testing whenever criteria 1 and 2 are met

C4d staining without evidence of rejection; all 4 features must be present for diagnosis

1. Linear C4d staining in peritubular capillaries (C4d2 or C4d3 by IF on frozen sections, or C4d>0 by IHC on paraffin sections)
2. Criterion 1 for active or chronic, active ABMR not met
3. No molecular evidence for ABMR as in criterion 2 for active and chronic, active ABMR
4. No acute or chronic active TCMR, or borderline changes

Category 3: Borderline changes

Suspicious (borderline) for acute TCMR

Foci of tubulitis (t > 0) with minor interstitial inflammation (i0 or i1), or moderate-severe interstitial inflammation (i2 or i3) with mild (t1) tubulitis; retaining the i1 threshold for borderline with t > 0 is permitted although this must be made transparent in reports and publications

- No intimal or transmural arteritis (v = 0)

Category 4: TCMR

Acute TCMR

Grade IA	Interstitial inflammation involving >25% of nonsclerotic cortical parenchyma (i2 or i3) with moderate tubulitis (t2) involving 1 or more tubules, not including tubules that are severely atrophic
Grade IB	Interstitial inflammation involving >25% of nonsclerotic cortical parenchyma (i2 or i3) with severe tubulitis (t3) involving 1 or more tubules, not including tubules that are severely atrophic
Grade IIA	Mild to moderate intimal arteritis (v1), with or without interstitial inflammation and/or tubulitis
Grade IIB	Severe intimal arteritis (v2), with or without interstitial inflammation and/or tubulitis
Grade III	Transmural arteritis and/or arterial fibrinoid necrosis of medial smooth muscle with accompanying mononuclear cell intimal arteritis (v3), with or without interstitial inflammation and/or tubulitis

Chronic Active TCMR

Grade IA	Interstitial inflammation involving >25% of the total cortex (ti score 2 or 3) and >25% of the sclerotic cortical parenchyma (i-IFTA score 2 or 3) with moderate tubulitis (t2) involving 1 or more tubules, not including severely atrophic tubules; other known causes of i-IFTA should be ruled out
Grade IB	Interstitial inflammation involving >25% of the total cortex (ti score 2 or 3) and >25% of the sclerotic cortical parenchyma (i-IFTA score 2 or 3) with severe tubulitis (t3) involving 1 or more tubules, not including severely atrophic tubules; other known causes of i-IFTA should be ruled out
Grade II	Chronic allograft arteriopathy (arterial intimal fibrosis with mononuclear cell inflammation in fibrosis and formation of neointima)

Category 5: Interstitial fibrosis and tubular atrophy

Grade
- I. Mild interstitial fibrosis and tubular atrophy (≤25% of cortical area)
- II. Moderate interstitial fibrosis and tubular atrophy (26% to 50% of cortical area)
- III. Severe interstitial fibrosis and tubular atrophy (>50% of cortical area)

Table 70.8 Banff Diagnostic Categories for Kidney Transplant Histopathology (Including 2017 Update) (Cont'd)

Category 6: Other changes not considered to be caused by acute or chronic rejection

- BK virus nephropathy
- Posttransplant lymphoproliferative disorders
- Calcineurin inhibitor nephrotoxicity
- Acute tubular injury
- Recurrent disease
- De novo glomerulopathy (other than transplant glomerulopathy)
- Pyelonephritis
- Drug-induced interstitial nephritis

include some combination of pulsed steroids, plasmapheresis, intravenous immunoglobulins, and anti-CD20 monoclonal antibody to suppress/remove DSA.[199] Other therapies that have been used for active ABMR include C1 inhibition, eculizumab, bortezomib, and bortezomib plus conversion to belatacept.[198,200]

There is also a growing body of literature regarding non-HLA antibody-mediated ABMR.[201] The best studied non-HLA antibody implicated in graft injury is probably the angiotensin II type 1-receptor activating antibody, the effects of which may be mitigated by angiotensin receptor blockade blocker therapy.[202]

Significance of Acute Rejection

Although acute rejection is frequently reversed, retrospective studies show that it is strongly associated with the development of chronic rejection and poorer allograft survival. Both increased severity (as evidence by histology and change in creatinine) and later timing (more than 6 months posttransplant) of acute rejection are associated with poorer long-term outcomes.[203] Whatever the outcome is in terms of allograft function, treatment involves exposing the patient to supplemental immunosuppression and its attendant risks.

Acute Calcineurin Inhibitor Nephrotoxicity. The CNIs, especially in high doses, cause an acute decrease in GFR by renal vasoconstriction, particularly of the afferent glomerular arteriole. This is manifested clinically as dose and blood concentration–dependent acute reversible increase in plasma creatinine. Because acute CNI nephrotoxicity is mainly hemodynamic in origin, histology is frequently normal. However, with prolonged CNI toxicity tubular cell vacuolization and hyaline arteriolar thickening may be seen.[204] The treatment of acute CNI nephrotoxicity is dose reduction.

Distinguishing Acute Calcineurin Inhibitor Nephrotoxicity and Acute Rejection. Distinguishing acute CNI nephrotoxicity and acute rejection clinically can be difficult. Low and high blood concentrations in the presence of rising creatinine suggest but do not imply rejection and drug nephrotoxicity, respectively. Both syndromes can coexist. Indicators of a diagnosis of acute CNI nephrotoxicity are severe tremor (neurotoxicity), a moderate increase in plasma creatinine (>25% over baseline), and high trough blood CNI concentrations (e.g., cyclosporine >350 ng/mL or tacrolimus levels >15 ng/mL). Indicators of a diagnosis of acute rejection are low-grade fever, allograft pain and tenderness (although with current drug regimens these symptoms or signs are uncommon), rapid, non-plateauing increases in plasma creatinine, and low drug concentrations. Fever and symptoms localized to the allograft do not occur with CNI toxicity but do not necessarily imply rejection; acute pyelonephritis must also be considered.

If acute CNI nephrotoxicity is suspected, our practice is to reduce the CNI dose and repeat a serum creatinine and drug level within 48 to 96 hours. If graft function has not improved or plateaued at this point we usually go on to kidney biopsy. The threshold for biopsy is lower in high-risk patients: those who are highly sensitized, have previously rejected an allograft, or are at high risk of early recurrent primary kidney disease (see later). In certain circumstances, where kidney biopsy is deemed high risk, we will empirically treat with a steroid pulse for a presumptive diagnosis of acute rejection. However, failure of graft function to rapidly improve with this strategy will usually prompt a biopsy.

Most transplant centers provide rapid biopsy and processing of tissue, with basic histology available within 5 to 6 hours. Since a delay of 6 hours in initiating specific therapy should not be detrimental to the graft, a biopsy-proven diagnosis is the preferred approach. In addition to determining the degree and type of rejection in the allograft, histology also occasionally reveals unexpected pathology such as TMA or polyoma virus infection. Biopsy results alone should not dictate management; rather, the combination of clinical and histologic findings should be used to shape a treatment plan.

Monitoring. Signs and symptoms of graft dysfunction often occur late, after significant graft injury. Thus, monitoring of serum electrolytes and immunosuppressive drug levels is an essential part of posttransplant management. Serum creatinine, basic chemistry panel, liver function tests, and CBC are routinely checked to screen for graft dysfunction and manifestations of drug toxicity. Drug levels for CNI, MMF, and mTORi are also monitored for adjustment of immunosuppressive drug dosing. The frequency of monitoring is greater immediately posttransplant, and gradually decreased. At our institution, we monitor routine blood levels twice weekly during the first month, once a week during the second month, and once every two weeks during the 3rd-6th months posttransplant. Thereafter, we require monitoring on a monthly basis. The frequency of monitoring is increased if there is graft dysfunction and subsequent treatment. BK plasma PCR monitoring is performed at set time points in the first two years posttransplant.

Most patients are on CNIs. Cyclosporine A (CsA) can be measured using trough (C0) levels, 2-hour post-dose (C2) or through abbreviated AUCs. C0 is the measured level after the dosing interval (e.g., 12 hours after dosing if given every 12 hours), C2 is the measured level 2 hours after dosing, and AUC is the area under the curve during the first 4 hours after dosing. C2 levels correlate more closely with AUC, but no significant differences have been found in the incidence of acute rejection, graft loss, or adverse events between patients monitored using C0 or C2 levels.[205] The standard target level for CsA is C0 of 150-300 ng/mL early and 100-200 ng/mL late posttransplant[206] or C2 1400-1800 ng/mL early and 800-1200 ng/mL later after transplantation.[207] Tacrolimus C0 levels correlate better with AUC[208,209] and measuring the C0 tacrolimus level is usual practice. The standard target level for tacrolimus C0 is 8-12 ng/dL in the early posttransplant months and 6-9 ng/dL in the first posttransplant year to year–and-a-half. We individualize our longer term tacrolimus targets aiming for a level at the lower or higher end of 5-8 ng/mL depending on immune risk, tolerability, and other clinical factors. For patients with evidence of CNI toxicity and who are otherwise stable, compliant, and at low immunological risk, a target of 3-7 ng/mL may be reasonable in the long term.[80,210]

There are two major weaknesses in transplant rejection monitoring and diagnosis in its current form. 1) Graft dysfunction (as evidenced by creatinine rise) may occur relatively late into a rejection episode, resulting in a delay in diagnosis and treatment. 2) Kidney biopsy, the gold standard for rejection diagnosis, is expensive, invasive and has real risks. The development of biomarkers that can aid in the early detection of graft dysfunction or even obviate the need for biopsy is therefore highly desirable. Biomarkers that have been investigated include cytokines, IL-2 receptor, CD30, adhesion molecules, and other inflammatory markers such as complement and acute-phase proteins.[211–213] A three-gene model of urinary 18S-normalized CD3ε messenger RNA, 18S-normalized IP-10 messenger RNA, and 18S ribosomal RNA showed good discrimination for acute rejection, with the test preempting acute rejection episodes by up to 10 days.[214]

Donor-derived cell-free DNA measurement is now commercially available. The technique utilizes the presence of single nucleotide polymorphisms (SNPs), which are nucleotides at specific positions in the human genome that commonly vary between individuals.[215,216] The most commonly used assay analyzes over 250 different SNPs, allowing the test to distinguish donor-form recipient cell-free DNA without a donor blood sample. The assay reports donor-derived cell-free DNA as a percentage of total cell-free DNA in the recipient's blood. A donor-derived cell-free DNA level in excess of 1% likely reflects underlying graft injury. The test is of limited utility in the diagnosis of TCMR, especially Banff 1A or less. However, testing appears more sensitive for microvascular inflammation and ultimately may prove useful in identifying those patients with subclinical ABMR.[217] The test cannot be reliably used in those patients with prior kidney transplants or other solid organ transplants.

Protocol posttransplant DSA monitoring is performed at many centers on all or select groups of patients. DSA may either be formed prior to transplant (preformed) or occur de novo posttransplant. De novo (versus preformed) DSA are associated with worse allograft outcomes.[218] DSA may be accompanied by 1) normal renal function and histology, 2) normal renal function and ABMR histology (subclinical ABMR), 3) overt ABMR, or 4) chronic active ABMR. DSA screening is certainly warranted in immunologically high-risk recipients and at the time of 'for clinical indication' biopsy. Treatment of DSA depends on accompanying clinical and histological findings. The detection of an isolated de novo DSA warrants intensification of immunosuppression, strong consideration for allograft biopsy (irrespective of graft function) and, in some centers, will trigger specific therapies such as IVIg. DSA may be further risk stratified by assessing their ability to bind complement (C1q assay) or by typing their (IgG) subclass. Treatment of DSA accompanied by ABMR is discussed separately.

The use of molecular diagnostic tools, such as the 'molecular microscope', which employs a microarray-based approach to measure differential gene expression in renal biopsy tissue, have promise, and may prove particularly useful in diagnosis and prognosis of ABMR.[219]

Many transplant centers perform elective protocol biopsies at set time points after transplant irrespective of graft function. The merits and demerits of such an approach are keenly debated.[220] To our knowledge, there are no data showing that widespread screening with protocol biopsies in the tacrolimus-MMF era results in better outcomes.[190] Protocol biopsies may, however, have a useful role in selected high immunological risk groups.[221]

Acute Thrombotic Microangiopathy

Acute de novo TMA after kidney transplantation is a rare but serious complication.[222] It usually occurs in the early post-transplant period and is accompanied by increasing plasma creatinine and lactate dehydrogenase levels, thrombocytopenia, falling hemoglobin level, schistocytosis, and low haptoglobin concentrations. This diagnosis can be overlooked because thrombocytopenia and anemia occur commonly after transplantation in the setting for ATG induction therapy. The diagnosis is confirmed by allograft biopsy, which shows endothelial damage and, in severe cases, thrombosis of glomerular capillaries and arterioles.

Causes include CNIs,[223] OKT3, ABMR,[224] viral infections such as cytomegalovirus (CMV), and the recurrence of a previously undiagnosed primary disease. The presence of hepatitis C and anticardiolipin antibodies increases the risk.[225] Early diagnosis of TMA is essential to salvage kidney function. There are no controlled trials of therapy for de novo TMA after transplant. Our initial measure is to switch CNI (either from tacrolimus to cyclosporine or cyclosporine to tacrolimus). Another reasonable approach is to switch off treatment with the CNI drug class completely and move to a belatacept or mTori-based immunosuppression regimen. We concurrently look for other potential underlying etiologies including ABMR. If the syndrome fails to improve with the above measures we initiate plasma exchange.[226,227] Eculizumab has also been employed successfully.[228,229]

Acute Pyelonephritis

Urinary tract infections (UTIs) may occur at any period but are most frequent shortly after transplantation because of catheterization, stenting, and aggressive immunosuppression.[230] Other risk factors for urinary tract infection (UTI) are anatomic abnormalities and neurogenic bladder. Fever, allograft pain and tenderness, and leukocytosis are usually more pronounced in acute pyelonephritis than in acute rejection. Diagnosis requires urine culture, but empiric antibiotic treatment should be started immediately. Delay in treatment can lead to rapid clinical decline in the immunosuppressed patient. The most commonly implicated microorganisms are gram-negative bacilli, coagulase-negative staphylococci, and enterococci. Kidney function usually returns to baseline quickly with antimicrobial therapy and volume expansion. Recurrent pyelonephritis requires investigation to exclude underlying urologic abnormalities. A voiding cystourethrogram (VCUG) should be considered to evaluate for reflux into the transplant allograft. High-grade reflux may warrant reimplantation of the transplanted ureter to the bladder.[231]

Acute Allergic Interstitial Nephritis

In the setting of kidney transplantation, acute allergic interstitial nephritis is a diagnosis of exclusion. Distinguishing acute allergic interstitial nephritis and TCMR is very difficult. In fact, the pathogenesis is somewhat similar in both cases, involving mainly cell-mediated immunity. While fever and rash after ingestion of a new drug favor the former, these clinical features are rarely seen. Mononuclear cell and eosinophil infiltration of the transplanted kidney may occur with either condition, but endothelialitis implicates rejection. Polyomavirus infection must also be considered in the differential diagnosis. Both acute allergic interstitial nephritis and TCMR usually respond to steroids; of course, the suspected drugs must be stopped. SMX-TMP is the drug most commonly implicated in causing allergic interstitial nephritis in kidney transplant patients; other antibiotics including penicillins, cephalosporins, and quinolones can also be implicated.

Early Recurrence of Primary Disease

Several kidney diseases may recur early and cause acute allograft dysfunction. These may be classified into three groups: 1) glomerulonephritides, 2) metabolic diseases such as primary oxalosis, and 3) systemic diseases such as hemolytic-uremic syndrome/thrombotic thrombocytopenia purpura (HUS/TTP). Primary focal segmental glomerulosclerosis (FSGS) is considered in more detail in the following section because of its relatively high frequency of recurrence and its propensity to cause severe graft injury.

Primary Focal Segmental Glomerulosclerosis. The recurrence of primary FSGS is variable. For the sporadic variety it is reported to be about 30%,[232] but recurrence of familial FSGS is rare. Risk factors for recurrence include white ethnicity, younger age, rapidly progressive FSGS in the recipient's native kidneys, and recurrence of disease in a previous allograft. Most cases become manifest (as proteinuria) hours to weeks after transplant. This rapidity of recurrence suggests the presence of a pathogenic circulating plasma factor.[233] Because of the poor prognosis of delayed treatment, patients with primary FSGS should be monitored after transplantation for new-onset proteinuria. Early biopsy is indicated in those who develop proteinuria; this may not show FSGS lesions per se, but the electron microscopic demonstration of diffuse foot process effacement. Treatment options include plasmapheresis or immunoadsorption, high-dose CNIs, ACE-inhibitors, high-dose corticosteroids, cyclophosphamide, adrenocorticotropic hormone gel and rituximab, but controlled studies are lacking.[234–237]

Antiglomerular Basement Membrane Disease. Before transplantation, patients with ESRD due to antiglomerular basement membrane (GBM) disease should generally be on dialysis for at least 6 months and have negative anti-GBM serology.[238] If these criteria are fulfilled, posttransplant recurrence is rare. De novo anti-GBM disease can occur in recipients with Alport syndrome. Here, the recipient with abnormal type IV collagen produces antibodies against the previously "unseen" normal α5 chain NC1 domain in the basement membrane of the transplanted kidney. Patients with allograft dysfunction should be treated with plasmapheresis and cyclophosphamide.[238]

Hemolytic-Uremic Syndrome/Thrombotic Thrombocytopenia Purpura (HUS/TTP). The causes of *de novo* TMA after kidney transplant have been discussed earlier. Recurrence of classic (diarrhea-associated) HUS/TTP is uncommon. However, transplantation should still be deferred until the disease is quiescent for at least 6 months. In contrast, recurrence of atypical (non–diarrhea-associated) HUS/TTP, particularly if inherited, has been reported to be as high as 80%.[239] Certain genetic disorders of complement regulation (such as of Factor H) are associated with high risks of severe recurrence, so it is very useful to define these risks, if possible, before proceeding with transplant.[240] One treatment option in patients with underlying mutations in the genes coding liver-produced proteins such as Factor H or Factor I is to perform dual liver kidney transplant, where the role of liver transplant is to supply functional factor H or I.[241] Eculizumab has been shown to successfully treat and prevent recurrent atypical HUS and post-kidney transplant 'prophylactic' eculizumab represents a viable, if expensive, option for preventing disease recurrence.[140,228]

POSTRENAL DYSFUNCTION IN THE EARLY POSTTRANSPLANT PERIOD

Most urologic complications are secondary to technical factors at the time of transplant and manifest themselves in the early postoperative period, but immunologic factors may play a role in some cases.

Urine Leaks

Leaks may occur at the level of the renal calyx, ureter, or bladder. Causes include infarction of the ureter due to perioperative disruption of its blood supply and breakdown of the ureterovesical anastomosis. Severe obstruction may also result in rupture of the urinary tract with leakage. Clinical features include abdominal pain and swelling; the plasma urea and creatinine levels increase due to resorption of solutes across the peritoneal membrane. If a perirenal drain is being used, however, a urine leak may present with high-volume drainage of fluid. Ultrasound may demonstrate a fluid collection (urinoma); aspiration of fluid from the collection (or from the drain bag) by sterile technique allows comparison of the fluid and plasma creatinine. When the excretory function of the kidney is good, the creatinine concentration in the urinoma greatly exceeds that in the plasma.

In cases in which ultrasound diagnosis is difficult, renal scintigraphy may be useful in demonstrating extravasation of tracer from the urinary system, provided there is adequate kidney function. Rough localization of the site of the leak is sometimes possible by this technique. Antegrade pyelography allows precise diagnosis and localization of proximal urinary leaks. Cystography is the best test to demonstrate a bladder leak.

The clinical features may mimic those of acute rejection. Whenever urine leakage is suspected, a bladder catheter should be immediately inserted to decompress the urinary tract. Selected patients may do well with a bladder catheter or endourologic treatment. Many cases, however, require urgent surgical exploration and repair. The type of repair depends on the level of the leak and the viability of involved tissues.

Urinary Tract Obstruction

Although urinary tract obstruction can occur at any time after transplantation, it is most common in the early postoperative period. Intrinsic causes include poor implantation of the ureter into the bladder, intraluminal blood clots or slough material, and fibrosis of the ureter due to ischemia or rejection. Extrinsic causes include an enlarged prostate in elderly men (causing bladder outlet obstruction) and compression by a lymphocele or other fluid collection. Rarely, calculi cause transplant urinary tract obstruction.

Urinary tract obstruction is often asymptomatic, and should always be considered in the differential diagnosis of allograft dysfunction in the early transplant period. Ultrasound often demonstrates hydronephrosis. However, some dilation of the transplant urinary collecting system is often seen in the early postoperative period, and serial scans showing worsening hydronephrosis may be needed to confirm the diagnosis. Renal scintiscan with diuretic washout is useful in equivocal cases. Percutaneous antegrade pyelography is the best radiologic technique for determining the site of obstruction and can be combined with interventional endourologic techniques. In expert hands, endourologic techniques (e.g., balloon dilation, stenting) may be effective in treating ureteric stenosis and stricture. More complicated cases require open surgical repair. Extrinsic compression requires specific intervention such as draining or fenestration of the lymphocele. Obstruction in the early postoperative period due to an enlarged prostate should be managed with bladder catheter drainage and drugs such as tamsulosin.

LATE POSTTRANSPLANT PERIOD

LATE ACUTE ALLOGRAFT DYSFUNCTION

The causes and evaluation of late acute allograft dysfunction (defined as occurring >6 months posttransplant) include those of early acute dysfunction. Acute prerenal failure and ATN may occur at any time, and the causes are similar to those seen with native kidneys, such as shock syndromes, and may be further exacerbated by the hemodynamic effects of concurrent ACE-I or nonsteroidal antiinflammatory drug (NSAID) use. Urinary tract obstruction must also be considered in the differential diagnosis. In contrast to the early posttransplant period, the causes of obstruction are similar to those associated with native kidney disease (e.g., stones, bladder outlet obstruction, and neoplasia). Ureteric obstruction due to BK virus infection has also been described. Several causes of late acute allograft dysfunction are reviewed in more detail below.

Late Acute Rejection

Acute rejection is less common after the first 6 months. Late acute rejection should alert the physician to prescription of inadequate immunosuppression or patient noncompliance.[242] Withdrawal of steroids or CNIs by the physician may be initiated due to side effects of these medications, but when carried out later in the posttransplant period, may be associated with a high risk of acute rejection;[243,244] therefore, plasma creatinine must be carefully monitored in this setting. ABMR is increasingly recognized as an important cause of late acute allograft dysfunction, especially many years posttransplant; in a recent study, over 50% of 173 subjects biopsied for acute graft dysfunction, a mean of 7 years posttransplant, had evidence of ABMR.[81]

Acute rejection can also occur when CNI levels are subtherapeutic in the setting of newly prescribed medications that are known to decrease CNI levels (Table 70.9). Common

Table 70.9 Agents That May Interact With Transplant Medications

Drugs That Interact With CNIs and Sirolimus		
Class of Drug	Increase Level	Decrease Level
Ca channel blocker	Diltiazem, verapamil	
Antibiotics	Erythromycin, azithromycin, clarithromycin	Nafcillin
Antifungals	Fluconazole, ketoconazole, itraconazole, voriconazole	
Antituberculin		INH, Rifampin, Rifabutin
Antiviral	Ritonavir, nelfinavir, saquinavir	Efvirez, Nevirapine
Anti-seizure		Phenytoin, phenobarbital, carbamazepine, primidone
Antidepressant	Fluoxitine, nefazodone, fluvoxamine	
Foods and Herbal Preparations That Interact With CNIs		
Food	Grapefruit juice/pomegranate juice	
Herbs		St. Johns's wort

agents that decreased CNI levels include antituberculosis medications and anti-seizure medications. CNI levels should be measured more frequently and doseadjustments should be made when prescribing or withdrawing these medications.

Treatment is the same as for early acute rejection (whether TCMR or active ABMR or both) but responses are poorer with greater negative impact on allograft survival than early acute rejection or DGF. Risk factors for noncompliance include adolescence, more immunosuppressant adverse effects, lower socioeconomic status, and psychological stress or illness.[245] Closer monitoring, simplification of the drug regimen, and social worker assistance may aid in the management of patients with high-risk of nonadherence.

Late Acute Calcineurin Inhibitor Nephrotoxicity

Although lower doses of CNIs are generally prescribed after the first 12 months, acute CNI toxicity may occur at any time after transplant. This occurs most often in the setting of taking new medications that impair metabolism of the CNIs (Table 70.9). Patients should be made aware of common medications that interact with CNI. CNI levels should be monitored closely and dose-adjustments should be made when such medications are prescribed.

Transplant Renal Artery Stenosis

Renal Artery Stenosis. Transplant renal artery stenosis (TRAS) is the most common transplant vascular complication and is associated with reduced long-term allograft survival.[246,247] TRAS can arise at any time after transplantation, although the mean time to diagnosis is 0.83±0.81 years posttransplant.[248] The reported incidence varies widely.[247] TRAS may be a consequence of an inadequate arterial anastomosis, inherent vascular disease of the recipient, or preexisting renovascular disease of the donor. Immune-mediated or infection-related damage to the transplant renal artery also plays and important role in some patients; new posttransplant DSAs and CMV infection are both associated with the development of TRAS.[249,250]

Luminal narrowing of more than 70% is probably required to render a stenosis functionally significant. The stenosis may occur in the donor or recipient artery or at the anastomotic site. Stenosis of the recipient iliac artery may also compromise renal arterial flow. Worsening or difficult to control hypertension, an unexplained deterioration in kidney function, or azotemia associated with the introduction of an angiotensin-converting enzyme or angiotensin-receptor blocker should raise suspicion of TRAS.[247,251,252] Clinical examination may also reveal a new vascular bruit over the graft. Ultrasound with Doppler, magnetic resonance, and CT-angiography can support a diagnosis but direct angiogram is usually required for confirmation. CO_2 or minimal contrast angiography can be used to reduce the risk of radiocontrast injury.[253] Good outcomes have been reported with both primary angioplasty and with stent placement in terms of blood pressure control and renal function.[254,255] If the diagnosis is made in the early postoperative period, surgical approach with revision of the anastomosis may be preferred.[247]

Infections Causing Late Acute Allograft Dysfunction

Human Polyomavirus Infection. The polyomaviruses are DNA viruses, the best known of which are BK virus, JC virus, and SV40 virus. BK virus causes a mild self-limiting upper respiratory tract infection in healthy individuals (mostly in childhood). Around 80% of the adult population have serological evidence of prior BK infection. Following primary infection, the virus remains dormant in the urothelium. In immunosuppressed states, the virus may reactivate and replicate. Viral reactivation may result in viral shedding into the urine, viremia, cystitis, ureteritis, or interstitial nephritis (BK nephropathy).

Over the past 25 years, BK virus has been increasingly recognized as an important cause of kidney allograft dysfunction and loss. This probably reflects both improved recognition and reporting of the disease and the increasing use of lymphocyte depleting induction therapies and more potent maintenance MMF- and tacrolimus-based immunosuppression regimens. BK nephropathy is most common in the first 2 years posttransplant. Around 30% to 40% of renal transplant recipients develop BK viruria, around 10% to 20% will become BK viremic, and about half of those BK viremic patients will manifest BK nephropathy on biopsy.[256] Because BK viruria and viremia almost always precede BK nephropathy, urine or plasma BK viral screening offers the opportunity to identify infection early and instigate measures to clear the virus before nephropathy occurs.[257] Many centers advocate biopsy when plasma viral titers reach a predefined threshold (usually $>10^4$ copies/mL), while others, such as our own, treat empirically and reserve biopsy for patients with evidence of new graft dysfunction. Caution is warranted in lowering immunosuppression without an allograft biopsy; underimmunosuppression may precipitate acute rejection especially in the early posttransplant period and patients with high risk for rejection.

Approaches to screening vary and are influenced by local prevalence and economic factors. Many transplant centers now screen all new transplant recipients at intervals over the first two years posttransplant. Protocols for screening include testing the urine by light microscopy for decoy (infected) cells or by polymerase chain reaction (PCR) quantification of urinary or plasma viral load. KDIGO guidelines recommend quantitative plasma PCR testing as follows: 1) monthly for the first 3-6 months after transplantation, 2) then every 3 months until the end of the first posttransplant year, 3) whenever there is an unexplained rise in serum creatinine, and 4) after treatment of acute rejection.[188]

Allograft biopsy is required for the diagnosis of BK interstitial nephritis. Adequate sampling is needed, as the interstitial nephritis can be patchy. The presence of intranuclear tubule cell inclusions by light microscopy should raise suspicion; diagnosis is confirmed by immunohistochemistry using antibodies against BK viral proteins (SV-40). The excellent performance of immunohistochemical stains and concurrent measurement of BK viral titers at the time of 'for-cause' biopsy mean that BK nephropathy is no longer frequently mistaken for and treated as TCMR.

The mainstay of treatment is reduction in immunosuppression.[257] Our usual practice has been to first discontinue the anti-metabolite (usually MMF) in response to significant viremia or biopsy evidence of nephropathy. If this measure fails to result in a favorable viral response, we reduce the dose of CNI by 30% to 50% and continue to monitor viral titers. There are a number of adjunct therapies that may be considered in subjects who either fail to respond to immunosuppression reduction or in whom very aggressive immunosuppression reduction is unattractive because of their high-risk immune risk status. Leflunomide is a tyrosine kinase inhibitor, approved for the treatment of rheumatoid arthritis, that has antiviral effects in vitro. A number of small series have suggested that treatment with leflunomide (usually as a substitute for an antimetabolite) results in enhanced BK viral clearing, although in the absence of clinical trials its efficacy remains contentious.[258,259] The antiviral cidofovir has also, by some reports, been associated with a reduction in BK viral load; prudent dosing is of importance in view of the drug's potential nephrotoxicity.[260] Successful treatment of BK virus with intravenous immunoglobulin has been reported.[261] Switching tacrolimus to cyclosporine or mTORi should also be considered (although this may simply represent a de facto reduction in immunosuppression).[107,262] Finally, hopes for fluoroquinolones as a therapy for BK virus were dashed by two randomized controlled trials that showed levofloxacin to be no better than placebo at 1) treating or 2) preventing BK viremia/viruria in kidney transplant recipients.[263,264]

Hepatitis C. Around 1% of the U.S. population is hepatitis C (HCV) positive. Left untreated, 10% to 20% of persons infected with HCV will develop end-stage liver disease (ESLD) over a period of 20 to 30 years. In the absence of advanced liver disease, the presence of hepatitis C is not considered a contraindication to kidney transplant. However, historically post-kidney transplant outcomes in HCV+ recipients have been significantly worse than those of HCV- recipients.[265] Prior to the emergence of the new direct-acting antiviral regimens, the options for treatment of HCV in patients with ESRD, either before or after transplant, were limited by relative contraindications to interferon-α therapy in kidney transplant recipients and to ribavirin treatment in patients on dialysis.[266–268] However, HCV treatment has undergone a revolution in the past five years.[269] The infection is now reliably curable in kidney transplant recipients with combination antiviral therapy, and we are optimistic that these new HCV therapies will translate into improved post-kidney transplant survival for HCV+ recipients.[270,271] Indeed, such is the confidence in the new HCV therapies that expansion of the deceased donor transplant pool by allowing HCV+ donor kidneys be transplanted into HCV- recipients is being investigated. A recent small study reported on 10 HCV- patients who received kidney transplants from HCV RNA positive donors. All recipients received HCV treatment, starting immediately before transplant and continuing for 12 weeks after transplant and had undetectable HCV RNA at 12 weeks.[272]

Drug and Radiocontrast Nephrotoxicity

Drugs that cause nephrotoxicity in the native kidney will also adversely affect the kidney allografts. Drug-related nephrotoxic effects that are more common in the setting of transplantation are listed in Table 70.9. Special attention should be paid to drugs that interact with CNIs. CNIs are metabolized by the cytochrome P-450 isoenzyme CYP3A5, and drugs that interact with CYP3A5 will affect its plasma level. When prescribing diltiazem, verapamil, ketoconazole, and the macrolide antibiotics, particularly erythromycin and clarithromycin, the dose of CNI should be reduced and levels should be followed. Conversely, rifampicin, phenobarbital, and phenytoin lower the CNI level, so the CNI should be increased.

Drugs with known nephrotoxic effects such as aminoglycosides, amphotericin, and NSAIDs probably have enhanced toxicity when used concomitantly with a CNI. Nevertheless, they are sometimes required in transplant recipients. Use of the liposomal preparation of amphotericin is preferable because it is less nephrotoxic than the standard preparation (amphotericin B).

Plasma creatinine can be increased with high-dose SMX-TMP by inhibiting tubule secretion of creatinine (GFR per se is not compromised and plasma creatinine decreases within 5 days of stopping SMX-TMP). Rarely, SMX-TMP can provoke allergic interstitial nephritis.

Not surprisingly, ACE-inhibitors or ARBs have been implicated in precipitating AKI in the presence of transplant renal artery stenosis. Overall, if carefully prescribed, these agents are well tolerated. The use of ACE-inhibitors or ARBs in the immediate posttransplant period, when volume status and CNI dosages are fluctuating, is not recommended.

The risk of developing AKI after administration of radiocontrast to kidney transplant recipients has not been well defined. Single-center studies point toward a higher prevalence in kidney transplant recipients.[273] Presumably, risk factors for contrast nephrotoxicity are similar to those in patients who have not undergone transplant surgery. Thus, the same preventive measures should be used.

Late Allograft Dysfunction and Late Allograft Loss

Preventing late allograft loss remains a major challenge. Death with a functioning graft accounts for approximately half of graft losses. Important causes of death-censored late graft loss include acute rejection, recurrent or de novo glomerular diseases, AKI related to sepsis or hypotension, and pyelonephritis.[274] The majority of the remaining late graft failures, with features of interstitial fibrosis and tubular atrophy (IF/TA) on histology, were conventionally attributed to an amalgam of immune and non-immune (CNI)-mediated fibrosis and vascular injury termed chronic/sclerosing allograft nephropathy. More recently improved pathologic and immunologic diagnostic tools (C4d stain and DSA) mean that underlying chronic ABMR is increasingly recognized in biopsies with IF/TA. Any biopsy with IF/TA may be further supplemented to include features suggestive of non-immune causes of IF/TA such as CNI toxicity, chronic hypertension, obstruction, pyelonephritis, viral infection, and recurrent or de novo glomerular disease (Table 70.8).

Fig. 70.6 illustrates factors that are thought to contribute to the development of late allograft failure, often referred to under the umbrella term 'chronic allograft injury'.

The causes of late dysfunction are summarized in Table 70.10 and further discussed in the following sections. Certain causes such as transplant renal artery stenosis and urinary tract obstruction have been discussed earlier.

Chronic ABMR. Several studies have shown that an antibody-mediated process is commonly implicated in late graft injury. In one study, 99 of 173 patients that underwent 'for-cause' biopsy, a mean of seven years posttransplant, had evidence of an ABMR, indicated by C4d positivity, DSA or both[81]. Another study that evaluated all death censored graft losses in 1317 kidney transplant recipients (KTRs), irrespective of time since

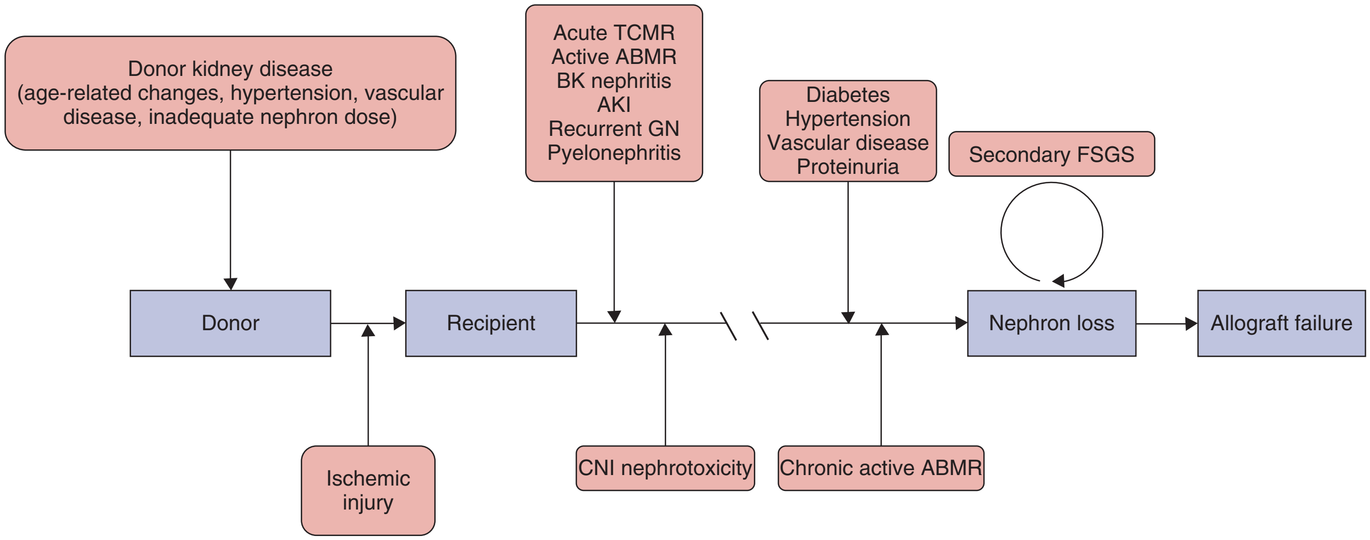

Fig. 70.6 The multifactorial pathogenesis of chronic allograft injury.

Table 70.10 Causes of Late Chronic Allograft Dysfunction

Prerenal
Transplant renal artery stenosis
Intrarenal
Acute rejection – TCMR or ABMR Chronic active ABMR Chronic active TCMR CNI nephrotoxicity BK nephropathy Recurrence of primary disease De novo primary kidney disease
Postrenal
Urinary tract obstruction

CNI, Calcineurin inhibitor.

transplant, found evidence for chronic ABMR in 18% (with histologic evidence of either transplant glomerulopathy or IF/TA). The probability of ABMR on 'for-cause' biopsies rises from around 10% at six months to 35% at five years posttransplant, is associated with medication nonadherence, and augurs poor graft survival.[275]

The Banff 2005 update added the term 'chronic active antibody-mediated rejection', currently requiring all three of the following 1) morphologic evidence of chronic tissue injury evidenced by transplant glomerulopathy, peritubular capillary basement membrane multilayering (on EM), and/or arterial intimal fibrosis of new onset; 2) evidence of recent endothelial-antibody interaction (peritubular capillary C4d staining); and 3) evidence of DSA.

Chronic active ABMR, most commonly accompanied by transplant glomerulopathy on histology, is clinically characterized by a slow decline in kidney function, hypertension, and (often heavy) proteinuria. Treatment options are limited. A single center study of 23 consecutive kidney transplant recipients with ABMR diagnosed at least 6 months posttransplant, nearly half of whom had some evidence of transplant glomerulopathy on biopsy, observed minimal benefit in response to therapies that included plasmapheresis, IVIg, rituximab and bortezomib.[276] A recent randomized controlled trial showed no benefit in terms of change in GFR or proteinuria for bortezomib treatment versus placebo in patients with chronic ABMR.[277] Another small randomized controlled trial showed no benefit for rituximab combined with intravenous immunoglobulin versus placebo for the treatment of chronic ABMR.[278] A recent small case series reported excellent allograft outcomes in patients with chronic ABMR treated with tocilizumab, suggesting that more rigorous studies of the IL-6 receptor antibody in this population are warranted.[279]

Our approach to patients with a diagnosis of chronic active ABMR is to optimize maintenance immunosuppression. We switch to a tacrolimus-MMF regimen if the patient has been taking CsA or AZA. We target tacrolimus levels around 8 ng/mL, maximize MMF dosing, and add low dose maintenance prednisone if the patient is not already on steroid.[280] We also add ACE-inhibitors or ARBs, control hypertension, and modify other cardiovascular risk factors. The lack of good treatment options for chronic ABMR underscores the importance of preventing de novo DSA formation and minimizing, to the greatest degree possible, the use of donors to whom the recipient has preformed DSA.

Calcineurin Inhibitor Toxicity. Over thirty years after CNI-associated nephropathy was first described in heart transplant recipients there remains debate about the importance of CNI in late graft failure.[67] Nankivell et al. reported longitudinal histologic data on 120 diabetic kidney transplant recipients [119 were simultaneous pancreas kidney (SPK) recipients] who underwent sequential kidney transplant biopsies from the time of transplant up to 10 years posttransplant. The study identified histologic evidence of CNI toxicity defined as the presence of 'striped cortical fibrosis or new-onset arteriolar hyalinosis (not from renal ischemia or preexisting hyalinosis in the allograft) supported by tubular microcalcification (without preceding acute tubular necrosis)' in over 50% of biopsies at five years and 100% of biopsies at 10 years posttransplant. Based on the pathological findings, the authors attributed the majority of late graft dysfunction to CNI toxicity.[65] In contrast, newer studies have suggested a much lower prevalence of interstitial fibrosis at five years post-biopsy and question the specificity of arteriolar hyalinosis for the diagnosis of CNI toxicity.[274,281] This, combined with increased evidence for immune-mediated injury in many cases of late allograft failure, has led many to deemphasize the importance of chronic CNI toxicity.

We remain believers in CNI toxicity as a cause of graft dysfunction. Indeed, the significant difference in GFR between belatacept- and cyclosporine-treated patients five years post-transplant in the BENEFIT trial lends credence to the nephrotoxic effects of CNIs and emphasizes the potential merits of developing CNI avoidance strategies.[282] If the clinical and histologic picture suggests a significant component of chronic CNI nephrotoxicity without evidence of rejection, we dose-reduce the CNI. Substituting CNI with either sirolimus or belatacept should also be given strong consideration.[102,112] mTORi treatment should probably be avoided in those with baseline proteinuria or GFR of less than 40 mL/min.[283]

RECURRENCE OF PRIMARY DISEASE

Diseases that recur early in the posttransplant period are addressed above. The incidence of late recurrence is difficult to estimate: the original cause of ESRD is often unknown, transplant kidney biopsies are not always performed, and most relevant studies are small and retrospective with variable follow-up periods. An Australian study of patients with biopsy-proven glomerulonephritis found a 10-year incidence of graft loss from recurrence of 8.4%.[284] However, patients whose primary kidney disease was biopsy-proven glomerulonephritis had allograft survival that was comparable to patients with non-glomerulonephritis etiologies. As kidney allograft survival improves, recurrent or de novo disease is being increasingly diagnosed and is recognized as an important cause of late graft loss.[274] Recurrence can present as decreasing GFR, proteinuria, or hematuria.

IgA Glomerulonephritis

Studies with longer follow-up times have shown that histologic recurrence of this condition is common. The reported incidence varies from 13% to 53%,[285] and likely reflects the varying threshold for biopsies among different centers. The estimated 10-year incidence of graft loss due to recurrence was 9.7%.[284] The risk of recurrence is higher if a previous graft was lost to recurrent disease.[286] In patients with proteinuria, ACE-inhibitors and ARBs should be initiated.[287] One large retrospective study showed an association between maintenance steroid use and a reduction in posttransplant IgA recurrence.[128] This should be taken into account when discussing the plan for maintenance immunosuppression with the patient.

Lupus Nephritis

Allograft and patient survival overall have been thought to be similar in patients with ESRD due to lupus nephritis compared with those with ESRD from other causes,[288] but analysis of the United States Renal Data System (USRDS) database showed worse outcomes of deceased-donor transplant recipients with ESRD due to lupus nephritis.[289] Recurrence of severe systemic lupus erythematosus (SLE), either systemically or within in the graft, is uncommon. Low rates of recurrence in SLE probably reflect patient selection, disease activity "burning out" on maintenance dialysis, and the effects of powerful posttransplant immunosuppression. Indeed, a recent analysis of nearly 7000 kidney transplant recipients with ESRD from SLE suggested that acute rejection (affecting 26%) had a far more important impact on graft failure than recurrent disease (affecting 3%).[126] As with other glomerular diseases that may recur, transplantation should be deferred until SLE is clinically quiescent. Many centers prefer a 6 to 12 month period of clinical quiescence before proceeding with transplantation to reduce the risk of recurrence. If the patient is receiving anticoagulation for antiphospholipid syndrome (APS) before transplantation, anticoagulation should be resumed as soon as safely possible (initially with intravenous heparin) after transplant surgery. This procedure is used to reduce the risk of thrombosis of the allograft or other sites.

Granulomatosis With Polyangiitis and Microscopic Polyangiitis

Renal and extrarenal recurrences of these diseases have been described. In a pooled series of 127 patients, antineutrophil cytoplasmic antibody (ANCA)–associated small vessel vasculitis recurred in 17% of cases; renal involvement recurred in 10% of cases.[290] A lower incidence (7%) of recurrence has been reported in a more recent study.[291] Positive ANCA serology at the time of transplant does not appear to predict relapse. However, patients with ESRD secondary to ANCA-associated vasculitis should not undergo transplant surgery until the disease is clinically quiescent. Treatment options for recurrence include cyclophosphamide and rituximab.[292]

Membranoproliferative Glomerulonephritis (MPGN)

MPGN has recently been reclassified based on the deposition of C3 alone (the C3 nephritides) versus C3 and immunoglobulin on immunofluorescence.[293] The C3 nephritides (dense deposit disease and C3 nephritis) are driven by an underlying defect in the regulation of the alternative complement pathway. The recent classification change means that there are limited data on the recurrence rate of the C3 nephritides; however, five-year graft survival in 75 patients with ESRD from MPGN type II (now termed 'dense deposit disease') was reported to be significantly worse than other pediatric kidney transplant recipients and recurrence was documented in 12 of the 18 patients biopsied.[294] Limited success has been reported in treating posttransplant recurrence of the C3 nephritides with the complement inhibitor eculizumab.[295,296] Membranoproliferative glomerulonephritis associated with glomerular deposition of C3 and immunoglobulin should trigger the evaluation and treatment of underlying causes such as infection (especially hepatitis C), autoimmune diseases, and plasma-cell dyscrasias.

Membranous Nephropathy

Membranous nephropathy may recur after transplant or arise de novo.[297] The associated clinical features vary from minimal proteinuria to nephrotic syndrome. As with native kidney disease, hepatitis B virus (HBV) infection and other conditions associated with membranous nephropathy such as malignancy should be excluded. A high proportion of patients with primary membranous nephropathy are positive for the anti-PLA2R1 antibody and monitoring antibody levels may be useful in tracking disease activity.[298–300] Management includes treatment of the underlying cause and RAAS inhibition. The recurrence of primary membranous nephropathy has been successfully treated with rituximab, and some investigators argue that treatment with rituximab (either 375 mg/m^2 x 4 doses or 1 g x 2 doses) should be offered to those patients with recurrent posttransplant membranous nephropathy whose proteinuria remains over 1 g/24 hours despite maximum tolerated ACEI/ARB therapy.[54,301]

Comparative Transplant Outcomes in Patients With ESRD From Glomerulonephritides

A recent study compared the posttransplant outcomes of U.S. patients with ESRD from different GN types. Using those patients with ESRD from IgA nephropathy as the referent group the adjusted hazard ratios for the 1) death and 2) allograft failure excluding death with a functioning graft, respectively, were 1.57 and 1.20 for FSGS, 1.52 and 1.27 for membranous nephropathy, 1.76 and 1.50, membranoproliferative GN, 1.82 and 1.11 for lupus nephritis, and 1.56 and 0.94 (not statistically significant) for vasculitis. [302]

Diabetic Nephropathy

Recurrence of diabetic nephropathy in the allograft has not been well studied. This reflects the relatively poor long-term survival of diabetic kidney transplant recipients; the duration of exposure to the diabetic milieu is often insufficient to allow development of severe diabetic nephropathy. PTDM can also result in diabetic nephropathy;[303] histologic evidence of this may occur surprisingly rapidly after transplant.[304]

ASSESSING OUTCOMES IN KIDNEY TRANSPLANTATION

The most convenient and widely used method for assessing outcomes after kidney transplantation is measurement of allograft survival. Other important measures include allograft function (typically measured by plasma creatinine), patient survival, number of rejection episodes, days of hospitalization, and quality-of-life indices. Registry data from the USRDS, the Scientific Registry of Transplant Recipients, the Collaborative Transplant Study (CTS), and Australia and New Zealand Dialysis and Transplant Registry (ANZDATA) have all proved extremely useful in assessing these outcomes.

ACTUAL AND ACTUARIAL ALLOGRAFT AND PATIENT SURVIVAL

Allograft survival is calculated from the day of transplantation to the day of reaching a defined end point (e.g., return to dialysis, re-transplantation, or death). The most widely accepted measure of outcome is the Kaplan-Meier probability estimate of patient and graft survival. One year, 5-year, and 10-year actuarial survival rates are frequently presented, but actual survival may ultimately not be as impressive as projected survival.[305] Another actuarial measure commonly used is allograft half-life (median allograft survival).

Traditionally, allograft survival is assessed under two distinct time phases: early and late. Early allograft loss refers to loss within the first 12 months, and late loss to any time thereafter. This distinction is empiric but makes clinical sense. In the first 12 months, allograft loss is relatively common, because of technical complications and rejection. After 12 months, the incidence is lower and generally stable over time. Usually, analysis of long-term survival is restricted to those allografts that have survived to 12 months after transplant. The causes of late allograft loss are also different and are discussed later. Patient death is in essence equivalent to allograft loss, but allograft survival is also sometimes calculated after censoring for patient death ("allograft failure excluding death with a functioning graft").

SURVIVAL BENEFITS OF KIDNEY TRANSPLANTATION

Comparison of survival between the general dialysis population and transplanted patients is greatly affected by selection bias, as only relatively healthy patients are referred (and listed) for transplantation. Thus, comparisons among patients on the waiting list who do or do not receive a transplant are usually performed instead. Of course, such analyses assume that the two groups (those who have undergone transplant surgery or those still on the list) can otherwise be matched; this is not necessarily true.

One USRDS study found that during the first 106 days after transplantation, the risk of death after transplantation was higher than the corresponding risk while remaining on the waiting list (on dialysis). This mainly reflected the risks associated with the transplant procedure itself. Thereafter, however, transplantation conferred a survival benefit. On the basis of 3 to 4 years of follow-up, transplantation reduced the risk of death overall by 68%.[306]

SHORT-TERM OUTCOMES IN KIDNEY TRANSPLANTATION

The acute rejection rate has fallen substantially over the past 25 years. The rate of acute rejection in the first year posttransplant is currently around 10%. Fig. 70.7 illustrates the decline in rate of acute rejection in the first posttransplant year between the years 1992 and 2010.[123] Short-term graft survival has improved. One-year graft survival in 2014 was 93% for deceased donor transplants and 97% for living donor transplants. Tables 70.11 and 70.12 illustrate one-year graft survival by donor type between 1998 and 2014.[13]

LONG-TERM OUTCOMES IN KIDNEY TRANSPLANTATION

There has also been an improvement in long-term allograft survival (Tables 70.11 and 70.12). This improvement is seen most prominently in the recipients of living kidney donor transplants.[13]

Beyond the first posttransplant year, the principal causes of kidney allograft loss are death with a functioning graft and chronic allograft injury (CAI); less common causes are late acute rejection and recurrent disease.[274] The primary cause of death remains cardiovascular disease, followed by infection and malignancy (Fig. 70.8).[123] In children, death is a much less common cause of allograft loss; conversely in the elderly, death is more common.

FACTORS AFFECTING KIDNEY ALLOGRAFT SURVIVAL

Prospective studies and analyses of registry data have shown that many variables influence kidney allograft survival. These can be considered as donor, recipient, or donor-recipient factors. Many of them contribute to the development of chronic allograft injury and have been discussed above.

DONOR-RECIPIENT FACTORS

Delayed Graft Function

DGF is associated with poorer allograft and patient survival and poorer allograft function.[307] Registry data show that DGF reduces allograft half-lives by 30%, a larger effect than early acute rejection.[308] Even though cold ischemia time has been steadily decreased from 24 hours to 18 hours over the past 20 years, the incidence of DGF in deceased donor transplants has remained at approximately 25%.[4]

Human Leukocyte Antigen Matching

The goal of HLA matching is to reduce the number of 'foreign targets' against which a recipient may develop an immune

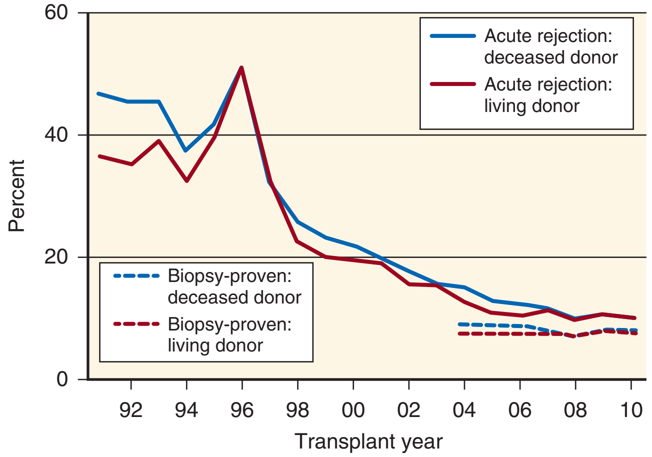

Fig. 70.7 Secular trends in acute rejection in the first year posttransplant.

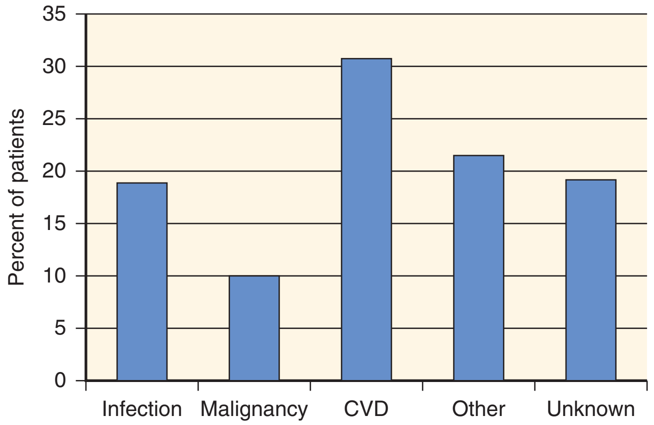

Fig. 70.8 Causes of death with allograft function (2007-2011) taken from the United States Renal Data System 2013 Annual Data Report. *CVD,* Cardiovascular disease.

Table 70.11 Secular Trends in 1-, 5-, and 10-Year Mortality and Graft Failure for Adult Deceased Donor Transplant Recipients in the United States

	1 Year Posttransplant			5 Years Posttransplant			10 Years Posttransplant		
Year	Probability of All-Cause Graft Failure (%)	Probability of Return to Dialysis or Repeat Transplant (%)	Probability of Death (%)	Probability of All-Cause Graft Failure (%)	Probability of Return to Dialysis or Repeat Transplant (%)	Probability of Death (%)	Probability of All-Cause Graft Failure (%)	Probability of Return to Dialysis or Repeat Transplant (%)	Probability of Death (%)
1998	12.6	8.9	5.5	33.8	24.1	18.2	56.7	40.6	37.9
1999	13.2	8.8	5.9	33.6	23.0	18.8	56.3	39.3	38.1
2000	12.7	8.1	6.4	33.9	22.7	19.6	56.3	38.3	38.9
2001	12.2	8.0	5.7	33.1	21.3	19.7	55.3	36.7	38.5
2002	12.3	8.3	5.6	32.8	22.1	18.8	53.5	35.9	37.0
2003	11.8	7.3	5.6	31.7	20.3	18.4	54.4	35.7	37.6
2004	11.1	7.1	5.4	31.3	20.5	18.2	53.2	35.4	36.7
2005	11.2	6.9	6.0	29.9	19.0	17.8	52.4	33.4	36.5
2006	10.4	6.6	5.1	29.3	18.6	17.1			
2007	9.5	5.9	4.6	28.2	17.7	16.8			
2008	9.4	6.0	4.5	26.8	16.1	16.3			
2009	9.3	5.5	4.9	26.9	16.4	16.2			
2010	8.8	5.4	4.4	26.6	16.0	16.5			
2011	7.4	4.4	3.9						
2012	7.8	4.7	3.8						
2013	7.7	4.7	3.5						
2014	6.9	3.8	3.7						

Table 70.12 Secular Trends in 1-, 5-, and 10-Year Mortality and Graft Failure for Adult Living Donor Transplant Recipients in the United States

	1 Year Posttransplant			5 Years Posttransplant			10 Years Posttransplant		
Year	Probability of All-Cause Graft Failure (%)	Probability of Return to Dialysis or Repeat Transplant (%)	Probability of Death (%)	Probability of All-Cause Graft Failure (%)	Probability of Return to Dialysis or Repeat Transplant (%)	Probability of Death (%)	Probability of All-Cause Graft Failure (%)	Probability of Return to Dialysis or Repeat Transplant (%)	Probability of Death (%)
1998	6.5	4.8	2.3	21.3	15.0	10.1	42.4	30.8	23.2
1999	6.3	4.6	2.1	21.0	14.9	9.4	41.0	28.9	22.4
2000	7.0	5.0	2.6	22.3	15.2	10.6	42.1	29.2	23.7
2001	6.7	4.6	2.5	21.7	14.8	10.2	41.4	28.1	23.7
2002	6.3	4.4	2.4	20.8	14.1	10.2	39.9	26.4	24.3
2003	5.5	4.0	1.8	20.1	13.8	9.4	39.3	26.0	23.0
2004	5.2	3.6	2.1	18.8	12.7	8.8	38.3	24.6	22.4
2005	5.4	3.7	2.0	18.7	12.7	8.8	38.4	25.1	22.2
2006	4.5	3.1	1.7	16.8	11.2	8.0			
2007	3.8	2.5	1.4	16.7	10.5	7.9			
2008	4.3	2.9	1.6	15.4	10.1	7.4			
2009	4.1	2.8	1.3	15.2	9.4	7.6			
2010	3.7	2.4	1.4	15.3	9.6	7.3			
2011	3.5	2.0	1.8						
2012	3.5	2.1	1.5						
2013	2.6	1.5	1.2						
2014	3.0	1.9	1.4						

response. While modern era immunosuppression can undoubtedly ‘overpower’ HLA mismatching, good HLA matching nonetheless remains desirable and beneficial in terms of graft survival. In a recent study of nearly 200,000 deceased donor kidney transplant recipients between 1987 and 2013, each additional HLA donor-recipient mismatch (out of 6 HLA A, B and DR loci) was associated with a reduction in allograft survival. A similar, but somewhat diminished, association was observed between HLA mismatch and allograft in an analysis restricted to the more modern transplant era (2009-2013).[309,310] For living donor recipients the receipt of a two-haplotype matched kidney still confers a significant survival advantage, haplo-identical graft half-lives being over 25-years.[311] The advent of HLA epitope-matching may prove important in the coming years.

Cytomegalovirus Status of Donor and Recipient

Registry data show a small but definite association of donor and recipient CMV serologic status with kidney allograft and recipient survival (hazard ratio 1.1).[312] Donor-negative–recipient-negative pairings have the best outcomes, whereas donor-positive–recipient-negative pairings have the worst. Cytomegalovirus probably affects graft outcomes through overt infection, but subclinical effects on immune function may also be important.

Timing of Transplantation

There is evidence that preemptive (before initiation of dialysis) transplantation is associated with a lower risk of acute rejection and allograft failure.[313] Other retrospective studies have shown that longer time on dialysis is independently associated with poorer graft and patient survival.[314,315] Between 2003 and 2012, 17% of kidney transplants performed in the United States were preemptive.[316] Minimizing time on dialysis has many potential benefits; this strategy should thus be pursued whenever possible.[4,317]

Center Effect

Not surprisingly, reported outcomes have varied from transplant center to center.[318] This reflects normal statistical variance as well as center expertise. Outcomes are confounded by many donor and recipient factors that differ across centers.[319] USRDS data suggest minimal difference in outcomes among small and large transplant centers in the United States.[320]

DONOR FACTORS

The quality of the kidney immediately before transplantation has a major impact on long-term graft function and the risk of developing chronic allograft injury.

Donor Source: Deceased Versus Living Donor

The donor source is one of the most important predictors of short- and long-term allograft outcomes. In general, living donor allografts are superior to deceased donor allografts (see Tables 70.11 and 70.12). This benefit applies across all degrees of HLA mismatching. The better outcomes reflect several factors: very healthy living donors, the absence of brain death, the benefits of elective as opposed to semi-urgent surgery, minimization of ischemia-reperfusion injury, higher nephron mass and the effects of a shorter waiting time on, or complete avoidance of, dialysis. This further emphasizes the importance of the “healthy transplant kidney” effect.

Allograft outcomes are superior from deceased donors with trauma as opposed to non-trauma being the cause of death.[321]

Donor Age

Older kidney donor age is associated with reduced kidney transplant survival. A donor age effect is apparent among deceased and living donor transplants.[4,322,323] These results are thought to reflect a higher incidence of DGF and of nephron "underdosing." Allografts from older donors have fewer functioning nephrons because of the aging process[324] and donor-related conditions such as hypertension and atherosclerosis. However, because of the organ donor shortage, older deceased donor kidneys are being increasingly utilized. Donor age younger than 5 years is also associated with poorer outcomes, reflecting higher rates of technical complications and probably nephron underdosing (see later). En bloc transplantation (two kidneys) from donors aged 0 to 5 years significantly improves survival, however.

Donor Sex

There is evidence that allografts from female donors have slightly poorer survival.[325] Again, this probably reflects a nephron underdosing effect (see later), because women have a smaller kidney mass than men. However, female recipients of male kidneys may have poorer graft survival that is related an immune response to antigens encoded by the Y-chromosome (H-Y antigens).[326–328]

Donor Race/Ethnicity

Deceased donor kidneys from African-Americans are associated with reduced allograft survival compared to those donated by non–African-Americans.[329] There is increasing evidence that much of the increased risk of graft loss that is associated with African-American donated kidneys is attributable to the presence of the APOL1 high-risk variant (present in around 15% of African-American donors). African-American donated kidneys from donors lacking the APOL1 high-risk variant have outcomes comparable to non-African-American donated kidneys. Integration of APOL1 genotyping (instead of race) into clinical prediction models such as the KDPI is warranted.[330,331]

Donor Nephron Mass

An imbalance between the metabolic/excretory demands of the recipient and the functional transplant mass has been postulated to play a role in the development and progression of chronic allograft injury (Fig. 70.6). Nephron underdosing, exacerbated by perioperative ischemic damage and postoperative nephrotoxic drugs, might lead to nephron overwork and eventual failure, similar to the mechanisms occurring in native progressive kidney disease. Thus, kidneys from small donors transplanted into recipients of large body surface area or large body mass index would be at highest risk of this problem. There is support for this hypothesis from animal[332] and retrospective human studies.[333–335]

Cold Ischemia Time

Prolonged cold ischemia time is associated with higher risk of DGF and poorer allograft survival.[175,336] Registry data suggest that more than 24 hours is particularly deleterious to the graft.

Expanded Criteria Donors and Kidney Donor Profile Index

As the discrepancy between the numbers of patients awaiting kidney transplant and available organs increases, many countries are now using ECD allografts. ECD kidneys are defined by donor characteristics that are associated with a 70% greater risk of allograft failure when compared to a reference group of non-hypertensive donors of age 10 to 39 years, whose cause of death was not cerebrovascular accident (CVA) and whose terminal creatinine is less than 1.5 mg/dL.[337] This includes donors who are 60 years or older and donors aged 50 to 59 with two of the following criteria: 1) CVA as cause of death, 2) history of hypertension, or 3) terminal creatinine > 1.5 mg/dL. Survival of ECD kidneys is, on average, shorter for two general reasons: first, the baseline GFR of these kidneys is likely lower, and second, ECD kidneys tend to be transplanted into older recipients, who have higher rates of posttransplant death. Older patients have higher risk of mortality while awaiting transplantation, and have been shown to benefit from ECD kidney transplantation.[338]

In the US, the new Kidney Allocation System has replaced the standard and expanded criteria deceased donor categories with a single pool of kidneys graded using the kidney donor profile index (KDPI). The KDPI score is calculated using 10 donor characteristics and is a modified version of the predictive tool first described by Rao et al.[329] The KDPI is expressed as a percentile score with 0% and 100% signifying excellent quality and marginal organs, respectively. In order to maximize the utility of the deceased donor organ supply, deceased donor kidneys with a KDPI < 20% are to be allocated to candidates with the highest posttransplant life expectancy, as judged by the 4-variable estimated posttransplant survival (EPTS) score. Older candidates, for whom long waiting times represent a barrier to transplantation, who would previously have agreed to receipt of an ECD kidney, may now choose to accept deceased donor kidneys with a high KDPI value (>85%).

Donation After Cardiac Death

There has been significant increase in the use of donation after cardiac death (DCD) kidneys.[339] DCD donors can be sub-classified as 'uncontrolled' or 'controlled'. Uncontrolled donors are either unsuccessfully resuscitated or present dead on arrival to hospital, while controlled donors suffer a cardiac arrest following the withdrawal of life support in the intensive care unit or operating room immediately prior to donation. The duration of warm ischemia time is likely to be significantly greater in the setting of uncontrolled donation.[340] Protocols for managing DCD kidneys vary from center to center. In uncontrolled donation, isolated perfusion of the kidneys with cold preservation solution can be achieved using double-balloon aortic catheterization (with balloons inflated in the aorta above and below the renal arteries) to minimize warm ischemia time. Short-term outcomes (such as rates of DGF and primary nonfunction) are inferior to those seen with brain dead donors. However, long-term outcomes of DCD organs (from donors <50 years old) are similar to those from standard deceased donors.[341]

RECIPIENT FACTORS

Recipient Age

In general, allograft survival rates are poorer in those at the extremes of age, that is, younger than 18 or older than 65 years of age.[4,342] In the very young, technical causes of graft loss such as vessel thrombosis are relatively more common. Acute rejection is also a more common cause of allograft loss; conversely, death with a functioning graft is relatively rare. Death with a functioning allograft is a much more common cause of graft loss in the elderly (responsible for more than 50% of graft failures). Conversely, acute rejection may be less common. Thus, although randomized controlled trials are not available to definitively inform practice, it seems reasonable, in general, to use less aggressive immunosuppression in the elderly.[343]

Recipient Race/Ethnicity

African-American recipients have poorer allograft survival compared with that of whites.[344] This probably reflects multiple factors including higher incidence of DGF, higher incidence of acute and late acute rejection, stronger immune responsiveness, a predominantly white donor pool (with resultant poorer matching of HLA and non-HLA antigens), altered pharmacokinetics of immunosuppressive drugs, and a higher prevalence of hypertension. In the United States, socioeconomic factors may also play a role.[345] Indeed, there is some evidence that African-Europeans have equivalent outcomes to whites in Europe.[346] Asian and Latino recipients have superior outcomes to whites; the reasons for this are unknown.[344] Strategies that may improve outcomes in African-American recipients include diligent use of higher doses of immunosuppression and the identification of barriers that may limit equitable access to healthcare.[347] Increasing living donation from African-American donors is also desirable. Pre-donation screening for the APOL1 high-risk genotype in donors of African descent may improve African-American living donor safety and recipient outcomes.[330] Further good data will hopefully come from the APOLLO Network research consortium, which has been recently established with a goal of prospectively identifying, genotyping, and following up African-American donors and recipients.[348]

Recipient Sex

Registry studies of the association of recipient sex with transplant outcomes have yielded differing results. In the CTS database, female recipients had slightly better allograft survival than male recipients of deceased donor kidneys or HLA-identical living donor kidneys.[325] Data from US transplant centers has shown better allograft survival in male as opposed to female recipients of living donor kidneys.[349] An important difference between female and male transplant candidates is the higher degree of sensitization of the former to HLA antigens, as well as to non-HLA antigens. Women tend to be more sensitized because of pregnancy and possibly because of more blood transfusions because of anemia related to menstruation. An immune response to H-Y antigens by female recipients may play a role, although the generally larger nephron dose in male donors may be a confounding factor in registry analyses.

Recipient Sensitization

Patients who are highly sensitized are generally considered at higher risk for adverse early and late graft outcomes compared with unsensitized recipients. We use pretransplant cPRA to help guide immunosuppression planning, although preformed DSA, rather than a high cPRA per se, probably has greater prognostic significance for graft outcome.[125] The principal reasons for sensitization are previous transplants, pregnancy, and blood transfusion. Indeed, allograft survival is poorer in recipients of subsequent transplants compared with recipients of a first transplant.[350] Highly sensitized patients have longer wait-times until transplantation and are usually given more intensive immunosuppression. Recent improvements made to the allocation system in the United States and the availability of desensitization protocols have offered such patients better access to transplantation.

Acute Rejection

Acute rejection remains a significant risk factor for allograft loss. Even when acute rejection is successfully treated, some irreversible graft injury/scarring likely ensues. Acute rejection refractory to steroids, acute rejection with a humoral component, and late acute rejection have particularly negative impacts on allograft and patient outcomes.[305]

Recipient Immunosuppression

Undoubtedly, improvements in the acute rejection rate and in allograft survival reflect the effectiveness of modern antirejection drugs such as the CNIs and MMF. The contribution of long-term CNI nephrotoxicity, particularly with currently used maintenance doses, to chronic kidney allograft dysfunction and loss remains controversial (see Late Allograft Dysfunction and Late Allograft Loss). For now, CNIs remain the cornerstone of immunosuppression.[351] Registry data show that the most immunosuppression regimens used at transplant centers in the United States include tacrolimus and MMF.

Recipient Compliance

Poor compliance with the immunosuppressive regimen markedly increases the risk of acute rejection (particularly late acute rejection) and allograft loss.[352] Allograft loss has been reported to be seven-fold higher in nonadherent patients.[353] Efforts are being made to improve strategies to prevent nonadherence.[354] Noncompliance is a particularly difficult problem in the pediatric adolescent transplant population.[352,355]

Recipient Body Size

Morbid (grade 2 or higher) obesity, corresponding to a Quételet's (body mass) index (BMI) of 35 kg/m^2 or greater, is associated with more transplant surgery–related complications, more DGF, and poorer allograft survival.[356,357] Indeed, even grade 1 obesity (BMI of 30 to 34.9 kg/m^2) is a risk factor for allograft failure. Poorer long-term graft survival probably reflects the effects of DGF, nephron overwork, and more difficult dosing of immunosuppressive drugs. Nevertheless, study of patients with BMI greater than 30 kg/m^2 suggest that transplantation provides a survival benefit over remaining on the waiting list (on dialysis), at least up to a BMI of 41 kg/m^2.[358] Bariatric surgery prior to transplantation is an option for morbidly obese patients.[359] The issue of nephron underdosing has been discussed in the donor factors above, but also relates to its interaction with the recipient's size. Body surface area has been used as a surrogate measure for both donor nephron mass, as well as recipient metabolic demand. Gross mismatch (i.e., low donor body surface area [BSA] and high recipient BSA) has been associated with poorer long-term allograft survival.[334,335]

Recipient Diabetes Mellitus

Diabetes mellitus as the primary cause of ESRD is a risk factor for allograft failure, due to death with a functioning graft.

Recipient History of Hepatitis C

Hepatitis C antibody positivity is a risk factor for allograft failure, due to both premature graft failure and death with a functioning graft.[265] The natural history of kidney transplantation in HCV+ recipients may, however, change (for the better) with the advent of new highly effective antiviral therapies.

IMPROVING KIDNEY ALLOGRAFT OUTCOMES: MATCHING KIDNEY AND RECIPIENT RISK

Maximizing the life span of donated organs is a key goal in kidney transplantation. The criteria used for allocation of deceased donor allografts can have an important effect on overall allograft survival. A purely utilitarian approach (to maximize allograft survival) would direct organs only to the youngest and healthiest, maximizing the "life years from transplant" (LYFT) gained from transplantation.[360] In practice, a balance must be struck between utility and equity (ensuring that anyone medically fit for a transplant has a reasonable chance of obtaining one).[361] In the US, the New Kidney Allocation system was instituted in late 2014. This system matches the 20% highest quality deceased donor kidneys with the top 20% of candidates, in the hope of maximizing the utility of the best organs (see section on ECD and KDPI).

MEDICAL MANAGEMENT OF THE TRANSPLANT RECIPIENT

More emphasis is being placed on the general medical management of transplant patients. Comprehensive practice guidelines on the care of kidney transplant recipients were published by both the American Society of Transplantation and the European Best Practice Guidelines Expert Group in 2000 and 2002, respectively.[362,363] KDIGO (Kidney Disease: Improving Global Outcomes) has more recently published its evidenced-based 2009 KDIGO Clinical Practice Guideline for the Care of Kidney Transplant Recipients.[188] Reflecting the paucity of quality evidence, only 25% of the KDIGO graded guidelines were level '1' ("we recommend") and 75% were level '2' ("we suggest"). The evidence supporting the guidelines was of low or very low quality in almost 85%. There is much opportunity for improving our understanding and management of transplant recipient care.

Transplantation is generally preferable to dialysis, but there is an increased appreciation that the posttransplant state is often one of chronic kidney disease and heightened cardiovascular risk. The management of common electrolyte, endocrine, and cardiovascular complications after transplant is discussed in the following sections.

ELECTROLYTE DISORDERS

HYPERCALCEMIA AND HYPOPHOSPHATEMIA

Hypercalcemia is common and is due mainly to persistent hyperparathyroidism or the overzealous administration of calcium and vitamin D. The management of posttransplant hyperparathyroidism is discussed later. Hypophosphatemia is also common in the early posttransplant period, particularly when allograft function is excellent. This is due to a combination of reduced phosphate absorption (vitamin D depletion is common) and urinary phosphate wasting which is a consequence of high fibroblast growth factor (FGF)-23 and PTH levels and the tubular effects of CNIs, sirolimus and high-dose steroids.[364] Rarely, phosphate depletion is severe enough to cause profound muscle weakness, including respiratory muscle weakness. Phosphate normalizes in the majority of patients by one year posttransplant, mirroring post-transplant declines in fractional excretion of phosphate, PTH, and FGF-23.[365] In the longer term, persistent negative phosphate balance likely contributes to posttransplant bone disease. Treatment involves a diet high in phosphorus (e.g., inclusion of low-fat dairy products) and vitamin D replacement. However, over-aggressive replacement of phosphate posttransplant can lower calcium and vitamin D levels and potentially exacerbate hyperparathyroidism.[366] In addition, acute phosphate nephropathy has been reported in posttransplant recipients on phosphate replacement.[367] Our practice is to reserve oral phosphate supplements for patients with a phosphate < 1-1.5 mg/dL or symptomatic hypophosphatemia.

Hyperkalemia

Mild hyperkalemia is common, even with good allograft function. The principal cause is CNI-induced impairment of tubular potassium secretion. Indeed, a recent study suggests that tacrolimus activates the thiazide sensitive renal sodium-chloride transporter resulting in hypertension and reduced renal potassium excretion.[368] Hyperkalemia may be exacerbated by poor allograft function, ingestion of excess potassium, hyperglycemia, and medicines such as ACE-inhibitors and β-blockers. Hyperkalemia can also be caused by an amiloride-like effect of trimethoprim, a component of TMP-SMX, frequently used for prophylaxis against *Pneumocystis jirovecii* (previously *carinii*). Because the hyperkalemia is usually not severe and typically improves with reduction in CNI dosage, additional treatment is often not required. Treatment with the mineralocorticoid fludrocortisone is usually effective in decreasing potassium (although sometimes at the expense of hypertension and edema) and may be considered in occasional cases.[369] Other medications that may be considered include a loop diuretic, a thiazide diuretic, sodium polystyrene sulfonate (Kayexalate), or patiromer.[370]

Metabolic Acidosis

Mild metabolic acidosis is also common and often associated with hyperkalemia. In most cases, it has the features of a distal (hyperchloremic) renal tubular acidosis. This reflects tubule dysfunction caused by CNIs, rejection, or residual hyperparathyroidism (and the effect of TMP, as above). Alkali repletion with oral bicarbonate may be necessary.

Other Electrolyte Abnormalities

Hypomagnesemia is common and due to a magnesuric effect of the CNIs, as well as to residual hyperparathyroidism, and is usually asymptomatic. Magnesium supplements are sometimes prescribed when the plasma magnesium level is less than 1.5 mg/dL. However, their effectiveness is limited, they can cause diarrhea, and they add more complexity to the multidrug regimen of the transplant recipient.

BONE DISORDERS AFTER KIDNEY TRANSPLANTATION

Bone disease in the ESRD patient is multifactorial and involves varying degrees of hyperparathyroidism (osteitis fibrosa cystica), vitamin D deficiency, low bone turnover, aluminum intoxication (osteomalacia), and amyloidosis (see Chapter 53). Unfortunately, bone disease can remain a problem after transplantation owing to persistence of the conditions discussed above and to the superimposed effects of immunosuppressants on bone.

HYPERPARATHYROIDISM

Residual hyperparathyroidism is very common in the first posttransplant year. However, hyperparathyroidism may persist for years; one study found elevated serum PTH in 23 (54%) of 42 normocalcemic patients more than two years after transplant who had plasma creatinine levels less than 2 mg/dL.[371] Not surprisingly, the main risk factors for posttransplant hyperparathyroidism are the degree of pretransplant hyperparathyroidism and the duration of dialysis.[372] Inadequate vitamin D stores and poor allograft function (de novo secondary hyperparathyroidism) probably contribute to persistence of the condition in some patients.

Typically, posttransplant hyperparathyroidism is manifest by a low plasma phosphate and a mild to moderate elevation in the plasma calcium. Serum PTH is inappropriately high for the level of plasma calcium. Posttransplant hyperparathyroidism is often asymptomatic and tends to improve with time. Posttransplant therapy with paricalcitol has been shown to increase the likelihood of resolution of hyperparathyroidism at 1 year posttransplant.[373] However, active vitamin D analogs must be used with caution and stopped if the plasma calcium rises above the normal range or complications of hypercalcemia occur. The calcimimetic, cinacalcet, has been shown safely and effectively lower serum PTH and calcium and raise serum phosphate concentrations in a placebo-controlled study of kidney transplant recipients with persistent hyperparathyroidism.[374]

There are two main indications for posttransplant parathyroidectomy: 1) severe symptomatic hypercalcemia (in the early posttransplant period, now rare, and usually managed medically with cinacalcet and/or bisphosphonate), and 2) persistent, moderately severe hypercalcemia (serum calcium 12.0–12.5 mg/dL or greater) for more than a year after transplantation or calcific uremic arteriolopathy (calciphylaxis), a rare complication following transplantation. Subtotal parathyroidectomy is the procedure of choice.

GOUT

The most important cause of hyperuricemia and gout after transplant are the CNIs, particularly cyclosporine. The CNIs impair renal uric acid clearance. Approximately 80% of CNI-treated kidney transplant recipients develop hyperuricemia, and about 13% develop new onset gout.[375] Diuretic use may exacerbate hyperuricemia and precipitate a gout attack.

Acute gout should be treated with colchicine or high-dose steroids; NSAIDs should generally be avoided. Colchicine-induced myopathy and neuropathy is more common in patients with impaired kidney function and in cyclosporine-treated (and presumably tacrolimus-treated) patients, due to an increase in colchicine levels. Therefore, the lowest effective dose of colchicine should be used and patients should be monitored for muscle weakness. For prevention of further gouty attacks, allopurinol is usually employed. Note that the metabolism of azathioprine is inhibited by allopurinol. Ideally, these drugs should not be co-prescribed. If azathioprine must be used in conjunction with allopurinol, then the azathioprine dose should be reduced to one quarter of the original dose and the complete blood count closely monitored. A safer alternative is to change azathioprine to MMF; no adjustment of MMF is required. The newer xanthine oxidase inhibitor, febuxostat, has also been successfully employed in hyperuricemic kidney transplant patients.[376] The packaging insert for febuxostat warns against concurrent azathioprine use. In cases of severe recurrent gout, it may be worthwhile to stop CNIs altogether. The uricosuric agent, probenicid, may be cautiously employed in kidney transplant recipients with excellent renal function. The uricase, pegloticase, remains untested in the transplant population.

CALCINEURIN INHIBITOR–ASSOCIATED BONE PAIN

A syndrome of severe bone pain in the lower limbs has been associated with CNI use. This is uncommon and thought to represent a vasomotor effect of the CNIs. Osteonecrosis and other common bone lesions should be excluded before the diagnosis is made. Symptoms usually respond to reduction in CNI dosage and administration of calcium channel blockers (especially nifedipine). Calcitonin has also been used successfully. Magnetic resonance imaging (MRI) of the involved bones may show bone marrow edema.[377]

OSTEONECROSIS

Osteonecrosis (avascular necrosis) is a serious bone complication of kidney transplantation. The pathogenesis is not well understood, but high doses of steroids are one risk factor. Up to 8% of kidney transplant patients develop osteonecrosis of the hips;[378] this figure is falling with lower dose steroid protocols.[379] The most commonly affected site is the femoral head; other sites are the humeral head, femoral condyles, proximal tibia, vertebrae, and small bones of the hand and foot. Many patients have bilateral involvement at the time of diagnosis. The principal symptom is pain; signs are nonspecific. Diagnosis is made by imaging studies; MRI is the most sensitive, plain radiography is the least sensitive, and scintigraphy is intermediate. However, MRI abnormalities do not always imply clinically significant osteonecrosis. Treatment remains controversial. Options include resting the joint, decompression, or joint replacement.

OSTEOPOROSIS

Osteoporosis is a common bone disorder characterized by a parallel reduction in bone mineral and bone matrix so that bone mass is decreased but is of normal composition.

The most commonly used definition is that based on the World Health Organization scoring system. Osteoporosis is defined as bone density greater than 2.5 SD below the mean of sex-matched young adults (T score); osteopenia, as 1.0 to 2.5 SD below the T score. The greater the reduction in bone density in the non-transplant population, the greater is the risk of fracture.

Reduction in bone mineral density is now recognized as a very common complication of kidney transplantation. Most bone loss occurs in the first 6 months after transplant.[380] Risk factors for posttransplant osteoporosis include steroid use, ongoing hyperparathyroidism, vitamin D deficiency or resistance, and phosphate depletion. Diabetes mellitus is also associated with an increased risk of posttransplant fracture. The risk of hip fracture in the kidney transplant population is high at 3.3 to 3.8 events per 1000 person-years; indeed, transplant recipients have a 34% increased risk of fracture early posttransplant when compared to transplant-waitlisted dialysis patients. The risk of hip fracture in transplant and waitlisted dialysis patients subsequently equalizes before 2 years posttransplant.[381]

Bone mineral density (BMD) as measured by dual X-ray absorptiometry has been shown to predict fractures in the general population.[382] Surprisingly, the effects of steroid avoidance and minimization protocols on BMD and fracture rates have been somewhat contradictory with some studies finding an association with reduced fracture rates[383,384] and improved bone mineral density while others have failed to identify a beneficial association.[379,385] Treatment with bisphosphonates early posttransplant has been shown to prevent bone loss when compared to control, but may exacerbate preexisting adynamic bone disease.[386] Perhaps as a consequence, bisphosphonates have not been shown to reduce fracture rates in the kidney transplant population.[387] Notably, bisphosphonates are not recommended in patients with a GFR < 30 mL/min/1.73 m^2. A number of studies have shown a favorable effect for vitamin D derivatives with or without calcium supplementation on BMD in kidney transplant recipients.[388,389] In patients with persistent posttransplant hyperparathyroidism and hypercalcemia, cinacalcet treatment may improve BMD, in addition to serum calcium and PTH levels.[390]

Current KDIGO guidelines recommend monitoring of calcium, phosphate, and PTH posttransplant. Dual-energy X-ray absorptiometry (DEXA) scanning is recommended in the first 3 months posttransplant for patients on steroid or with risk factors for osteoporosis and GFR > 30 mL/min/1.73 m^2. For kidney transplant recipients within a year of transplant, with GFR > 30 mL/min/1.73 m^2 and low BMD, treatment with vitamin D, calcitriol/alfacalcidiol, or bisphosphonate should be considered.[188]

POSTTRANSPLANT DIABETES MELLITUS

Posttransplant diabetes mellitus, formerly referred to as new-onset diabetes after transplantation (NODAT) is common after kidney transplantation. First-time kidney transplant recipients in the US between the years 2004-2008 had a 3-year cumulative incidence of posttransplant diabetes of over 40%. Risk factors include older age, obesity, positive hepatitis C antibody, CMV infection status, non-white ethnicity, family history, steroids, CNIs (especially tacrolimus), and episodes of acute rejection. Strategies to prevent and treat PTDM include steroid minimization, avoidance of tacrolimus, and lifestyle modification. The adoption of steroid-free immunosuppression regimens at discharge after kidney transplantation has been shown to reduce the odds of PTDM.[391,392] A small Austrian study reported a reduction in PTDM at one year in patients whose blood glucose was strictly maintained with insulin immediately posttransplant versus those that received standard management of blood glucose.[393]

Unfortunately, PTDM is associated with reduced graft and patient survival.[394] PTDM may require treatment with oral agents or insulin. Metformin is probably the drug of choice in kidney transplant recipients with adequate GFR because it is the most effective in reducing complications of type 2 diabetes mellitus.[303] For most patients with type 2 diabetes mellitus the American Diabetes Association recommends an HbA1C target of 7%. There are certainly benefits of tight glycemic control, especially in terms of the reduction in microvascular complications.[395,396] However, these benefits must be weighed against the risks of overzealous glycemic control as evidenced by the ACCORD trial where intensive glycemic control (targeting a HbA1C <6%) was associated with an increased risk of mortality and severe hypoglycemia.[395,397] For most patients a 'personalized' HbA1C target based on carefully considered risks and benefits is a good strategy.

CARDIOVASCULAR DISEASE

Cardiovascular disease is a leading cause of death in kidney transplant recipients.[398] Despite pretransplant screening, the cumulative incidence of myocardial infarction (MI), stroke, and de novo peripheral arterial disease are 11%, 7%, and 24%, respectively.[399,400,401] De novo congestive heart failure is also common.[402] The kidney transplant recipient population is enriched with traditional cardiovascular risk factors such as smoking, diabetes mellitus, and hypertension and are also burdened with nontraditional CKD-related and transplant-associated risk factors.[403,404] Aspirin is an effective therapy for secondary prevention of cardiovascular disease in the general population.[405,406] However, a recent study showed marked variability in the use of cardioprotective medications, including aspirin, in kidney transplant recipients.[407]

SMOKING

Tobacco smoking should be strongly discouraged; there is evidence that it affects allograft function as well as recipient survival.[408]

HYPERTENSION

The prevalence of hypertension in the CNI era is at least 60% to 80%.[188] Causes include use of steroids, CNIs, weight gain, allograft dysfunction, native kidney disease, and transplant renal artery stenosis. The complications of posttransplant hypertension are presumed to be a heightened risk of cardiovascular disease and allograft failure.[409] A post hoc analysis of the FAVORIT trial (which investigated the effect of lowering homocysteine levels on cardiovascular outcomes in kidney transplant recipients) found that each increase of 20 mm Hg in baseline systolic blood pressure was associated with 32% and 13% increases in the adjusted relative risks of cardiovascular events and death, respectively. In contrast, each 10 mmHg drop in diastolic blood pressure below 70 mm Hg was associated with a 31% increase in the relative risks of both cardiovascular events and death. The highest risk was seen in patients with a systolic blood pressure > 140 mm Hg and diastolic blood pressure < 70 mm Hg.[410]

There has been some controversy over blood pressure treatment targets in recent years. For transplant patients we follow the 2017 American College of Cardiology/American Heart Association Hypertension Guidelines for adults with CKD and hypertension to *initiate antihypertensive drug therapy at SBP ≥130 mm Hg or DBP ≥80 mm Hg and treat to goal of SBP <130 mm Hg and DBP <80 mm Hg.*[411]

Nonpharmacologic measures such as weight loss, reduced sodium intake, reduced alcohol intake, treatment of obstructive sleep apnea, and increased exercise should be encouraged. The dosage of steroids and CNIs should be minimized. However, antihypertensive drug therapy is still frequently required. There is little evidence for one antihypertensive drug class over another. In stable transplant recipients (good allograft function and more than 6 to 12 months out from transplant) we treat with the 2017 American College of Cardiology/American Heart Association Hypertension Guidelines recommended first-line antihypertensive drugs, namely thiazide diuretics, calcium channel blockers, and ACEIs or ARBs.

We routinely use ACEIs and ARBs, especially in patients with proteinuria, diabetes, or other cardiovascular indications. However, in the early posttransplant months, we generally avoid ACEIs or ARBs as they may exacerbate hyperkalemia or result in a creatinine increase that may be confused for rejection. Clinical factors that influence our choice of antihypertensive therapy in kidney transplant recipients are shown in Table 70.13.

PROTEINURIA

Proteinuria, even when modest, is associated with poorer allograft survival.[412] We initiate ACEI or ARB treatment in our kidney transplant recipients with proteinuria, although we note that the only randomized controlled trial of ACEI or ARB therapy (Ramipril) in proteinuric kidney transplant patients was negative (no difference in doubling of creatinine, ESRD, or death compared to placebo). Indeed, the only between group difference at the end of the study was a significantly lower hemoglobin in the ramipril treated group.[413]

Table 70.13 Antihypertensive Agents in the Transplant Recipient

Indication	β-Blocker	ACE-I/ ARB	Ca-Channel Blocker	Thiazide Diuretic
Hypertension only		X	X	X
CHF	X	X		
Post-MI	X	X		
CAD	X	X		
Diabetes		X		
Proteinuria		X		

ACE-I, Angiotensin-converting enzyme inhibitor; *ARB,* angiotensin receptor blocker; *CAD,* coronary artery disease; *CHF,* congestive heart failure; *MI,* myocardial infarction.

HYPERLIPIDEMIA

The prevalence of hyperlipidemia after transplant is very high.[414] Steroids, CNIs (cyclosporine more than tacrolimus), and sirolimus are the principal causes. Some studies suggest that hyperlipidemia is associated with poorer allograft outcomes, although no causal relation has been established. The American Heart Association Guidelines (for the general population) have deemphasized the importance of cholesterol concentration on the decision to treat with statin and highlighted the importance of assessing cardiovascular risk. High-intensity statin is recommended for all patients over 21 years of age with 1) atherosclerotic cardiovascular disease, 2) LDL >190 mg/dL, and 3) persons aged 40 to 75 years with diabetes mellitus and an estimated 10-year cardiovascular risk > 7.5%. Moderate intensity statin is recommended for 1) persons aged 40 to 75 years with an estimated 10-year cardiovascular risk < 7.5% and diabetes mellitus and 2) non-diabetics 40 to 75 years of age with an estimated 10-year cardiovascular risk > 7.5%.[415] The application of a risk-guided statin treatment algorithm in the kidney transplant population seems like a reasonable approach. The only caveat is that the starting dose of a statin should be reduced in kidney transplant recipients as the CNIs (especially cyclosporine) increase statin blood levels and may predispose to statin-related toxicities.[416]

The main randomized controlled trial of statin therapy in kidney transplant recipients showed no benefit in the primary outcome (composite of cardiac death, nonfatal myocardial infarction or coronary intervention procedure).[417] However, follow-up after an additional two years of statin therapy showed a significantly reduced risk of major cardiac events.[418] As in the general population, statin use is associated with an increased risk hyperglycemia in kidney transplant recipients.[419]

Hypertriglyceridemia can be a problem posttransplant. Strategies for lowering triglyceride levels include, lifestyle modification, substitution of sirolimus for an alternative agent (e.g., MMF), and treatment with ezetimibe.

The high prevalence of metabolic syndrome, a consequence of dysregulation of glucose and vascular metabolism, is being increasingly recognized in kidney transplant recipients.[420] Better recognition and management of metabolic syndrome, including control of obesity, glucose, hyperlipidemia, and blood pressure, are expected to decrease the morbidity and mortality of transplant recipients.

HYPERHOMOCYSTEINEMIA

As in the general population, hyperhomocysteinemia has been proposed as a risk factor for cardiovascular disease in kidney transplant recipients. Plasma homocysteine concentrations typically fall after transplant but do not normalize. Based on the results of well-performed randomized controlled trials in both the general and CKD populations, current interventions to lower homocysteine cannot be recommended.[421]

CANCER AFTER KIDNEY TRANSPLANTATION

A number of recent studies have linked transplant and cancer registry data, giving good insight into posttransplant

Table 70.14 Cancers Categorized by SIR for Kidney Transplant Patients and Cancer Incidence

	Common Cancers in Both General and Transplant Populations: Incidence ≥10 Per 100,000 People	Common Cancers in the Transplant Population: Incidence in General Population <10 Per 100,000, but Estimated Incidence in Transplant Population ≥10 Per 100,000 People	Rare Cancers Incidence in Both General and Transplant Populations <10 Per 100,000 People
High SIR >5	Kaposi's Sarcoma (with HIV)	Kaposi's sarcoma Vagina Non-Hodgkin's lymphoma Kidney Non-melanoma skin Lip Thyroid Penis Small intestine	Eye
Moderate SIR (1-5)	Lung Colon Cervix Stomach Liver	Oro-nasopharynx Esophagus Bladder Leukemia	Melanoma Larynx Multiple myeloma Anus Hodgkin's lymphoma
No Increased Risk	Breast Prostate Rectum		Ovary Uterus Pancreas Brain Testis

SIR, Standard Incidence Rate.
Adapted from the KDIGO practice gudelines.[183–188]

cancer incidence. Data from the US and UK show that kidney transplant recipients are at considerably greater risk for certain cancers when compared to the general population (Table 70.14).[422–424]

There are several reasons why reported cancer incidence has increased. First, immunosuppression inhibits normal tumor surveillance mechanisms, allowing unchecked proliferation of "spontaneously occurring" neoplastic cells. There is also experimental evidence that cyclosporine has tumor-promoting effects mediated by its effects on TGF-β production[425] and expression of the angiogenic cytokine vascular endothelial growth factor (VEGF).[426] Second, immunosuppression allows uncontrolled proliferation of oncogenic viruses (Table 70.15). Third, factors related to the primary kidney disease (analgesic abuse, certain herbal preparations, HBV or HCV infection) or the ESRD milieu (acquired renal cystic disease) might promote neoplasia.

It is believed that the cumulative amount of immunosuppression rather than a specific drug is the most important factor increasing the cancer risk. However, there is evidence that the routine use of CNIs has increased the risk of skin cancers,[427] fortunately, these are usually not fatal. The long-term impact of the currently employed powerful immunosuppression regimens on cancer incidence is unknown, but it is certainly of concern. The single most important measure to prevent cancers is to minimize excess immunosuppression. A general rule is that when cancer occurs, immunosuppression should be significantly decreased. In some cases, rejection of the allograft may result, but the risks and benefits of immunosuppression must be judged on a case-by-case basis.

Table 70.15 Viral Infections Associated With Development of Cancers in Kidney Transplant Patients

Virus	Neoplasm
HBV	Hepatocellular cancer
HCV	Hepatocellular cancer
EBV	PTLD
HPV	Squamous cell cancers of anogenital area and of mouth
HHV-8	Kaposi's sarcoma

HBV, Hepatitis B virus; *HCV,* hepatitis C virus; *EBV,* Ebstein-Barr virus; *HPV,* human papillomavirus; *HHV-8,* human herpes virus-8, *PTLD,* posttransplant lymphoproliferative disorder.

SKIN AND LIP CANCERS

Squamous cell carcinoma, basal cell carcinoma, and malignant melanoma are all more common in kidney transplant patients. The incidence of non-melanomatous skin cancer in kidney transplant recipients is 13 times greater than in the general population.[423] Risk factors include time after transplant, cumulative immunosuppressive dose, exposure to ultraviolet light, fair skin, and human papillomavirus infection. Primary and secondary prevention is important: Patients should be specifically counseled on minimizing exposure to ultraviolet light and to self-screen for skin lesions.[188] Retinoids are sometimes used in high-risk patients.[428] Suspicious skin lesions should be surgically excised. In patients with posttransplant squamous

Table 70.16 Clinical and Pathologic Spectrum of PTLD and Summary of Its Management

	Early Disease (50%)	Polymorphic PTLD (30%)	Monoclonal PTLD (20%)
Clinical features/ symptoms	Infectious mononucleosis-type illness	Infectious mononucleosis-type illness ± weight loss, localizing symptoms	Fever, weight loss, localizing symptoms
Pathology	Preserved architecture, atypical cells infrequent	Intermediate transformed cells	High-grade lymphoma with confluent and marked atypia
Clonality	Polyclonal	Usually polyclonal	Monoclonal
Treatment	Reduce immunosuppression	Reduce immunosuppression ±rituximab	Reduce immunosuppression ± rituximab ± chemotherapy
Prognosis	Good	Intermediate	Intermediate/poor

PTLD, Posttransplant lymphoproliferative disorder.

cell carcinoma, switching from a CNI to sirolimus reduces the risk of further such tumors developing by 44% (although places patients at risk for other important metabolic effects).[106]

ANOGENITAL CANCERS

Cancers of the vulva, uterine cervix, penis, scrotum, anus, and perianal region are significantly more common in kidney transplant recipients. Furthermore, these cancers tend to be multifocal and more aggressive than in the general population. Infection with certain human papillomavirus strains is an important risk factor. Secondary prevention measures include yearly physical examination of the anogenital area and, in women, yearly pelvic examinations, and cervical histology. Suspicious lesions should be excised, and patients should be closely followed for recurrence.

KAPOSI'S SARCOMA

The incidence of Kaposi's sarcoma in transplant and nontransplant patients depends greatly on ethnic background. Those of Jewish, Arab, and Mediterranean ancestry are at much higher risk. Other risk factors are cumulative immunosuppressive dose and human herpes virus-8 infection. Visceral (lymph nodes, lungs, gastrointestinal tract) and nonvisceral (skin, conjunctivae, oropharynx) involvement may occur. The prognosis for the former is poor, but for the latter it is good. Treatment involves various combinations of surgical excision, radiotherapy, chemotherapy, and immunotherapy. Immunosuppression, of course, should be reduced or modified. Substitution of sirolimus for CNIs has been reported to stop progression of the disease without sacrificing allograft function.[105]

POSTTRANSPLANT LYMPHOPROLIFERATIVE DISORDER

Posttransplant lymphoproliferative disorder (PTLD) is one of the most feared complications of transplantation because it can occur early after transplant and it carries a high morbidity and mortality. The cumulative 3-year incidence of PTLD is 0.5% in adult and 1.5% in pediatric kidney transplant recipients. More than 90% are non-Hodgkin's lymphomas, and most are of recipient B cell origin.[429] Risk factors include 1) Epstein-Barr virus (EBV)–positive donor and EBV-negative recipient; 2) CMV-positive donor and CMV-negative recipient; 3) pediatric recipient (in part because children are more likely to be EBV naive); and 4) intensity of immunosuppression.[430] Clinical trials of belatacept saw high rates of PTLD in EBV-negative subjects; the drug is now contraindicated in patients who are seronegative for EBV.[86]

Important in the pathogenesis of PTLD is the infection and transformation of B cells by EBV; transformed B cells undergo proliferation that is initially polyclonal, but a malignant clone may evolve. Thus, the clinical and histologic spectrum of PTLD at presentation and its treatment can vary greatly (Table 70.16). Extranodal, gastrointestinal tract, and central nervous system involvement is more common than in non-transplant lymphomas. The kidney allograft may be involved. Treatment options for PTLD include immunosuppression reduction, rituximab and CHOP [cyclophosphamide, doxorubicin (Adriamycin), vincristine, prednisone]. Screening high-risk patients (EBV+ donor/EBV- recipient) for EBV viremia in the first year posttransplant with institution of rituximab treatment for persistently viremic patients has been shown to reduce PTLD incidence in one small study.[431]

CANCER SCREENING

Since life expectancy in the kidney transplant population is significantly improved compared to patients receiving dialysis, we recommend that patients follow cancer screening guidelines (cervical smear, mammogram, colonoscopy) for the general population. We also recommend annual dermatologic screening. Screening for kidney cancer is not advised in the most recent KDIGO guidelines.[188]

INFECTIOUS COMPLICATIONS OF KIDNEY TRANSPLANTATION

The transplant procedure itself and subsequent immunosuppression increase the risk of serious infection. The principal factors determining the type and severity of infection are exposure (in the hospital and community) to potential pathogens and the state of immunosuppression.[432] Factors affecting the net state of immunosuppression include the cumulative amount of immunosuppression, recipient comorbidities (e.g., diabetes, UTIs), infection of viruses that affect the immune system (e.g., EBV, CMV, HIV, hepatitis C), and the integrity of mucocutaneous barriers.

The patterns of infection after kidney transplantation can be roughly divided into three time periods: 0 to 1 month, 1 to 6 months, and more than 6 months after transplant.[432] These divisions of time serve as guidelines only. Maintenance immunosuppressive regimens are becoming more powerful, and more elderly patients are undergoing transplant surgery;

on the other hand, antimicrobial prophylaxis is becoming more effective. An outline of the pattern of infection after transplantation according for time after transplantation is shown in Table 70.17.

A general point is that at any time when life-threatening infection occurs, immunosuppression should be reduced to an absolute minimum or stopped altogether (so-called stress-dose steroids often are required). Prompt diagnosis (e.g., bronchoscopy in patients with pneumonitis) and therapy are essential.[432]

Table 70.17 Infections After Transplantation According to Time After Transplantation

Time After Transplant	Type of Infection
<1 month	Infection with antimicrobial-resistant species:
	MRSA, VRE, Candida species (non-albicans)
	Aspiration
	Catheter infection
	Wound infection
	Anastomotic leaks and ischemia
	Clostridium difficile colitis
	Donor-derived infection (uncommon):
	HSV, LCMV, Rhabdovirus (rabies), West Nile virus, HIV, *T. cruzi*
	Recipient-derived infection (colonization):
	Aspergillus, pseudomonas
1-6 months	With PCP and antiviral (CMV, HBV) prophylaxis:
	Polyomavirus, BK infection, nephropathy
	C. difficile colitis
	HCV infection
	Adenovirus infection, influenza
	Cryptococcus neoformans infection
	Mycobacterium tuberculosis infection
	Without prophylaxis:
	Pneumocystis
	Infection with herpesviruses (HSV, VZV, CMV, EBV)
	HBV infection
	Infection with Listeria, Nocardia, toxoplasma, stongyloides, leishmania, *T.cruzi*
>6 months	Community-acquired pneumonia, UTI
	Infection with aspergillus, typical molds, mucor species
	Infection with Nocardia, rhodococcus species
	Late viral infections:
	CMV infection
	Hepatitis (HBV, HCV)
	HSV encephalitis
	Community-acquired (SARS, West Nile virus infection)
	JC polyomavirus infection (PML)
	Skin cancer, lymphoma (PTLD)

CMV, Cytomegalovirus; *EBV,* Epstein-Barr virus; *HBV,* herpes virus B; *HCV,* herpes virus C; *HSV,* herpes simplex virus; *LCMV,* lymphocytic choriomeningitis virus; *MRSA,* methicillin-resistant Staphylococcus aureus; *PCP,* Pneumocystis carinii pneumonia; *SARS,* severe acute respiratory syndrome; *UTI,* urinary tract infection; *VRE,* vancomycin-resistant enterococcus.

Adapted from Fishman JA. Infection in solid-organ transplant recipients. *N Engl J Med* 2007;357:2601-2614.

INFECTIONS IN THE FIRST MONTH

The majority of infections in the first month are standard infections, as would be seen in non-transplant patients after surgery. Thus, infections of surgical wounds, the lungs, and the urinary tract, and infections related to vascular catheters predominate. Bacterial infections are much more common than fungal ones. Preventive measures include ensuring that donor and recipient are free of overt infection before transplant, good surgical technique, and SMX-TMP prophylaxis to prevent UTIs.

INFECTIONS FROM 1 TO 6 MONTHS AFTER TRANSPLANT

Weeks of intensive immunosuppression increase the risk of opportunistic infections. Infections with CMV, EBV, *Listeria monocytogenes, Pneumocystis jirovecii,* and *Nocardia* spp. are relatively common. Preventive measures include antiviral prophylaxis (for 3 to 6 months) and SMX-TMP prophylaxis (for 6 to 12 months).

INFECTIONS MORE THAN 6 MONTHS AFTER TRANSPLANT

With gradual reduction in immunosuppression, the risk of infection long-term usually diminishes. However, patients can be roughly divided into two groups based on risk. Those with good ongoing allograft function and no need for late supplemental immunosuppression are at low risk for developing opportunistic infections unless exposure is intense (e.g., to *Nocardia* spp. from soil). In contrast, those with poor allograft function are at higher risk for opportunistic infection. This probably reflects both poor allograft function and that many of these patients have received large cumulative doses of immunosuppression.

Late amplification of immunosuppression may increase the risk of opportunistic infection in any patient. Therefore, any patient receiving a “late” steroid pulse or antithymoglobulin therapy should be restarted on SMX-TMP +/- anti-CMV prophylaxis (if donor or recipient were CMV positive). The role of EBV infection in causing PTLD has been discussed earlier; CMV and *P. jirovecii* pneumonia infection are discussed later.

CYTOMEGALOVIRUS

Exposure to CMV (as evidenced by the presence in serum of anti-CMV IgG) increases with age; more than two thirds of adult donors and recipients are latently infected before kidney transplantation. Infection may arise from 1) reactivation of latent recipient virus, 2) primary infection with donor-derived virus (transmitted in the allograft or less commonly via blood products), or 3) reactivation of latent donor-derived virus. CMV *disease* means that there is infection with symptoms or with evidence of tissue invasion, or both. The risk of CMV infection or disease is highest in CMV-positive donor/CMV-negative recipient pairings, followed by CMV-positive donor/CMV-positive recipient pairings and then CMV-negative donor/CMV-positive recipient pairings. The risk is lowest with CMV-negative donor/CMV-negative recipient pairings. OKT3/polyclonal antibody therapy, particularly

Table 70.18 Manifestations of CMV Disease in the Kidney Transplant Recipient

Tissue Affected	Clinical/Pathological Features	Comment
Viremia (blood)	Fever, malaise, myalgia	Nonspecific, may occur after cessation of prophylaxis
Bone marrow	Leukopenia	May require azathioprine or MMF dose reduction. Valganciclovir may also cause leukopenia
Lungs	Pneumonitis	May be life-threatening; exclude coinfection with other organisms
GI tract	Inflammation and ulceration of esophagus or colon	May be life-threatening.
Liver	Hepatitis	Rarely severe
Retina	Blurred vision, flashes, floaters	Relatively rare in kidney transplants; if occurs, usually late.
Kidney	Viral inclusions	Association with FSGS

CMV, Cytomegalovirus; *GI,* gastrointestinal; *MMF,* mycophenolate mofetil.

when prescribed for treatment of rejection, significantly increases the risk of subsequent CMV disease.

CMV disease usually arises 1 to 6 months after transplantation, although gastrointestinal and retinal involvement often occur later. Typical clinical features are fever, malaise, and leukopenia; there may be symptomatic or laboratory evidence of specific organ involvement (Table 70.18). Urgent investigation and immediate empiric treatment are needed in severe cases. Confirmation of presumed CMV disease is by demonstration of the virus in body fluids or solid organs. CMV infection after transplantation is most commonly associated with viremia. CMV viremia may be detected by culture, antigenemia assay, or quantitative PCR. The PCR assay is fast, sensitive, and allows the monitoring of viral load in response to therapy. However, viremia may be absent in certain CMV infections, most notably the retina or gastrointestinal tract. In such instances the CMV infection may be diagnosed by ophthalmologic exam +/- vitreous humor CMV PCR and immunohistochemistry or culture of gastrointestinal tissue, respectively. Other procedures such as lumbar puncture and bronchoscopy should be aggressively pursued according to symptoms and signs. A “tissue diagnosis” is also required to exclude coinfection with other microbes such as *P. jirovecii.* In addition to its direct effects, CMV may have indirect effects after transplant, including an increased risk of infection, rejection, and PTLD.[433]

The treatment of posttransplant CMV infection depends on disease severity and organ systems involved. Low-grade CMV viremia usually responds to a reduction in immunosuppression (usually the antimetabolite). The VICTOR trial demonstrated that oral valganciclovir was non-inferior to intravenous ganciclovir in the treatment of non-life-threatening CMV infection in solid organ transplants.[434] Therefore, patients with symptomatic or significant viremia but relatively mild symptoms may be treated as an outpatient with oral valganciclovir. However, patients with severe systemic CMV infection or organ-specific disease [i.e., pneumonitis, gastrointestinal, or central nervous system (CNS)] should be treated initially with intravenous ganciclovir with a transition to oral valganciclovir as symptoms improve. CMV viral load should be monitored with therapy. For viremic patients with mild to moderate disease we generally advocate continuing maintenance treatment for 2 weeks after resolution of viremia. Treatment of severe or invasive disease may benefit from longer courses of induction treatment followed by maintenance dose therapy.[435] Dose adjustment is required in kidney dysfunction for both valganciclovir and ganciclovir. Although supportive data are unavailable, it is reasonable to add CMV hyperimmune globulin in severe cases.[436]

The prevention of CMV disease is of great clinical importance. One strategy is to give prophylaxis to all patients at risk (D+/R-, D-/R+, D+/R+). Another strategy is to monitor for CMV viremia and begin prophylaxis only when there is evidence of active viral replication.[433] Prophylactic and preemptive strategies using valganciclovir have been shown comparable in terms of cost and prevention of symptomatic CMV.[437] Valganciclovir has proven to be an excellent therapy both for the prophylaxis and treatment of CMV; however, myelosuppression, especially leukopenia/neutropenia, is a common and sometimes difficult to manage side effect. Letermovir is currently being trialled as CMV prophylaxis in kidney transplant patients and may prove to be a useful alternative agent.[438]

PNEUMOCYSTOSIS

Antimicrobial prophylaxis is very effective in preventing pneumonia due to *P. jirovecii* (previously *carinii*). The preventive agent of choice is SMX-TMP: It is generally well tolerated and quite inexpensive; furthermore, it prevents UTIs and opportunistic infections such as nocardiosis, toxoplasmosis, and listeriosis. Alternative preventive agents include dapsone and pyrimethamine, atovaquone, and aerosolized pentamadine.[439] Typical symptoms of pneumonia due to *P. jirovecii* are fever, shortness of breath, and cough. Chest radiography characteristically shows bilateral interstitial-alveolar infiltrates. Diagnosis requires detection of the organism in a clinical specimen by colorimetric or immunofluorescent stains. Because the organism burden is usually lower than in human immunodeficiency virus (HIV)–infected patients, the sensitivity of induced sputum or bronchoalveolar lavage specimens is lower in kidney transplant recipients; tissue should be quickly obtained if these tests are negative and the clinical suspicion remains high.

The treatment of choice remains SMX-TMP.[439] High-dose SMX-TMP may increase the plasma creatinine without affecting GFR. There is no firm evidence to support the use of higher dose steroids during the early treatment phase of pneumocystosis in kidney transplant recipients.

Table 70.19 Recommended and Contraindicated Vaccines in Transplant Recipients

Recommended	Comments	Contraindicated[a]
Influenza	Annually	Live varicella vaccine[b]
PCV13	Once	BCG
PPSV23	Repeat every 5 years, up to 3 doses/lifetime including one dose in those aged 65 years and older	Smallpox
Tdap/Td	Td booster every 10 years	Intranasal influenza
9vHPV	Only for ≤ 26 years of age	Live Japanese B encephalitis
HepA		Oral polio
HepB	Give 3-dose series if anti-HBS<10 mIUL/mL	MMR
Asplenia or complement inhibitor therapy		Live oral typhoid Ty21a
MenACWY	2-dose series with booster every 5 years	Yellow fever
MenB	2-dose series, no booster	
Hib	If not already given	

[a]Not an exhaustive list, all live vaccines are generally contraindicated.
[b]Awaiting data on the non-live 'Shingrix' (HZ/su) zoster vaccine.
PCV13, Pneumococcal conjugate vaccine (13-valent); *PPSV23,* pneumococcal polysaccharide vaccine (23-valent); *MenACWY,* meningococcal (quadrivalent) conjugate; *MenB,* serogroup B meningococcal vaccines; *Hib,* Haemophilus influenzae type b conjugate vaccine; *Tdap,* tetanus and reduced diphtheria toxoid, and acellular pertussis vaccine; *Td,* tetanus and reduced diphtheria toxoids; *HepA,* hepatitis A vaccine; *HepB,* hepatitis B vaccine; *9vHPV,* human papillomavirus vaccine; *MMR,* measles, mumps, and rubella vaccine.

IMMUNIZATION IN KIDNEY TRANSPLANT RECIPIENTS

Important general rules concerning immunization in kidney transplant patients are the following: 1) Immunizations should be completed at least 4 weeks before transplantation; 2) live vaccines are generally contraindicated after transplantation. We follow the *IDSA clinical practice guideline for vaccination of the immunocompromised host.*[440] We pay special attention to pretransplant vaccination in asplenic patients and those who will or may require eculizumab therapy. Vaccines that are recommended and contraindicated are summarized in Table 70.19. We await further data on the Shingrix (HZ/su) non-live vaccine in kidney transplant recipients but expect that it may soon be recommended.[441] Household contacts of transplant recipients should receive yearly immunization against influenza.

SUMMARY

Infections are a predictable complication of kidney transplantation. Minimizing infections requires meticulous surgical technique, antiviral prophylaxis for the first 3 to 6 months, SMX-TMP prophylaxis for the first 6 to 12 months, vaccination and, of course, avoidance of excess immunosuppression. A substantial increase in immunosuppression, no matter what the time period after transplant, should trigger resumption of SMX-TMP and probably antiviral prophylaxis.

TRANSPLANT ISSUES IN SPECIFIC PATIENT GROUPS

TRANSPLANTATION IN PATIENTS WITH DIABETES

Although the survival rate of diabetic patients after kidney transplantation is lower than that of nondiabetic patients, the benefits of transplantation versus staying on dialysis remain.[306] Cardiovascular disease is highly prevalent in the diabetic ESRD population and aggressive cardiovascular risk modification should be undertaken. A subset of diabetic ESRD patients is also suitable for kidney-pancreas transplantation (see the following discussion).

KIDNEY-PANCREAS TRANSPLANTATION

The benefits of pancreas transplantation are 1) freedom from insulin therapy and the metabolic derangements of type 1 diabetes mellitus and 2) potentially slowing or reversal of the progression of end-organ damage from this condition. The disadvantage of pancreas transplantation is higher risk of surgical complications, and the need for higher levels of immunosuppression. Thus, patients must be carefully selected for this procedure. Although select patients with type 2 diabetes can undergo pancreas transplantation, the vast majority of pancreas transplantations are performed in patients with type 1 diabetes. For diabetic ESRD patients deemed suitable candidates for kidney plus pancreas transplantation, options include simultaneous kidney and pancreas (SPK) transplantation or pancreas after kidney (PAK) transplantation (the latter allows living donor kidney transplantation). Pancreas transplant alone (PTA) was also performed in a minority of patients without ESRD. In contrast to SPK transplantation, there is some concern as to whether pancreas transplant alone/later improves patient survival. However, PAK transplantation offers the advantages of preemptive living donor kidney transplantation and better kidney allograft outcomes. Rates of pancreas transplantation have decreased somewhat over the past decade from a peak of around 1500 procedures in 2004 to just over 1000 in 2017.

Complications of pancreatic transplantation include thrombosis, infection, rejection, and problems related to drainage of the exocrine secretions.[442] Safe drainage of the exocrine secretions is vital. Drainage of exocrine secretions into the bladder affords the advantages of sterility and of serial measurement of urinary amylase concentrations that can aid in early detection of pancreatic allograft dysfunction. Important disadvantages include severe cystitis, hypovolemia, and acidosis (the last two due to large losses of bicarbonate-rich fluid). Because rates of technical complications from enteric drainage have decreased, this technique is becoming more popular.[442]

The graft half-life for pancreas transplants is over 12 years in the US. Pancreas graft survival rates are better for recipients of SPK allografts than for PTA or PAK recipients.[443] The major causes of graft failure are technical (40%) and acute rejection (15%) in the early posttransplant period, and chronic rejection (25%) in the late posttransplant period.[444]

KIDNEY TRANSPLANTATION IN PATIENTS WITH HUMAN IMMUNODEFICIENCY VIRUS INFECTION

HIV infection was traditionally considered an absolute contraindication to kidney transplantation. This reflected fears that immunosuppression would promote severe infections and that the short survival of HIV-positive patients undergoing transplant surgery would waste valuable allografts. With the introduction of antiretroviral agent therapy in 1996 and subsequent improvements in the survival of HIV-positive patients, many centers accept patients with HIV for transplantation. Indeed, very good allograft and recipient survival has been reported.[445,446] A recent study, which included 510 patients transplanted with HIV, found that HIV+ kidney transplant recipients had a lower 5- and 10-year graft and patient survival when compared to HIV- controls. Interestingly, when those 362 HIV+ patients not coinfected with HCV were compared to HIV- controls 5- and 10-year graft and patient survival were identical suggesting HCV coinfection was the driver of worse outcomes in the HIV+ group as a whole.[447] Interestingly HIV+ patients undergoing kidney transplant appear to be at higher risk of acute rejection and likely benefit from ATG induction.[448] Generally, HIV+ patients must be stable on HAART and have CD4+ T-cell counts of at least $200/m^3$ and undetectable HIV RNA levels before being considered for transplant. Patients with HIV should be referred to centers with specialized HIV+ transplant programs because of the complexity of posttransplant management; e.g., the potential for interactions between the multiple antiviral medicines, some of which inhibit and some of which induce the cytochrome P450 system. Successful transplantation of kidneys from HIV-positive donors to HIV-positive recipients has been reported.[449]

PREGNANCY IN THE KIDNEY TRANSPLANT RECIPIENT

Fertility is often improved after kidney transplantation.[450] Although the risks of pregnancy and childbirth are greater in kidney transplant recipients compared to the general population, it is generally considered safe for the mother, fetus, and kidney allograft if the following criteria are met before conception: good general health for more than 18 months before conception, stable allograft function with a plasma creatinine level less than 2.0 mg/dL (preferably less than 1.5 mg/dL), minimal hypertension, minimal proteinuria, immunosuppression at maintenance doses, and no dilation of the pelvicalyceal system on recent imaging studies.[451]

A recent systematic review and meta-analysis of 50 studies that included data on 4706 pregnancies in 3570 KTRs, between the years 2000 and 2010, gives insight into the relative risks of pregnancy post-kidney transplant. The rate of live births was actually higher than that of the general population. However, KTR pregnancies were more likely to end with cesarean sections, preterm births (35.6 vs 38.7 weeks), and lower birth weights (2.4 kg vs 3.3 kg). Preeclampsia and gestational diabetes were both more common in KTRs than the general population. More than 50% of KTRs were hypertensive during pregnancy. Overall, pregnancy did not appear to associate with an increased risk of acute rejection or subsequent graft loss.[452]

All pregnant kidney allograft recipients should be managed as high-risk obstetric cases with nephrology involvement. Throughout the pregnancy, regular monitoring of blood pressure, proteinuria, kidney function, and urine cultures is advised. Significant kidney dysfunction occurs in a minority of cases; the principal causes are severe preeclampsia, acute rejection, acute pyelonephritis, and recurrent glomerulonephritis. Distinguishing these causes clinically may be difficult. Initial investigations should include plasma creatinine, creatinine clearance, 24-hour urinary protein excretion, urine microscopy, urine culture, and kidney ultrasound. Acute rejection should be confirmed by allograft biopsy before instituting antirejection therapy. Pulse steroids are used to treat rejection.

There are no transplant-specific reasons to perform cesarean section; if it is performed (for obstetric reasons), care should be taken to avoid damaging the transplant ureter. Kidney function should be monitored closely for 3 months postpartum because of the increased risk of HUS and possibly acute rejection.

Short-term and long-term data indicate that children born to transplant recipients using cyclosporine, steroids, or azathioprine do not have a significant increase in morbidity. Short-term data on tacrolimus are similarly reassuring. Dosages of cyclosporine and tacrolimus may need to be increased to maintain pre-pregnancy trough concentrations. CNI levels typically decrease during the first trimester due to increased volume of distribution. Therefore, CNI levels should be closely monitored and doses should be adjusted to avoid acute rejection from subtherapeutic levels. MMF is considered to be teratogenic, and it should not be used in women contemplating pregnancy.[453] Sirolimus should also be avoided.

SURGERY IN THE KIDNEY TRANSPLANT RECIPIENT

ALLOGRAFT NEPHRECTOMY

Allograft nephrectomy is not commonly required. Indications include 1) allograft failure with ongoing symptomatic rejection causing fever, malaise, and graft pain; 2) infarction due to thrombosis; 3) severe infection of the allograft such as emphysematous pyelonephritis; and 4) allograft rupture. The morbidity associated with allograft nephrectomy is relatively high. Ongoing rejection in a failed allograft can sometimes be controlled with steroids, but prolonged immunosuppression of a patient who is on dialysis is obviously not ideal. Rejection in this context is less likely to be controlled by small doses of steroids when it is acute and when the transplant is recent. Whether nephrectomy of the allograft after a patient returns to dialysis offers a survival advantage remains controversial. A retrospective analysis of registry data noted a survival advantage in transplant recipients who underwent allograft nephrectomy after returning to dialysis.[454] However, any potential benefits of nephrectomy must be weighed against the heightened risk of HLA-sensitization around the time of allograft nephrectomy.[455] This may be of great importance for those patients relisted repeat kidney transplant. We generally only perform nephrectomy for symptomatic late rejection.

NONTRANSPLANT-RELATED SURGERY OR HOSPITALIZATION

Many kidney transplant recipients are hospitalized or undergo surgery for non-transplant reasons. Precautions for procedures and surgery in transplant recipients are listed in Table 70.20. Common-sense measures such as maintenance of adequate volume status, avoidance of nephrotoxic medicines (including NSAIDs), and proper dosing of immunosuppressive drugs are advised. Whenever possible, immunosuppressive drugs should be given by the enteral route. When not possible we use intravenous steroid and intravenous MMF or AZA, we prefer when possible to use sublingual tacrolimus, usually at half the oral dose, instead of intravenous tacrolimus. When necessary, intravenous tacrolimus should be started at one fifth of the total daily dose. Intravenous cyclosporine should be prescribed in slow infusion form at one third of the total daily oral dose.

THE PATIENT WITH THE FAILING KIDNEY

As for native CKD, management of anemia, hyperparathyroidism, hypertension, preparation for dialysis, and creation of appropriate dialysis access are important. If there are no contraindications, patients should be relisted for transplant. Once the patient returns to dialysis, immunosuppression should be weaned gradually. In patients who are likely to be re-transplanted in the near future, we maintain immunosuppression to avoid HLA sensitization. Active communication between the primary nephrologist and the transplant physician is critical to ensuring a smooth transition for the patient. In cases where management of the patient has been primarily through a transplant center, early referral to a primary nephrologist is important for preparation and initiation of dialysis. Conversely, patients who are managed primarily by a general nephrologist should alert the transplant center to coordinate the modification and weaning of immunosuppressive medications.

Table 70.20 Precautions for Procedures and Surgery in Kidney Transplant Recipients

- Caution with radiocontrast exposure
- Maintain hydration
- Avoid nephrotoxic antibiotics and analgesic
- “Stress-dose steroids” is not always necessary
- If enteral route of medication is contraindicated, give CNI via IV route (1/3 total oral dose)
- Monitor allograft function, plasma potassium and acid-base balance daily
- Consider wound-healing impairment with sirolimus

TRANSPLANTATION/ IMMUNOSUPPRESSION: THE FUTURE

Major areas of ongoing investigation include expansion of the donor pool,[456,457] optimization of immunosuppression regimens (with particular focus on individualizing immunosuppression), induction of tolerance,[458–460] and xenotransplantation. There is also growing interest in therapies that modulate B cell and plasma cell function, thus preventing or treating ABMR.

CONCLUSION

Improvements in short- and long-term kidney allograft survival have been encouraging. This reflects multiple influences, including more effective immunosuppression, more frequent use of living donors, and better medical and surgical care. The focus is likely to shift somewhat toward improving other posttransplant outcomes such as complications of immunosuppression, chronic allograft dysfunction, and morbidity from cardiovascular disease. Availability of adequate numbers of organs for transplantation remains an ongoing problem.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. A 26-year-old male patient who is 9 months posttransplant undergoes kidney transplant biopsy because of an increase in serum creatinine from his baseline of 1.2 mg/dL to 1.6 mg/dL. The biopsy shows prominent tubulointerstitial inflammation and SV40 positive inclusions seen throughout the cortex and medulla consistent with BK nephritis. His BK plasma PCR is 350,000 IU/ml. His current immunosuppression medication regimen consists of tacrolimus, MMF, and prednisone. What is the most appropriate first step in his management?
   a. Start monthly intravenous immunoglobulin (IVIG).
   b. Add oral levofloxacin.
   c. Start low dose cidofovir infusions every 2 weeks.
   d. Stop MMF.
   Answer: d

2. A 50-year-old woman with end stage renal disease from polycystic kidney disease undergoes kidney transplant. She receives induction immunosuppression with basiliximab and is started on tacrolimus, MMF and a steroid taper. She initially enjoys prompt allograft function but on day 4 posttransplant her urine output starts to drop and her creatinine goes up. Her laboratories are remarkable for worsening thrombocytopenia, anemia with schistocytes on her blood smear, low haptoglobin and elevated LDH. What is the next step in her management?
   a. Start antithymocyte globulin.
   b. Start plasmapheresis.
   c. Stop tacrolimus and start cyclosporine.
   d. Start eculizumab.
   Answer: c

3. Belatacept should be avoided in:
   a. HLA identical donor recipient pairs
   b. EBV seropositive donor/EBV seronegative recipient pairs
   c. CMV seropositive donor/CMV seronegative recipient pairs
   d. Recipients older than 65 years of age
   Answer: b

4. Which drug is most likely to significantly increase tacrolimus levels?
   a. Azithromycin
   b. Diltiazem
   c. Phenytoin
   d. St. John's wort
   Answer: b

5. Posttransplant lymphoproliferative disorder is:
   a. More common in adults than children
   b. Associated with EBV infection
   c. Most commonly T cell lymphoma
   d. Usually unresponsive to rituximab
   Answer: b

6. Features suggestive of antibody mediated rejection include:
   a. Positive peritubular capillary C4d staining
   b. Donor specific antibody
   c. Glomerulitis and peritubular capillaritis
   d. All of the above
   Answer: d

7. Which of the following is not a lymphocyte depleting induction therapy?
   a. Basiliximab
   b. OKT3
   c. Alemtuzumab
   d. Anti-thymocyte globulin
   Answer: a

8. In HIV-infected patients with end stage renal disease, kidney transplantation may be considered in individuals:
   a. With a CD4 lymphocyte count less than 200 cells/mm^3
   b. With a HIV viral load of 1000 copies/mL
   c. On a stable regimen of highly active retroviral therapy
   Answer: c

9. The development of post-transplant diabetes mellitus is most commonly associated with which treatment?
   a. Tacrolimus
   b. Sirolimus
   c. Cyclosporine
   d. MMF
   Answer: a